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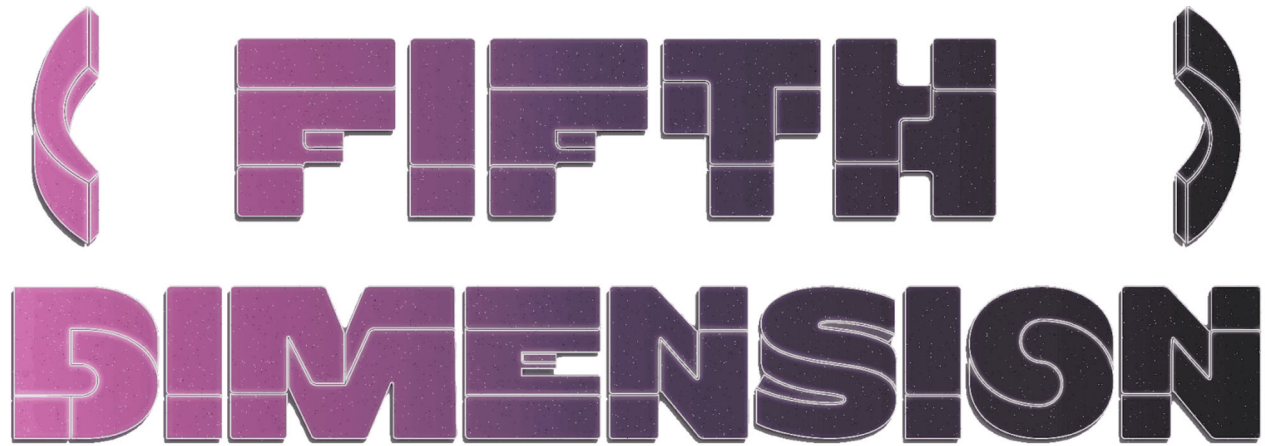
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Business Plan - Section C1, Team 1

Virtual Reality Entertainment Service

This business plan is the original work of the undersigned. All facts and figures are authentic. All contributions from others have been appropriately acknowledged. We have not reviewed or used any past Core plans in any way in the development of our plan. We did not misrepresent ourselves to potential suppliers, potential consumers, or to anyone else who contributed information to this plan.

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EXECUTIVE SUMMARY

Fifth Dimension is a virtual reality entertainment service with educational and informative offerings. We specialize in immersive VR experiences centered around traveling to historic locations around the world, through space and time. For \$40 customers can rent a VR headset for a two-hour period and freely use any of our pre-installed experiences during their time. Through Fifth Dimension, adventurous innovators, friends, and families, can have unforgettable experiences without ever leaving Boston. These features differentiate us from traditional entertainment services through numerous benefits, like our unique opportunities at an affordable price, immersive and interactive technology, and accessibility for individuals and groups alike.

Building this innovative service from the ground up has involved careful customer-centric planning. Through various research studies, we narrowed our target segment from the Greater Boston population to individuals with income above \$35,000, aged 20 to 50, who are adventurous and interested in VR but don't own a headset, totaling 810,000 potential customers. To generate more specific insights we designed and conducted a robust marketing survey totalling over 200 responses, 146 of which fell within our target segment. Based on their reported purchase intent at various price points, we created a demand curve to compute our chosen retail price of \$40 per 2-hour session. Respondents were also particularly interested in support for group activities leading to our 10% to 15% discount for parties of five or more. Once our target customers were set and understood, we began building brand awareness through in-depth IMC schedules for five years. These include custom advertisements from online, cable, transit, guerilla marketing, influencers, website, social media, and public relations vehicles, leading to awareness growth from 10% in year one to over 32% in year five.



Finally, we estimated an expected 25% repeat rate with five repeat occasions in years one and two, dropping to four occasions beyond that. Combined these metrics form our BASES model which projects a total revenue peak of \$3.7 million in year three before sustaining over \$3 million in the following years.

To fully evaluate the profitability of our business we must estimate our forecasted costs for the scope and duration of this plan. Our startup year costs consist of staff training, information systems expenses, and premarketing expenses, which total \$880,000 after a tax deduction. We expect approximately \$630,000 of that to be financed by outside investors. In year two we plan to expand our floor space by 1800ft² to match increasing demand, leading to an additional \$70,000 investment needed. After analyzing our operations expense, we concluded that most of our costs come from SG&A, mainly composed of rent and employee salaries. These are expected to substantially grow in year two with the aforementioned expansion before slowing considerably in years three to five. Beyond these costs we have capital expenditure of high-quality enterprise VR headsets, light inventory costs, among others, which are dwarfed by SG&A and the direct labor of IT Assistants. As such, the vast majority of our expenses are fixed, rather than per-unit variable costs, which combined with our high revenues makes us highly profitable. This is evident in our computed NPV of \$238,116 and IRR of 27.46% in year five, well above even our aggressive risk premium of 9% and required rate of return of 19.96%.

Our financial performance makes us viable, but our commitment to quality, reliability, and efficiency will make us long-lasting. We performed key ratio analysis to assess how well our business handles liability obligations, profitability, and operating efficiency. Fifth Dimension is projected to have a current ratio of 10.76, which surpasses the average entertainment industry current ratio of 0.68 by well



over tenfold. This indicates that our business can easily meet short-term liability obligations. Our gross margin of 85.75% in year five also outperforms the industry average of 77.26%, suggesting that we manage costs of goods sold considerably better than our competitors.

To ensure stability at these financial levels we thoroughly investigated both possible qualitative and quantitative risks associated with our business, including a global recession, more affordable VR headsets, new competing technology, and increasing GPU prices. With recent rate hikes by the Federal Reserve, a looming recession may have short-term detrimental impacts to our bottom line, but is unlikely to cause lasting damage. We still plan to diversify our model by researching tangential business opportunities (ex. Augmented Reality), locations, services, and offerings to decrease our systematic risk exposure in the market. Another possible concern is the increasing prices of GPUs due to the boom of cryptocurrency mining and artificial intelligence. To mitigate, we will maintain an updated database of VR headset prices and trends for related manufacturers, along with selecting secondary headset suppliers, in the unlikely event that an astronomical price change prohibits our bi-yearly headset purchase.

Finally, a sensitivity analysis concluded that our high-impact and high-probability quantitative risks are target segment size, repeat rate, purchase intent, and awareness, all marketing variables. Consequently, our mitigation plans follow this trend, as we embrace innovation to thrive. We plan to launch CSR initiatives and group discounts, raising purchase intent, monthly new experiences to attract and retain target customers, and unique guerilla marketing campaigns to increase awareness, all already accounted for in our current expenses. Above all we intend to remain as we began: customer-centric, and in doing so maintain high profitability and entertainment disruption throughout these five years and beyond.



OVERVIEW

Provides general Fifth Dimension information like features, benefits, and trends

Attributes and Benefits

Summary of features which differentiate our business from competitors

Fifth Dimension is a virtual reality entertainment service with educational and informative offerings. We specialize in immersive VR experiences centered around traveling to historic locations around the world, through space and time. For \$40, customers can rent a VR headset for a two-hour period and freely use any of our pre-installed experiences during their time. Through Fifth Dimension, adventurous innovators, friends, and families can have unforgettable experiences without ever leaving Boston. These features differentiate us from traditional entertainment services through numerous benefits, like our unique opportunities at an affordable price, immersive and interactive technology, and accessibility for individuals and groups alike.

Mission Statement

Our summary of mission which highlights our main goals

“To provide immersive and educational experiences that allow our adventurous customers to travel the world and beyond, without ever having to leave Boston.”

Trends in the Entertainment Industry

Unique behavioral and economic trends we noticed during our research process

The entertainment industry as a whole is a rapidly changing landscape with yearly macro and micro trends. Due to the in-person and group nature of these services, the impacts of the COVID-19 pandemic have been particularly felt, impacting countless aspects like operations, customer relations, and drastically reducing profitability. As such, we have carefully conducted research to build our



service from the ground up with these trends in mind so that we may proactively plan around the necessities of this modern industry.

Firstly, there is a trend of employee layoffs for leisure activity establishments expecting to increase due to heightened regulations and financial effects of the pandemic.¹ At Fifth Dimension, we will heavily value our employees and will do everything possible to decrease employee layoffs to as low as possible. Therefore, it is important to ensure that our service continues to be profitable with a positive NPV and IRR through controlled costs and a reliable revenue stream.

Given our inherently technological nature, trends in that space are particularly applicable. Entertainment services rely heavily on the use of credit card transactions, making cybersecurity and protecting guest data major priorities.² At Fifth Dimension, we will invest in quality POS systems that will protect consumer data and credit card transactions while streamlining our payment processing.

Recreation and entertainment services are generally viewed as discretionary spending, so consumers may spend less during periods of economic uncertainty.³ Our service has a dependence on consumer spending in order to make a profit. Currently as the Boston area is recovering from the COVID-19 pandemic and inflation is on the rise, many individuals are limiting their discretionary spending in order to ensure financial stability. To combat this, we carefully planned our surveys to collect accurate data and price our service as affordably as possible, maximizing potential customers.

¹ Zeidner, Rita. "Cutting Staff in Times of Crisis." SHRM. SHRM, May 5, 2020. <https://www.shrm.org/hr-today/news/all-things-work/pages/cutting-staff-due-to-coronavirus-fallout.aspx>.

² Uzialko, Adam. "Why Your Business Should Accept Credit Cards." Business News Daily. Accessed December 5, 2022. <https://www.businessnewsdaily.com/4394-accepting-credit-cards.html>.

³ Chung, Heidi. "Survey: Consumers Already Cutting Entertainment Spending amid Economic Woes." Variety. Variety, August 15, 2022. <https://variety.com/vip/survey-consumers-already-cutting-entertainment-spending-amid-economic-woes-1235318061/>.



Arts and entertainment organizations must also ensure that venues are accessible to customers, especially customers with disabilities.⁴ At Fifth Dimension, we took into consideration that many people with disabilities who can not go up the stairs may still want to participate in our service. During our renovations, the entrance from our lobby to the main area on the first floor contains both a ramp on one side and a small set of stairs on the other side. Customers may also request to have their experience on our first main floor rather than going down the stairs to our basement.

Institutions in the entertainment industry must invest in information systems to keep track of members and donors, and to plan renewal campaigns as trends show.⁵ At Fifth Dimension, we will be investing in a POS system, ERP system, AT&T Internet Access, a business phone line, an Apple iMac, website design, and cordless phones.

There appears to be rising internet penetration, user-friendly interface of web portals, enhanced discounts and offers, changing consumer patterns and purchasing power that are driving the growth of the e-retail market.⁶ Therefore, Fifth Dimension will invest in making a website that is both easy to follow and attractive to consumers. Much of our advertising will consist of online advertisements, influencers (nano, micro, mid-tier, macro), our website, and our own social media. We expect to use the expected rising internet penetration of consumers from our target segment to create more awareness of our service and

⁴ Greene, Leigh. "How to Make Your Events More Accessible and Inclusive | Splash." Splash, May 2, 2018. <https://splashtat.com/blog/accessible-event-planning>.

⁵ "ProQuest, Better Research, Better Learning, Better Insights.," accessed September 15, 2022, <https://www.proquest.com/>.

⁶ "EBSCOhost Research Platform: EBSCO." EBSCO Information Services, Inc., www.ebsco.com. Accessed September 15, 2022. <https://www.ebsco.com/products/ebscohost-research-platform>.



inform these consumers of the benefits we provide with these online advertisements.

Trends show that entertainment and recreation companies face great competition with various attractions that have bigger resources and are long-established.⁷ When creating our sales forecast, we took into account the amount of revenues that would be taken away by our competitors and made this percentage increase as the years went by. We expect to face serious competition, which is why in our marketing strategies we will try to differentiate ourselves from competitors as much as possible. Some of our main competitors include: Key to Amaze, Lucky Strike, Fenway Park, and AMC. As shown in *Exhibit 1*, Fifth Dimension is competing not only with other Virtual Reality companies but also with the entertainment services industry as a whole in the Greater Boston area. Compared to our competitors, Fifth Dimension appears to be very unique for the fact that none of our competitors provide the immersive experience of being able to time travel around the world and beyond while interacting with your friends. Our service also appears to be more educational than many of our competitors, while still providing a fun experience for consumers.

⁷ "Market Research." Market Research. Accessed December 5, 2022.
<https://www.marketresearch.com/Consumer-Goods-c1596/Travel-Leisure-c90/Entertainment-c227/>.

Exhibit 1: Competitor Comparison Table

Describes Fifth Dimension's competitive advantage over other entertainment services

Company					
Price⁸	\$40	\$100 ⁹	\$24 ¹⁰	\$98 ¹¹	\$36.50 ¹²
Service Type	VR Time Travel	Variety of VR Games	Bowling & Arcade	Live Events (Sports)	Movie Theater
Strengths	-VR offered -Location: Newbury Street -Time Travel -Educational-Immersive -Interactive -Group Discount	-VR offered -Location: Kenmore Square -Game Based Experience	-Location: Fenway -Premium bowling -Food and cocktails -Affordable -Group Discount	-Location: Fenway -Red Sox Games -Concerts-Large Space -Food and drinks	- Location: -Boston Commons -Immersive visual and auditory experience -Food and drinks
Weakness	-No Food -No Games	-No Food -Headset Requires Desktops -Expensive	-No VR -Long wait times -Very Loud	-No VR -Expensive -Crowded -Very Loud	-No VR -Less Interactive -Less Educational

⁸ per 2 hours

⁹ "Experience Amazing: Virtual Reality Experiences: Boston," Key To Amaze VR, accessed December 5, 2022, <https://www.keytoamaze.com/>.

¹⁰ "Boston," Lucky Strike, December 2, 2022, <https://www.luckystrikeent.com/locations/fenway/>.

¹¹ "Fenway Park: Boston Red Sox," MLB.com, accessed December 5, 2022, <https://www.mlb.com/redsox/ballpark>.

¹² "AMC Boston Common 19." Boston, Massachusetts 02111 - AMC Theatres. Accessed December 5, 2022. <https://www.amctheatres.com/movie-theatres/boston/amc-boston-common-19>.

CORPORATE SOCIAL RESPONSIBILITY INITIATIVES

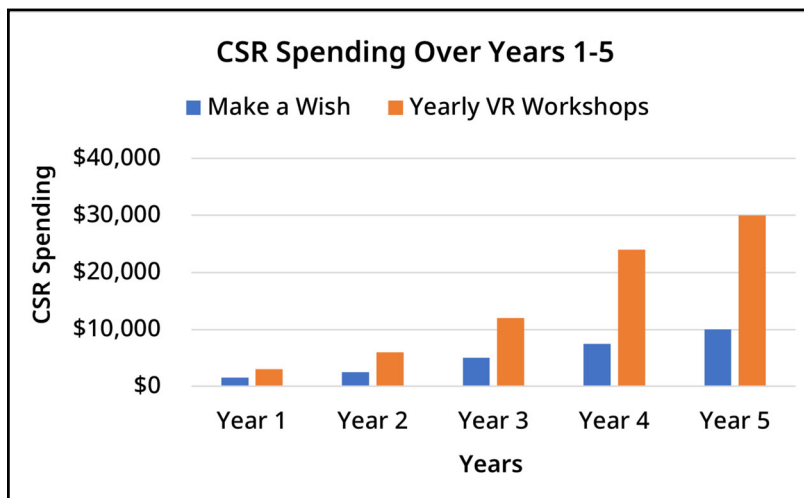
Provides information on two CSR initiatives of Fifth Dimension

VR Workshop for Underprivileged Communities

Implementation plan of our first CSR initiative as well as the cost and benefits it generates

For our first CSR Initiative we plan to work with Boston public schools and local non-profits to teach modern technology usage to those with limited access. Our IT Assistants are highly knowledgeable of advancements in recent technology and in-depth information regarding virtual reality, allowing them to design and run these workshops. During the workshops, we also aim to inform about global issues and incentivise sustainable habits, like offering a recycling VR simulation.

Our VR workshop for underprivileged communities will strengthen our educational aspects, as denoted on our positioning grid. These workshops are expected to cost \$3,000 each, covering staffing, training, asset creation, marketing expenses, and a small venue if needed. This CSR initiative may also attract new customers as new segments become more aware of virtual reality and our service. Responses to our questionnaire indicated that potential customers already familiar with VR are also more likely to purchase. Based on these results, the total CSR lift



Visualizes higher cost for workshop CSR initiative

Exhibit 2: CSR Spending Over Years 1-5

for the VR workshop for underprivileged communities would be 1.4% (*Appendix 1: VR Educational Workshop*). In year one we will host a pilot session of this workshop for \$3,000, increasing linearly to ten



sessions in year five for \$30,000 (*Exhibit 2*).

By implementing this initiative our purchase intent each year is expected to rise by 1.4%, increasing our revenues by around \$100,000 in year one and \$160,000 in year five. These increases in revenues are substantially higher than our expected costs to fund and advertise the workshops. Therefore, we will move forward with this initiative and will implement this into our IMC plan.

Make-A-Wish VR Sessions

Implementation plan of our second CSR initiative throughout the span of all five years

For our second CSR initiative, Fifth Dimension will be creating unique experiences for kids with terminal illnesses with the well known organization Make-a-Wish. In order to do this, we will be modifying our existing offerings to suit each child's requests. Our goal is to restore a sense of childhood fun and levity to these children while keeping true to the experience we hope to deliver to our customers.

These Make-A-Wish VR sessions reflect our focus on immersion shown in our positioning grid and are well within our budget to execute. Similarly to the VR workshop CSR, the purchase intent of prospective customers may increase due to this initiative. Based on our survey results the total lift for Make-A-Wish sessions would be 1.5% (*Appendix 2: Make A Wish VR Sessions*). In year one, Fifth Dimension will host sessions for the Make-A-Wish foundation that will cost us \$500 each, totaling \$1,500. These will also increase linearly up to 20 sessions totaling \$10,000 in year five. With this initiative, the purchase intent each year rises by 1.5%, increasing our revenues by around \$110,000 in year one and \$170,000 in year five. This increase is slightly more than the increase in revenues from the VR Educational Workshops, our other CSR initiative. Similarly, this raise in revenues is



far greater than the costs associated with holding these sessions, leading to the implementation of these Make-A-Wish VR Sessions into our IMC plan.

CSR Conclusion

Our incentives of implementing CSR initiatives and ensuring it is being accounted for

We believe in our responsibility as an innovative and technological business to give back to our communities and use our infrastructure for good. With the CSR initiatives detailed above we expect to increase brand awareness and raise purchase intent, all while contributing to local Boston groups. Finally, all CSR-related costs are reflected in our IMC schedule under the *VR Educational Workshops* line in the *CSR Initiatives* section (*Appendix 18: IMC Schedule Year 1 to 22: IMC Schedule Year 5*) and, consequently, billed under the *Marketing Expenses* line in our Income Statement.

MARKETING

Information on marketing overview, segment, awareness, location, pricing strategies

Marketing Overview

Detailed marketing plan as well as specific objectives to be achieved

Successfully marketing an entertainment service is crucial in order for the business to create brand awareness, attract consumers, and ultimately generate revenues. For these reasons, we set specific and measurable objectives, defining a marketing vision for our first five years. Our first objective was to narrow and target at least 10% of the Greater Boston population, equalling around 500,000 individuals. For our IMC plan, we set out to achieve a minimum awareness of 10% in year one and sustained growth of at least 2% year-to-year. In regards to location analysis, our goal was to only open a different second location if the cannibalization fell below 25% and would not require start-up marketing costs greater than \$20,000. Finally, our main pricing objective was to profitably provide our service at or below the \$50 price point our competitors charge (*Exhibit 1*). Combined these objectives created a clear road map for our marketing vision, accomplished through the plans and strategies detailed below.

Target Segment

Thought process and calculation justification of our target segment size

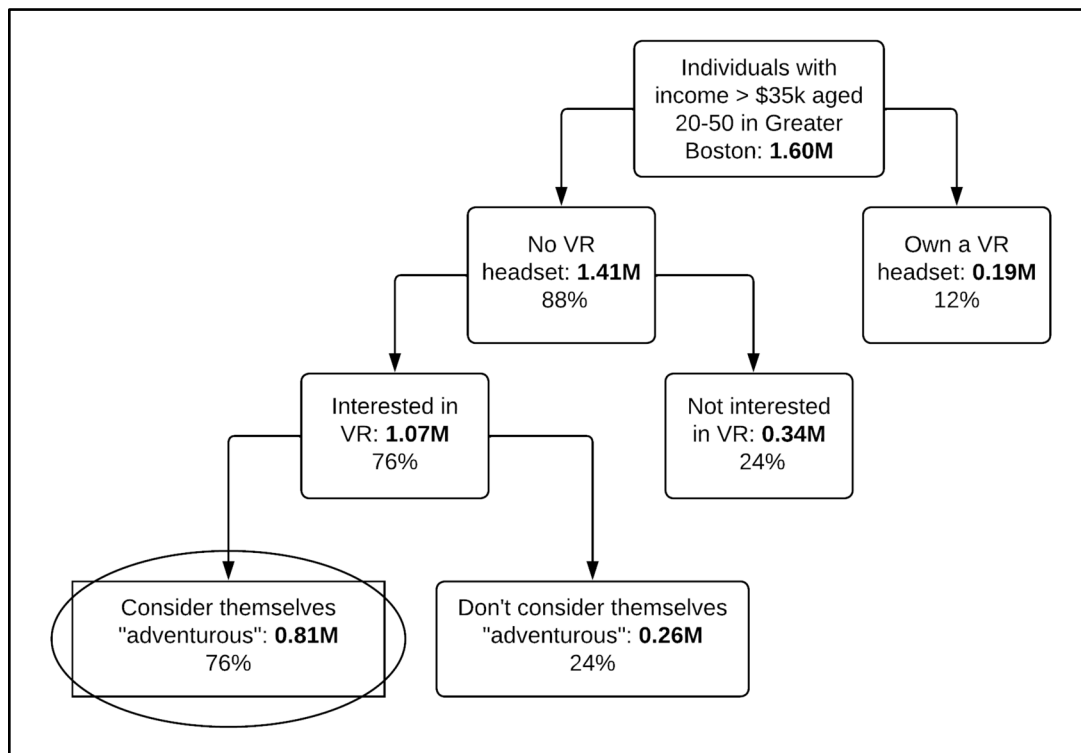
Segmentation is an integral step in the service life cycle as it decides our target audience and demographics, the implications of which are far reaching and affect almost every other marketing decision. We began our segmentation tree with the Greater Boston population of 4.9 million people¹³, as we are a Boston

¹³ Duffin, Erin. "Boston Metro Area Population 2021." Statista, September 30, 2022.

<https://www.statista.com/statistics/815215/boston-metro-area-population/#:~:text=Boston%2DCambridge%2DNewton%20metro%20area,in%20the%20U.S.%202010%2D2021&text=In%202021%2C%20the%20population%20of,also%20about%204.94%20million%20people.>



based service. We chose to include an income floor at the top of our tree by targeting individuals with an income above \$35,000, based on US poverty guidelines¹⁴, as it is unlikely that individuals below that could afford Fifth Dimension. Individuals under the age of twenty likely also find us unaffordable due to their financially dependent status as students or entry level workers. Those fifty and older tend to be less interested in virtual reality, comprising only around 10% of VR users, and may be at a higher risk of experiencing health issues in VR¹⁵. As such, our team decided to target individuals aged 20-50¹⁶, concluding the top of our tree. This census data narrowed our segment considerably to 1.6 million, at



Shows our targeted segment size of 0.81 million

Exhibit 3: Segmentation Tree

¹⁴ "Poverty Guidelines." ASPE. Accessed December 5, 2022.

<https://aspe.hhs.gov/topics/poverty-economic-mobility/poverty-guidelines>.

¹⁵ Blagojevic, Ivan. "Discover the World's Best Service & Software Providers." 99firms. Accessed December 5, 2022. <https://99firms.com/blog/virtual-reality-statistics/#gref>.

¹⁶ Bureau, U.S. Census. Explore census data 2020. Accessed December 5, 2022. <https://data.census.gov/all?q=Boston%2Bcity%2C%2BMassachusetts>.



which point we began adding segmentation variables more specific to our service (*Exhibit 3*).

We chose to target individuals who don't already own a virtual reality headset (88%)¹⁷ as those who do will prefer to use their own hardware at home rather than pay for our service. Intuitively, we also decided to only target people who have an interest in virtual reality, as those who do not are unlikely to visit our service. Lastly, we are basing the last part of our tree on a psychographic variable by targeting consumers who consider themselves “adventurous” (76%)¹⁸, as our differentiating factor from other VR services is our experience catalog offering travel to historic and iconic locations around the world and beyond. Our target segment ultimately ended up to be 810,000 people in year one, thereby meeting our marketing objective of at least 500,000 individuals. We created a segmentation grid which highlights our target segment compared to our non-targeted segments in terms of demographics, psychographics, key influencers, etc. (*Appendix 3: Segmentation Grid*).

When building our segmentation tree, our team considered several other possible variables such as education level, access to a car, visits to a museum in the past year, etc. However, we ultimately decided not to use these variables as they did not provide as accurate of a segmentation tree as the alternatives listed above (*Appendix 4: Unused Variables in Segmentation Tree*). We believe the targeting

¹⁷ Gilbert, Nestor. “74 Virtual Reality Statistics You Must Know in 2021/2022: Adoption, Usage & Market Share.” Financesonline.com. FinancesOnline.com, November 7, 2022. <https://financesonline.com/virtual-reality-statistics/>.

¹⁸ Schnitzer, Kyle. “This Is the Surprising Age When You Are Most Adventurous.” Ladders. Ladders, August 21, 2019. <https://www.theladders.com/career-advice/this-is-the-surprising-age-when-you-are-most-adventurous>.



strategy detailed above provides a clear market segment and strong foundation for our overall marketing plan.

Exhibit 4: Research Summary Table

Learnings from research via interviews, online, social media listening, and location visits

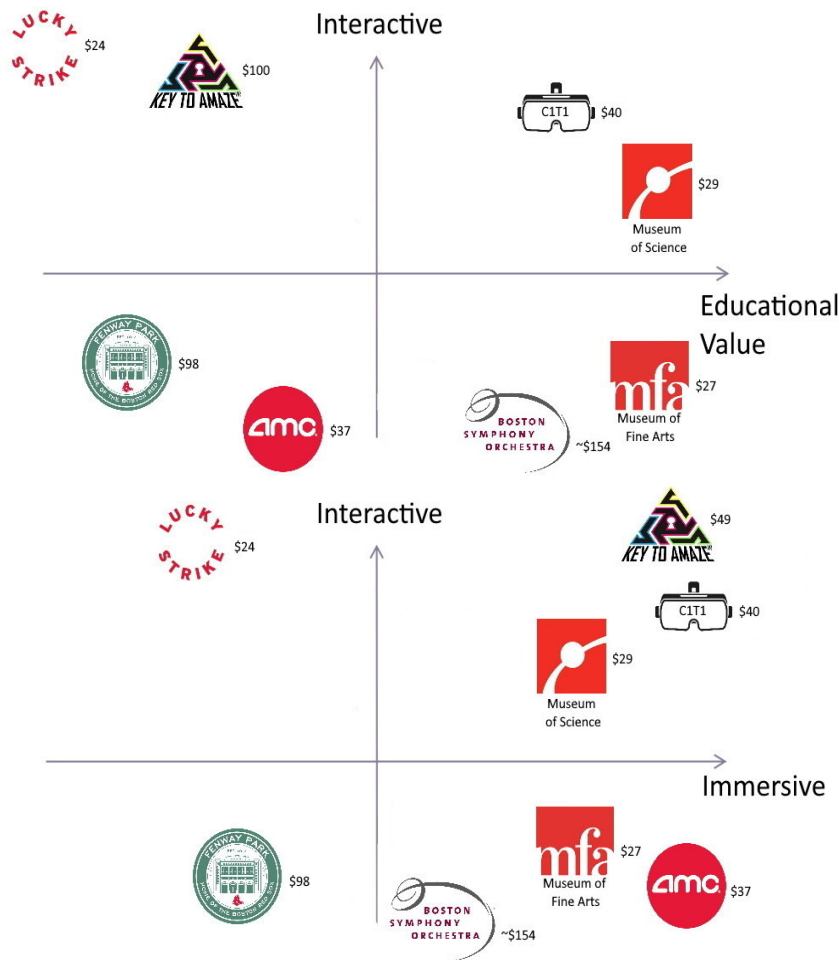
	Summary
One-on-One Interviews	- Majority have used VR before with mixed interest. - Emphasis on immersion, accessibility, comfort, & uniqueness. - High price strongly dissuades potential customers (\$60). - Some would go alone but most in groups (3-5 people). - Peak hours during weekend afternoons/evenings. - Must be within 35 min travel distance.
Industry Experts Interviews	- CEO and Staff at Key to Amaze: - Do not provide food due to legal hurdles - Importance of an open floor plan - Emphasis on social aspects of VR - Precautions to avoid safety accidents
Social Media Listening	- Importance of large physical space for group activities - Emphasis on customer service and high quality - Headsets must be kept clean
Location Visits	349 Newbury Street: - Very high foot traffic - Close proximity to shopping and food options - Draws Boston residents and tourists alike
Online Research	- VR is associated with leisure activities/gaming - Pleasant virtual environments shown to increase relaxation - Potential customers may seek stress and anxiety relief - VR has unique potential for new experiences - More popular with younger audiences



For the customer attributes in our positioning maps, we chose interaction, educational value, and immersiveness, which are three top-rated attributes in our House of Quality (*Appendix 27: House of Quality*). When it comes to VR experiences, interaction is crucial to make users engaged and fully immersed in the virtual

Exhibit 5: Positioning Maps

Fifth Dimension relative to entertainment services



prices per 2 hours

For adventurous residents in the Greater Boston Area with over \$35k income who are interested in VR but don't own a headset, Fifth Dimension is a VR entertainment service which is immersive and educational because it offers interactive and informative content, and is unlike Key to Amaze.

world. According to results from our research via one-on-one interviews, industry expert interviews, social media listening, location visits, and online research people are looking for more immersive experiences in entertainment services (*Exhibit 4*). Educational value has also been emphasized in customer interviews, and we believe that we can enhance our competitive advantages by incorporating educational and informative contents in

our service offerings (*Exhibit 5*).

In the segmentation variable test on *Income*, we excluded people with household income less than \$35,000. After this income adjustment, our revenue increased by approximately 31%, along with increased purchase intent and awareness (*Exhibit 6*). As a result, we decided to exclude people with household income less than \$35,000 from our target customers (*Appendix 7: Pivot Tables for Segmentation Variable Test on Income and Appendix 8: Bases Models for Segmentation Variable Test on Income*).

Exhibit 6: Segmentation Variable Test on the Income Variable

Revenue is expected to increase if we exclude people with income less than \$35,000

Income	
Before Excluding <\$35,000 PI = 16.15% Awareness = 8.00% Revenue = \$1,185,750	After Excluding <\$35,000 PI = 17.56% Awareness = 12.00% Revenue = \$1,555,071

In the segmentation variable test on *Age*, we excluded people younger than 20 years old and older than 50 years old. Although our revenue decreased by 28% after this age adjustment, we still decided to exclude people younger than 20 years old and older than 50 years old because the model we were using to calculate the revenue does not account for the real world factors (*Exhibit 7*). The model assumes everyone below the age of 20, including new born babies, and everyone above the age of 50, including people with impaired mobility, has a chance of coming to our business, which is not realistic and thus the model is not accurate and we cannot account for the result (*Appendix 9: Pivot Tables for Segmentation Variable Test on Age; Appendix 10: BASES Model for Segmentation Variable Test on Age*).

Exhibit 7: Segmentation Variable Test on the Age Variable

Revenue is expected to decrease if we exclude people between 20 - 50 years old

Age	
Before Excluding Age <20 & 50+ PI = 16.15% Awareness = 7.00% Revenue= \$2,075,063	After Excluding Age <20 & 50+ PI = 16.83% Awareness = 12.00% Revenue = \$1,490,668

Awareness

Our methods of generating awareness within our target segment and their impacts

Given the inherently innovative audience that our business targets, growing awareness was a fundamental part of our marketing strategy. Due to this, we targeted specific marketing objectives with our IMC plan, including a minimum awareness of 10% in year one, maximum of 33% in year five, and sustained growth of at least 2% year-on-year. These three marketing objectives were ultimately met throughout our layout of the five years of IMC schedules. As such, IMC vehicles



Online ads with our name, color scheme, and slogan

Exhibit 8: A/B Test of Two Online Ads

were carefully considered based on their reach, scalability, cost, and how applicable they were to our specific audience. These were separated into seven main categories, each described below and detailed in the included IMC schedules (*Appendix 18: IMC Schedule Year 1 to 22: IMC Schedule Year 5*).

Online Advertising: As we began building a plan to reach our technological audience, it quickly became clear that Online Advertising would play a central role due to both its fitting digital model and low CPM relative to other vehicles. Examples of two of our online advertisements can be seen in *Exhibit 8*. Using *Meta's Instagram and Facebook Ads Manager*, our segment size, and a 0.1% conversion rate from total impressions to people aware, we predicted awareness growth from approximately 2% in year one to approximately 6% in year five from online ads. These came at a CPM of around \$6 and approximately \$12 in those years respectively, or about \$8,000 per 1% awareness (*Exhibit 18-22*), making this one of our most efficient IMC vehicles, despite not being the highest in absolute cost or generated awareness.

Traditional Advertising: We explored three main vehicles in this category by researching their Boston-based opportunities and, like all vehicles, analyzing their CPM and applicability to our audience. Regional Magazines, like the *Boston Business Journal* and *Boston Globe Magazine*, had unfortunately limited information publicly



Digital draft of Fifth Dimension medium billboard
Exhibit 9: Medium Billboard Mockup

available, leading to uncertain CPM estimates in the tens of thousands per ad¹⁹, magnitudes more expensive than alternatives. Newspapers, like the *Boston Herald* and *Daily Free Press*, had similarly unavailable data but considerably lower estimated CPMs in the

¹⁹ "Bluegrass Unlimited to Boxing 2007 - Magazine Advertising Costs." Gaebler.com Resources for Entrepreneurs. Accessed December 4, 2022.
<https://www.gaebler.com/Magazine-Rates-for-Bluegrass+Unlimited++Boxing+2007+-+26359+-+26428>.



hundreds²⁰, still far too expensive. Finally, local Cable TV stations, like *WBZ-TV* and *WCVB-TV*, were found to be the most affordable and efficient of these vehicles with CPMs ranging from \$15 to \$50 for a thirty-second ad²¹. Consequently, Magazines and Newspapers were discarded from our marketing plan in favor of mixed Cable ads running nearly year-long through the duration of our service. These provide awareness ranging from approximately 2% in year one to approximately 4% in year five, costing approximately \$4,000 to \$6,000 respectively, which makes Cable by far our most efficient IMC vehicle at roughly \$2,000 per 1% awareness and a core driver of our IMC plan (*Exhibit 18-22*).

Public Advertising: Despite our clearly digital-leaning service, our public advertising vehicles account for the second highest cost and awareness in our plan. This is due to the inherently higher costs from physical assets and executions compared to digital ones. However, our main location targets foot traffic on Newbury, and survey results showed most target customers will be arriving on foot or by public transportation, making public advertising a worthwhile investment (*Appendix 24: Mode of Transportation*). Three vehicles make up this spending with each generating 1% to 3% awareness proportionate to their cost. Most of the public advertising cost was allocated to exterior bus transit ads at \$5,000 each²² for one to three weeks, twice a year, to capture public transportation riders. Just under

²⁰ "Cost of Newspaper Advertising in Boston, MA." Gaebler.com Resources for Entrepreneurs. Accessed December 4, 2022.

<https://www.gaebler.com/Cost-of-Newspaper-Advertising-In-Boston---MA>.

²¹ Pearce, Sean. "Boston Massachusetts TV Advertising Rates. Call 888-449-2526." Low Advertising Rates | TV, Radio, Digital | Local or National Remnants, November 18, 2022.

<https://howtobuyradioadvertising.com/boston-massachusetts-tv-advertising-ad-rates-888-449-2526/>.

²² "Boston Outdoor / out of Home Static and Digital Ads." Outdoor Advertising in Boston, MA - Get Boston Static and Digital Out of Home Ads (OOH & DOOH) Here. Accessed December 4, 2022.

<https://www.bluelinemedia.com/outdoor-advertising/boston-ma>.



half the budget went toward medium-sized billboards at \$2,500 per billboard²³ for two to three weeks, twice a year, to capture pedestrians (*Exhibit 9*). Signage accounted for the remaining approximately 10% of the budget, made up of a single sign for \$2,000²⁴. All in all, public advertising generates approximately 3% awareness for \$22,000 and approximately 5% awareness for \$46,000 in years one and five respectively, equalling about \$8,000 per 1% awareness (*Exhibit 18-22*). These vehicles were all designed and drafted with our main color scheme and style in mind, mirroring our logo's font, website's landing page, and overall tone (*Appendix 13: Signage*).

Guerilla Marketing: To match our progressive service and audience, two equally innovative marketing campaigns were developed. The first modified our billboard design with lenticular printing to create a headset image that shows different scenes depending on the angle viewed (*Exhibit 10*). This exemplifies the dynamic nature of VR in a unique way that is sure to catch the eye of potential



Digitally animated draft depicting dynamic lenticular printed headset billboard
Exhibit 10: Guerilla Marketing Lenticular Printing Billboard Mockup

customers walking by. We expect each billboard to cost around \$5,000²⁵ and correspondingly generate 0.8% awareness. Including a \$4,000 yearly city-wide overhead, we expect these billboards to generate 1.2% awareness for \$12,000 in

²³ "Billboard Advertising in Boston, MA (Suffolk County, MA)." Billboard Advertising in Boston, MA (Suffolk County, MA) - Rent Boston Billboard Ads Here. Accessed December 4, 2022.
<https://www.bluelinemedia.com/billboard-advertising/boston-ma>.

²⁴ "Signicade®." VistaPrint. Accessed December 4, 2022.
<https://www.vistaprint.com/signs-posters/signicade>.

²⁵ "Lenticular Printing, 3D Holograms." World3D Lenticular Printing, August 16, 2022.
<https://world3d.com/>.



year one and 3.5% awareness for \$36,000 in year five, increasing in count from one to four billboards respectively. The second guerilla marketing campaign involves placing giant plastic VR headsets on famous statues around Boston, like the dinosaur outside the *Museum of Science*. These will be 3D printed with cost-effective PLA plastic²⁶ and include an adjustable strap for a total estimated cost of ~\$8,000 per headset, generating ~0.8% awareness. Including a \$3,000 city-wide overhead, these statue headsets are expected to generate 1.1% awareness for \$11,000 in year one and 3.5% awareness for \$35,000 in year five, similarly scaling in count from one to four respectively. Combined, these guerilla marketing initiatives are our largest cost and awareness generator in the IMC plan, ranging from \$23,000 for 2.3% awareness in year one to \$71,000 for 7.0% in year five. This is roughly \$10,000 per 1% awareness (*Exhibit 18-22*), around average relative to our other vehicles.

Influencers: To build our online presence we chose to hire four influencers ranging in size from Nano, Micro, Mid-Tier, and Macro, with follower counts from under 10,000 to over 600,000 and costs per post of \$55 to \$3,250 respectively (*Appendix 23: Influencers Table*). These influencers post on two of the most popular social media platforms, Instagram and YouTube, and were chosen due to their existing VR content and related fanbase (*Appendix 12: Influencer Post*). In year one we expect a total of 66 posts made by two influencers, totalling approximately \$20,000 and 1.5% awareness. This scales linearly until year five when we expect 110 posts across 4 influencers, totalling approximately \$50,000 and 4.5% awareness. This results in roughly \$12,000 per 1% awareness (*Exhibit 18-22*), below

²⁶ "Material Guide." Craftcloud®. Accessed December 5, 2022.
<https://craftcloud3d.com/material-guide>.

average efficiency for our vehicles, leading to our choice to limit hiring to only four influencers over all five years.

CSR Initiatives: We have two main CSR Initiatives explained in the Corporate Social Responsibility section included before this Marketing section. These do not impact awareness, but rather purchase intent, and as such their impact cannot be directly compared to other IMC vehicles. Their cost, however, can be, and as it ranges from \$4,500 to \$40,000 in years one and five respectively the CSR Initiatives come out as one of the cheaper vehicles in the plan. For a more detailed breakdown, see the section referenced above.

Owned Media: This section does not generate awareness and as such, like the CSR Initiatives, cannot have its impact compared to other IMC vehicles. Despite this, its purpose is clear and deliberate, aiming not to bring in new customers but to retain existing ones by providing a sleek customer experience and increasing satisfaction. Our website lists available experience offerings as well as providing booking and payment capabilities so interested parties can easily browse and book our options. This will also decrease on site delays as less of our customers will need to book and pay in person. The website will cost \$15,000 yearly for upkeep



Depicts user experience of browsing available VR destinations on our website

Exhibit 11: Website Experiences Mockup

and maintenance costs, which is worthwhile for the value it adds. Our Customer Relations Manager will also run Fifth Dimension social media accounts on



Instagram and YouTube, regularly posting images, videos, and promotions to increase fan base interaction and engagement (*Appendix 11: Social Media Post*).

Earned Media: This section describes awareness gained indirectly through word of mouth, public relations, and carried over from the previous year. Word of mouth is expected to steadily grow to 3% awareness in year three and remain there for the following years for no cost, making it incomparable to other vehicles but important and effective. A fixed yearly budget of \$10,000 has been allocated to public relations, handled by our Customer Relations Manager. We expect this expense to be largely unnecessary and has been allocated as a contingency in the case that damage control after an incident is appropriate. Carryover from year to year follows a similar trend to the aforementioned word of mouth in which we expect its awareness to steadily grow to 2% in year two and maintain that for the duration of the IMC plan.

Awareness Conclusion: The IMC plan has been carefully built with the sections above to increase our brand awareness, create sustainable growth, and meet our marketing objectives. The cost and awareness impacts of these decisions can be summarized in the table below (*Exhibit 12*), highlighting the cost effectiveness of our traditional advertising driven by cable and, conversely, cost ineffectiveness of paid influencer posts by comparison. These ratios can be used moving forward to dictate the focus of future spending so as to minimize costs, maximize awareness, and maintain a sustainable business model past inception.

Exhibit 12: IMC Efficiency Table

Lists IMC categories by their average cost per 1% awareness over years one through five.

IMC Efficiency				
IMC Category	Average Cost	Average Awareness	Cost/1% Awareness	Cost/1% Rounded
Online Advertising	\$32,473	4.18%	\$7,760	\$8,000
Traditional Advertising	\$5,865	2.98%	\$1,968	\$2,000
Public Advertising	\$35,200	4.24%	\$8,302	\$8,000
Guerilla Marketing	\$50,200	5.00%	\$10,043	\$10,000
Influencers	\$29,999	2.50%	\$12,000	\$12,000
Earned Media	\$11,000	3.00%	\$3,667	\$4,000

Location

Displays the analysis that went into selecting the optimal location for Fifth Dimension

For entertainment services, location can be crucial as it defines the accessibility, rent cost, and overall audience exposed to the business. As such, we carefully researched available spaces in the Greater Boston area and rated them based on the factors listed below (*Exhibit 13*), with cost being considered the most important and aesthetics the least. Of the four locations listed, 349 Newbury Street edged ahead with incredible foot traffic and accessibility.

Exhibit 13: Locations Factor Rating Table

Shows potential locations ranked and weighted by relevant factors

	Weights (1-10)	39 Harvard Street	1054 Beacon Street	65 Sprague Street	349 Newbury Street
Size (in ft2)	8	4	2	9	6
Cost (per ft2 per year)	10	8	6	10	2
Aesthetic	4	5	7	1	8
Accessibility	5	9	10	3	9
Foot Traffic	7	5	8	1	10
Total		212	210	198	215



349 Newbury is a proven location previously occupied by the retail company GNC. Located in Back Bay, it is within the premiere shopping and dining destination in Boston, with over 450 shops and 200 restaurants in close proximity. Each year more than 50 million visitors spend over \$5 million each day in Back Bay, making this a great place to attract people within our target market²⁷. When developing our main location, we also considered the mode of travel that our target segment preferred. According to our market research survey, out of 145 respondents over 60% preferred to get to our location through public transportation or by walking (*Appendix 24: Mode of Transportation*). As a result, 349 Newbury Street, located within the most walkable part of Boston and with extensive infrastructure nearby, is the ideal location for our service.

For future expansion we considered 39 Harvard Street, 1054 Beacon Street, and 65 Sprague Street as potential second locations. 39 Harvard Street and 1054 Beacon Street are centrally located in downtown Boston, like our main location, and as such showed high amounts of cannibalization (*Appendix 25: Cannibalization*). Consequently, these options were deemed infeasible and eliminated from our analysis. 65 Sprague St, located a 25-minute drive from Back Bay, is considerably further removed. As such, while it has the far lower foot traffic and accessibility scores seen above (*Exhibit 2*), it showed only 29% cannibalization. We assumed that 50% of the customers in the overlapping 30 minute radius would switch to the new location. Because of this, despite the relatively lower cannibalization in this location, the extra cost for awareness, large initial investment, and objectively worse locale make it infeasible. Following our marketing objective, since 65 Sprague Street has more than 25% cannibalization rate, we concluded not to open

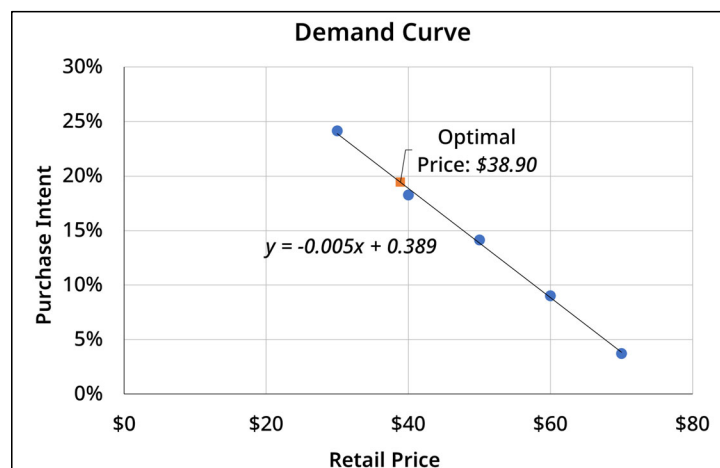
²⁷ Yardi Systems, Inc. "349 Newbury St - Space Availability," 349 Newbury St, Boston - Retail Space For Lease, accessed December 5, 2022, <https://www.propertyshark.com/cre/commercial-property/us/ma/boston/349-newbury-st/>.

this second location. Instead, we chose to expand our main location into the basement, thereby increasing our capacity by 1800ft² with no additional cost of overhead beyond rent and renovations.

Pricing

Discusses how we developed the pricing model of our business

Pricing is notoriously one of the hardest factors to correctly estimate, and yet one of the most crucial revenue drivers for a business. As such, we prioritized creating a robust pricing strategy based on consumer preferences, target costing, and required returns for our service. From a carefully constructed marketing survey, we were able to compile consumer purchase intents at various price ranges (*Appendix 15: Price vs Purchase Intent Tables*). From this pivot table we predicted that out of the individuals who answered that they would probably and definitely buy



Visualizes linear relationship between PI and price

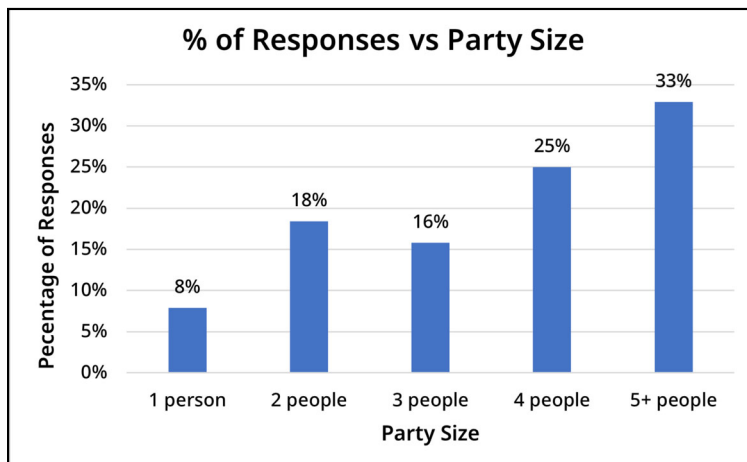
Exhibit 14: Demand Curve

our service only 30% and 80% respectively would ultimately purchase. From this, we were able to calculate the cumulative purchase intent and, subsequently, the quantity sold at each price point based on our overall target segment of 810,000 in year one. Finally, we multiplied

the quantity sold at each retail price to compute predicted revenues (*Appendix 16: Revenues Corresponding to Retail Price*). By plotting this adjusted purchase intent at the retail price points and applying a linear line of best fit we created our demand curve (*Exhibit 14*). The equation generated by this fit allowed us to use Excel Solver and compute our optimal retail price of \$38.90. As such, we decided on our final

retail price of \$40 for a 2-hour experience per person and calculated the total purchase intent to be 21.1% at this price point. (*Appendix 17: Adjusted PI at Optimal Price Table*). As competition increases in years three and four we expect to drop this price to \$38 and \$36 respectively to prevent new entrants from undercutting our service and maintain a high purchase intent.

Group Discount: After setting our retail price, our team began exploring discount options. Our marketing questionnaire survey results indicate that a group discount for party sizes of five or more would be the most applicable as an estimated 32.86% of our target customers fall into that category (*Exhibit 15*). Due to the discounted price these groups will pay, purchase intent will rise in accordance with our demand curve (*Exhibit 3*) shown above. In year one we decided to offer a 10% discount as we expect individual innovators to largely make up our customer



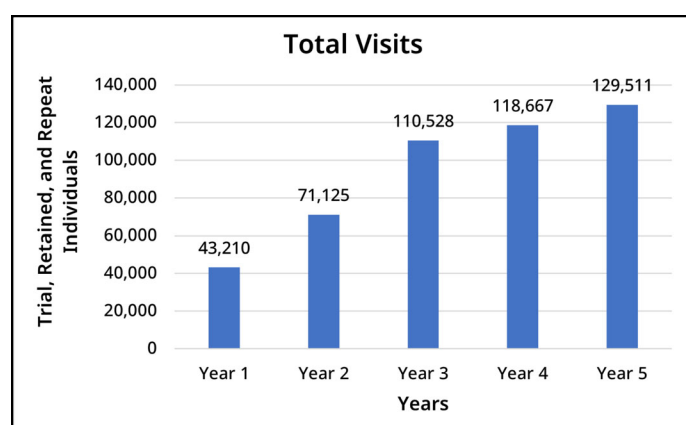
33% of demand is will qualify for the group discount

Exhibit 15: % of Responses vs Party Size Graph

base. In years two and after this discount increases to 15% for groups of five and more. This group discount may also encourage the 25% of target customers who planned to attend our service with a group of four to potentially add another member to their party

to qualify for the group discount (*Exhibit 15*). We plan to instate this discount in year one and advertise it on social media, like Instagram and YouTube, through our Customer Relations Manager. Overall, we expect this group discount to encourage larger party sizes, raise purchase intent, combat potential competitors with similar retail prices, and run throughout the duration of our service.

BASES Model: We generated a dynamic BASES model to represent the marketing strategy detailed above. Our target segment starts at 810,000 in year one and has a growth rate of 2% each year based on the Boston population growth rate²⁸. Adjusted purchase intent is calculated as a weighted average between the purchase intent at the regular retail price and at the group discounted price weighted by the percentage of customers expected to receive the discount. The retail price is also a weighted average of the regular retail price and the group discounted price with the same weighting. Based on this we calculated our trial



Trend of total visits plateauing in year four

Exhibit 16: Total Visits Graph

and retained individuals for each year, increasing somewhat linearly from around 20,000 in year one to 80,000 in year five. Our repeat rate is set as 25% for all five years based on data from our marketing survey as we intend to frequently introduce new experiences to

ensure customer retention. Similarly, the number of repeat occasions for each individual was dependent on those survey results, predicted at five in the first three years and four for the remaining two. Our trial, retained, and repeat individuals sum to our total yearly visits, expected to rise linearly before slowing in year four (*Exhibit 12*) due to the aforementioned decrease in repeat occasions. This decrease is attributable to some of our customers being less likely to attend our service on as many occasions as competition enters and our novelty wears off. Starting in year two we expect competition to decrease our revenue by 10%,

²⁸ Open data network. Accessed December 5, 2022.

https://www.opendatanetwork.com/entity/1600000US2507000/Boston_MA/demographics.population.change?year=2018.



increasing by 10% each year thereafter, up to a 40% competition adjustment in year five. This rate hike is particularly aggressive as virtual reality is a growing industry in the entertainment service sector, making it likely our service will face heavy competition that will increase as time passes. Our total revenue peaks at year three at \$3.7 million and decreases slightly in years 4 and 5, with the revenue staying above three million. Overall, it is clear that we were able to meet our objective of being able to provide our service for \$40 (the optimal price (*Appendix 16: Revenues Corresponding to Retail Price*)) and provide a group discount simultaneously, while still achieving high revenues.

Marketing Conclusion

Provides summary of the relevant components of Fifth Dimension's marketing plan

From calculating our target segment and generating awareness, to analyzing different second locations and pricing our service, our marketing plan contains many crucial decisions and fundamental objectives. We are confident that this strategy will sustain our service for the first five years and provide a base to build further into the future. Fifth Dimension was able to meet all of its established goals by targeting the correct consumers, making those consumers aware of our service, and providing a price that they would be willing to pay, concluding our marketing plan.



OPERATIONS

Breaks down operations through the service blueprint, design challenges, supply chain structure, location analysis, staffing plan, resource scaling, inventory, and logistics

Operations Overview

The master plan of our business's operations as well as all the contributing factors

The design process of operations addresses considerations taken into account while developing and growing our business, its inventory, and logistics. The main areas addressed consist of design challenges, supply chain structure, location factor analysis, layout design, staffing plan, resource scaling, and capacity utilization. Design challenges are discussed and addressed along the way as our business is projected to grow throughout the five years and beyond. During the startup phase of development we performed location factor analysis to determine where our facility would be the most optimally located. Our service layout design for that facility will greatly affect our business model and the number of customers that can be served, making it a high priority. Our business did not have many supply chain structure concerns because our operations are not heavily reliant on our suppliers besides Meta, a reliable tech giant which supplies our VR headsets. We plan to target an average capacity utilization rate of 85% in the long-term, and our staffing plan and resources will be scaled accordingly to meet our target demand.

Service Blueprint Overview

Utilizing our service process flow to ensure customer satisfaction and task efficiency

To effectively satisfy customer needs, we accept both walk-ins and online/phone reservations. As shown by our service blueprint (*Appendix 32: Service Blueprint*), when customers arrive, receptionists greet them and check the availability for them. Under the assistance of the POS system, receptionists calculate total expenses and receive payment from the customers. After IT



assistants bring customers to the main floor and explain basic rules and safety precautions, customers pick an experience of their choice. The IT assistants help the customer put on the headset to begin the experience. During the experiences, IT assistants regularly check on customers and assist them with switching experiences. After customers complete the two-hour period, receptionists thank customers for coming in, and IT assistants and janitors sanitize the headsets.

Design Challenges and Trade-Off

Space allocation, place of location and maintaining immersiveness for our business

When considering design challenges we emphasized on making sure our customers have enough space to interact and get immersed within the VR program, while preventing customers from running into each other. In our floor plan design (*Appendix 26: Facility Layout*), we prioritized maximizing the ft² per headset to allow customers to experience the maximal range of movement and walking space, while minimizing collision risks. In order to maximize the amount of floor space for our customers, we will be offering open-floor design rather than individual rooms. This design not only maximizes floor space for our customers, but also allows us to oversee the whole facility without the restrictions of walls. As such, we can respond to any accidents as soon as possible and further prevent incidents from happening in the first place. As a VR business, we are aware that some customers may feel visually distorted during our virtual experience, which may lead to fall risks. One of our business objectives was to ensure that we maintain a server to customer ratio of at least 1:13 even during our busiest times (*Appendix 30: Peak Weekends Scheduling Year 1*). By maximizing the server to customer ratio, we can ensure that the safety of our customers is carefully

	T / Bus	Walk	Car	Uber/Lyft
Count	49	41	31	21
% of total	33.79%	28.28%	21.38%	14.48%

Popularity and preference of the modes of transportation

Exhibit 17: Transportation Preferences



monitored at all times. Our programs are also designed to minimize the walking space and movement required for the experience.

After analyzing our marketing survey results, our data suggests that 75.86% of our customers arrive via the T-Transit, bus, walking, or Uber, and only 21.38% of our customers will be commuting by driving (*Exhibit 17*). We concluded that our customers preferred that our location be easily reachable through public transportation or walking (*Appendix 24: Mode of Transportation*). Therefore when deciding upon 349 Newbury St as our business location, we prioritized accessibility as a main factor. There is also an abundance of parking options within a 15 minute radius near our service, which resolves our parking concerns.

Immersiveness is a key aspect in our value proposition and one important component of that is sound. As some of our customers may include children, one of our design challenges was to ensure that families with kids do not disturb the experience of other customers. Our service experience requires that the user be at a minimum age of 13+ due to Oculus's safety standard of use²⁹. Families accompanying toddlers and young children are advised to stay near the sitting area by the reception with at least one accompanying adult. Since our VR experience and reception areas are separated, any excess noise should not affect our customers.

Bill of Material

Provides with a reason and explains why we do not have a bill of materials

Since our service model does not require us to make or assemble any of our final products. Therefore, there is no bill of materials for the Fifth Dimension.

²⁹ "Oculus Safety Center." Oculus. Accessed December 6, 2022.

<https://www.oculus.com/safety-center/#:~:text=Designed%20for%20Age%2013%2B,or%20use%20Meta%20VR%20Systems>.



Supply Chain Structure

Structure of our supply chain and describes and our process of maintaining sufficiency

The overall structure of our supply chain includes e-commerce suppliers and Meta. As our service does not require us to make or assemble anything, we decided to purchase all of our raw materials directly from suppliers. Our main and sole supplier for VR headsets is Meta, the parent company of Oculus, and a renowned leader in the virtual reality space. Our inventory consists of bottled water, cleaning supplies, and other miscellaneous items, which we intend to purchase from a large seller, like ecommerce giant Amazon, or wholesaler, like Target. Due to the simplicity and abundance of our inventory items we predict it is unlikely we will face serious shortages or price changes. Further, as these items are available from so many outlets, we can easily minimize transportation costs and even switch suppliers with little to no switching costs.

Location Analysis

Factors we've considered for analysis and our justification of location selection

When researching our location we considered foot traffic, square footage, rent per square foot, and accessibility by transportation as our main concerns. After considering these variables, we narrowed our options to 349 Newbury St, 39 Harvard St, 1054 Beacon St, and 65 Sprague St. We conducted a factor analysis on these locations and concluded that 349 Newbury was the best option after weighing various factors (*Exhibit 13*).

Operations Production Process

Shows the basis of our production process

Unlike most businesses, Fifth Dimension does not have any operations production process because we do not produce any finished products. Since we



have no working stations, our business is not concerned with throughput time or cycle time. To maintain satisfaction and sanitation standards we have decided to allocate one bottle of water, one Lysol disinfectant wipe, and 0.16fl oz of hand soap for each customer that is served, comprising all our direct materials.

Our capacity planning is based on our service blueprint, which is constant through all five years. It is important for us to be efficient with the use of space, so we will not be over the long-term targeted capacity of 85%. Each customer action from the service blueprint should result in flexible adjustments of on-site staff members and other key resources.

Layout Design and Process Breakdown

Provides an overview of what our space will look like for our customers and employees

As we just started in year zero, our location will be on the ground floor on Newbury Street with a total area of 2645 ft² (*Appendix 6: Assumptions*).

As laid out on our location design in *Appendix 26: Facility Layout*, when customers first walk into our location, they will find themselves in the main lobby where they can stay and wait for everyone in the group to show up, but if everyone is present, they can head straight to the small stairs at the entrance and they will check in with our receptionist located on their right. The stairs at the entrance are a combination of half stairs and half ramp, for easier transportation as well as accounting for customers with impaired mobility. Besides where the receptionists are located, the front right section of the business will serve as the offices for our main staff as well as the storage area.

The front part of the main office is used for the receptionists to greet our customers as well as having a VR headset storage right behind the receptionist, handing our customers the number of headsets they needed for the experience. While the middle part of the main office is where our main storage room is located.



We will be storing the hygienic supplies used for the business, as well as all other miscellaneous inventories to keep the business in good shape. Last but not least, the further back section of the main office will be our back office where the manager and the CEO will be resting and performing tasks.

Moving back to our main floor, we have fully utilized our floor space and offered the most amount of space for our customers to have fun in. While you will find our bathroom at the far back right of our business where it is the all-gender bathroom that everyone has the full ability to use. As we projected in the second year of operating, we will expand to the basement of our current location. That will give us an additional 1800 ft² of floor space solely meant for the fun area, where we will be fully utilizing the whole basement specifically for the customers. And in order to proceed to the basement of the business, our customers would just have to take the stairs right across from where the receptionists are.

Staffing Plan

Discusses how we developed our location's staffing requirements

In order to make sure that all of our customers are being served and our demands are being fulfilled, not only will we need enough floor space for all the customers, but also ensure that we have enough IT assistants for the customers. The method we used to determine the number of IT assistants is the projected average non-peak demand for year one, then accounting for the amount of time one IT assistant needs to wipe the headset and the startup time to get the headset running. By using this method, we were able to determine the number of IT assistants needed for both peak days as well as non-peak days during the year. For year one, due to low demand for our newly developed business, our calculation model suggested that we should hire two IT assistants in year one for both peak and non-peak days (*Appendix 29: Capacity Analysis*). While we took another step



forward and planned out the demand schedule for peak and non-peak weekdays, and peak and non-peak weekends. At the service rate of handling 20 customers per hour, the two IT assistants we hired will just be enough to cover all of our demands even during our peak days. As we continue the planning process for year five, we used the same calculation model for year five with peak and non-peak days, we will need to hire four IT assistants during non-peak weekdays and non-peak weekends at the service rate of 17 customers per hour. While needing 6 IT assistants to meet the full demand of peak weekdays and weekends at the service rate of 11 customers per hour (*Appendix 31: Peak Weekends Scheduling Year Five*).

Resource Scaling

This section mainly goes over the expansion of our location's footprint

The key resources in our business include VR headsets, physical space, and utilities that concern energy consumption. We are not concerned about the supply of bottled water and cleaning wipes because there are various distribution channels and the switching cost is low between suppliers. The capacity is limited by the amount of space available for customers to move around because we want to guarantee each customer at least 49ft² to minimize collision and fall risks. Our long-term goal is to meet an average capacity utilization of 80% to 85% because we want to keep the wait time low in ques. In year one we plan to lease 2645 ft² which allows us to install 21 headsets and reach a planned average capacity utilization of 42.69% in year one (*Appendix 29: Capacity Analysis*). We decided to expand our business by leasing an additional 1800 ft² in year two, which increases our total square footage to 4445 ft² and our average capacity utilization to 41.82% in order to meet our forecasted demand growth. The demand growth is calculated by subtracting the most likely demand uncertainty in year one from year two and

dividing the difference by the most likely demand uncertainty in year two (*Appendix 28: Demand Uncertainty Table*). The decision to expand is due to our forecasted demand growing 64.60% from year one to two, which would have increased our average capacity utilization to 70.27% (*Appendix 29: Capacity Analysis*). In order to decelerate the growth of the average capacity utilization we decided on additional space. We forecast the planned average capacity utilization to be 65.93% in year three, growing steadily and reaching 76.14% in year five

Year	Year 5
Capacity (yearly)	170,089
Average Demand Fulfilled	129,511
Average Capacity Utilization	76.14%

Calculation of year five utilization ratio

Exhibit 18: Year Five Capacity and Utilization

(*Exhibit 18*). Our target number of headsets at our business will increase based on the peak demand fulfilled hourly (*Exhibit 13*). Fixed investments on headsets will increase to meet our annual demand growth. We have chosen to acquire a number of headsets based on the peak demand fulfilled to accomplish our objective of minimizing wait time.

Managing Inventory

Explains the relevant details around our inventory needs

Our inventory consists of non perishables: bottled water, Lysol disinfectant wipe, Softsoap hand soap, and other miscellaneous items that are mainly use for cleaning and maintenance of the parameter (*Appendix 40: Inventory Year One to Appendix 44: Inventory Year Five*). Due to relatively small order quantities, we have decided to order via Amazon or other online Ecommerce websites. In the event of an emergency, we can easily switch to local distributors such as Target or Star Market with minimal to no switching costs. Our support services include many



online subscription based services that are mostly associated with our overhead operations. These costs include the Payment Operating System, Enterprise Resource Planning system, internet, and business phone line. Our virtual reality experiences are leased from a VR program maker, which eliminates the need for us to develop our own programs. Since these costs are relatively insignificant compared to our total cost, we do not expect a change to affect our financial viability in a meaningful manner.

Logistics Considerations

Provides considerations and estimates for our costs on logistics

Since our business solely deals with inbound logistics of inventory goods, we do not need to consider any measures for outbound logistics. Inbound logistics are handled by the receptionist, where they will track our order and sign when received. The logistic companies used to transport our goods will most likely be Amazon or a company chosen by the vendor. We used 10% of the total cost of our inventory goods as an estimate for logistics cost.

Defining Quality

This section defines our standard of quality for service based on our house of quality

From the interviews we conducted, we were able to conclude that by providing quality service that our potential customers are expecting, they will be more likely to come back and become retained customers. Some of the most important quality attributes they are looking for in our business are a safe, comfortable, and immersive environment where they can spend their quality time (*Appendix 27: House of Quality*). By safe service, our potential customers can be confident that employees of our service will constantly be checking on them, making sure to prevent accidents from happening in the first place. Workers are aware of potential safety issues that can happen and know how to prevent them, in



case of occurrence instructed on how to act in these situations. Elaborating on the comfortable quality, customers do not have to worry about the issue of having enough space for each individual who is experiencing our service. By having comfortable and cleaned headsets, our customers will not be affected by low quality headsets which could cause potential discomfortness and end the experience earlier. By immersive experience our customers are demanding a smooth and uninterrupted experience without program errors which will let them be fully integrated and sinking into the process (*Appendix 5: Summary of Interviews Table*).

In order to ensure the quality is always on the top of everything for our service, we will train our employees and have a clear set of instructions for them of what they need to do during both routine work and under unusual circumstances. To provide maximum comfortability and reduce overcrowding of the customers, we will be offering 75% of our total floor space to ensure all our customers will have at least 49 ft² of movement space, even at max capacity. Our company will also ensure to keep the headsets up-to-date available in terms of its hardwares and software systems. By carefully wiping and cleaning after each session with antiseptic wipes, our customers can ensure no contamination will occur from headsets to headsets and not worrying about hygienic issues. At the same time we understand that it is important to maintain technology from malfunctioning, we will have IT assistants who are responsible at all times for any unexpected hardware maintenance and ensure all headsets are ready to be used at any time. We will work our very best to ensure that we do not miss any of the service gaps, and we will ensure that our employees are fulfilling any knowledge, standards, delivery, and communication gaps.



Company Organization Breakdown

Presents the organization structure of the Fifth Dimension from years one through five

Our organization structures for five years are shown in (*Appendix 33: Organization Chart Year One - 37: Organization Chart Year Five*), which outlines the relationships of various roles in our company. We will pay all employees on an annual basis, and employee training takes place one year before their onboardings. In year one, Fifth Dimension consists of one CEO, CIO, VP of Operations, janitor, receptionist, and two IT assistants (*Appendix 38: OM Labor Year One*). The total administrative expenses in year one, including both salaries and training costs, amount to \$546,000. Overhead costs include general operating expenses, marketing expenses, IS expenses (*Appendix 47: IS Expense Table*), and software rental expenses, which equal \$680,000. In year two, we expanded the workforce by adding one CFO, expense manager, janitor, customer relationship manager, and two IT assistants. The total administrative expenses and overhead costs increase to \$859,000 and \$1,090,000 respectively. In year three, we added two CMO and IT assistants to our company, increasing total administrative expenses and overhead costs to \$958,882 and \$1,135,000. Our organizational structure in year four remains the same as that in year three, and the growth rate in employee salaries accounts for the slightly increasing total administrative expenses (\$1,011,000) and overhead costs (\$1,198,000). In year five, we added one more IT assistant (*Appendix 39: OM Labor Year Five*), and total administrative expenses and overhead costs are \$1,031,000 and \$1,239,000 respectively (*Exhibit 19*).

	Year 1	Year 2	Year 3	Year 4	Year 5
Administrative Expenses	\$545,650	\$858,220	\$958,882	\$1,010,360	\$1,030,144
Overhead Expenses	\$669,277	\$1,089,387	\$1,134,363	\$1,197,816	\$1,238,986

Shows Administrative and overhead expenses from year one to five

Exhibit 19: Expenses Table

Operations Conclusion

Summarizes the most critical aspects of operations for our business

In conclusion, our business is most concerned with location factor analysis, layout design, staffing plan, resource scaling, and capacity utilization, but we address them through five years. Our capacity utilization will continue to grow throughout the five years and beyond. We expect to reach an average capacity utilization of 76.14% in year five, which would require us to have between 4 to 6 IT assistants onsite to assist our customers at all times. Our operation would require us to have 63 VR headsets in year five available for customers in order to fulfill our expected peak hour demand and we will ensure to have enough headsets to meet the demand.



FINANCE

Information on financial metrics and projections for the next five years of Fifth Dimension

Finance Overview

Overview of the capital needed for the business before going over our investment viability

As founding and continuing to run Fifth Dimension will require significant funds, our main priority is to ensure we minimize costs wherever possible while maintaining a high quality service. During our start-up period of 12 months, we will be preparing the facility, hardware, and staff for our first customers. This period will incur numerous costs such as an initial investment for modern renovations, keeping cash on hand, inventory on hand, a down-payment on rent, software fees, information system expenses, initial marketing expenses, handling fees for utilities and insurance, and training costs for year one employees (*Appendix 48: FE Assumptions*). These initial start-up costs total approximately \$880,000 after a tax deduction. Of this sum, \$250,000 is expected to be funded by friends and family. Therefore, we will require a capital investment of the remaining \$630,000. In year two, we plan to expand our facility by renting and renovating the basement of our original location. We will be financing this large investment with our net income from year one and approximately an additional \$70,000 to allow the expansion to be possible. As our business concludes in year five we calculated a terminal value using a declining perpetuity of \$694,000. Finally, we predict a positive NPV of \$238,000 and positive IRR of 27.5% (*Appendix 51: Cash Flow*), which is greater than our business discount rate of 19.9%, making us cash flow positive and a good investment.



Start-Up Period Costs

Different requirement and the cost to get the business prepared for our first customers

In order to meet customer expectations we have chosen to invest in quality by refurbishing our facility with modern furniture, matching decor, and ample space to experience virtual reality. Through six to twelve months of careful renovation, we expect this vision to give us a competitive advantage over similar companies. In the long term, this will also ensure a standard of high-quality work and smooth business operations. To offer our experiences, we will be installing software of multiple makes, models, and vendors. As such, we must ensure these programs consistently work with our headsets with little to no incompatibility. We expect that the process of testing the headsets will take us one month, a worthwhile commitment to minimize issues in the future. All in all, we expect our innovative model, clean designs, and commitment to quality will capture potential customers and retain them for years to come.

Capital Expenditure: On average, the cost of renovating a new business can range anywhere from \$30 to \$450³⁰ per ft². We believe our costs will lie in the middle of this range with \$150 per ft² for the 2645 ft² floor space of our location in

Renovation Cost	\$396,750
VR Headsets	\$55,781
Yearly Total	\$452,531

Year zero capital expenditure costs by expense item

Exhibit 20: Year Zero Capital Expenditure Expense

year zero. For the remainder of our capital expenditure, we will purchase the latest high-end enterprise VR headsets from Meta, our main supplier of headsets and equipment. The

Meta Quest Pro, which launched in October of 2022, will cost us \$1499 per

³⁰ Patel, Jeel. "Commercial Renovation Cost Estimation." InvoiceOwl, September 28, 2022.

<https://www.invoiceowl.com/estimating-guide/commercial-renovation-cost/#:~:text=Commercial%20renovation%20cost%20per%20square,the%20building%20type%20and%20location.>



headset³¹. To fully meet our peak demand in year one, we plan to purchase 35 headsets in year zero (*Appendix 29: Capacity Analysis*), totalling \$55,500 (*Exhibit 20*). Combined with our renovation costs, we expect to spend around \$450,000 (*Appendix 45: Capital Expenditure Tables*) in capital expenditure costs in year zero, financed largely by our initial investment.

Initial Inventory: As a contingency to account for uncertainties within the first year of business, we plan to keep a quarter of year one's total marketing expenses as cash on hand amounting to \$24,600 to be paid out of our initial investment. Due to the uniquely digital model of our business, we do not rely heavily on inventory for daily operations. However, we will still require some inventory on the first day of business to ensure stock stability and availability. Our inventory consists largely of small items, like sanitary alcohol wipes and cleaning supplies for the headsets. As such, we do not expect to require large quantities of these to start in year one. With that being said, we decided to purchase and store 50% of the projected year one inventory supplies in year zero, costing us a total of approximately \$27,000.

Software Rental: As a VR entertainment service provider, it is not enough to just have quality headsets. Our backend systems and software being run are just as critical and, consequently, have justifiable expenses of their own. Our research suggested that a \$8000³² monthly cost for software renting and maintenance would suffice for our business. Since the software is being rented on a yearly basis, we will incur a total of \$96,000 in year zero to not only rent, but also run and test the software beforehand to ensure compatibility with our headsets. Giving our IT

³¹ "Meta Quest pro: Our Most Advanced New VR Headset: Meta Store." Meta. Accessed December 5, 2022. <https://www.meta.com/quest/quest-pro/>

³² Agile, Author: Anastasia Kushnir Writer at Bamboo, Author: and Writer at Bamboo Agile. "Software Maintenance Cost: What Is It and Why Is It so Important?" Insights, October 11, 2021. <https://bambooagile.eu/insights/software-maintenance-costs/>.



Assistants enough time to fix bugs and adjust settings in year zero will ensure a high quality experience for our customers at launch.

IS and Marketing Expenses: Regarding equipment, our business will incur some technical information system equipment expenses as we need our own computers and backend systems to maintain data coherence and integrity. Further, we plan to invest in a high quality internet service to run our wireless connectivity and a business phone line to handle customer inquiries. We will also be building and maintaining a custom designed website matching our branding to provide online booking capabilities for our customers. All of those costs will be considered as information system expenses, and will total approximately \$11,000³³, funded by our initial investment, to lay the technical foundation for our business. To ensure an effective marketing campaign in year one, we also plan to spend 30% of that year's marketing expenses in year zero, totalling approximately \$30,000, also to be paid out of our initial investment.

Utilities: Before the business can be properly started, we will need to have fully functional water and electricity running at our location. Furthermore, we need to buy insurance to safeguard against potential disasters. Paying rent for the location is also an unavoidable expense necessary for business operations. We expect a handling fee to connect utilities and

Indirect Labor	Training Cost
CEO	\$33,000
CIO	\$28,500
Facility Supervisor	\$27,000
Customer Relations Manager	\$18,000
Expense Manager	\$18,000
IT Assistant	\$33,000
Total Training Expense	\$157,500

Training cost for all year one employees

Exhibit 21: Year Zero Training Cost

³³ Nicolette Kier As a small business owner. "How Much Does a POS System Cost? Avoid Hidden Fees for Your Small Business." Merchant Maverick, August 1, 2022

<https://www.merchantmaverick.com/how-much-does-a-pos-system-cost/#:~:text=A%20point%20of%20sale%20system,month%20for%20your%20POS%20system.>

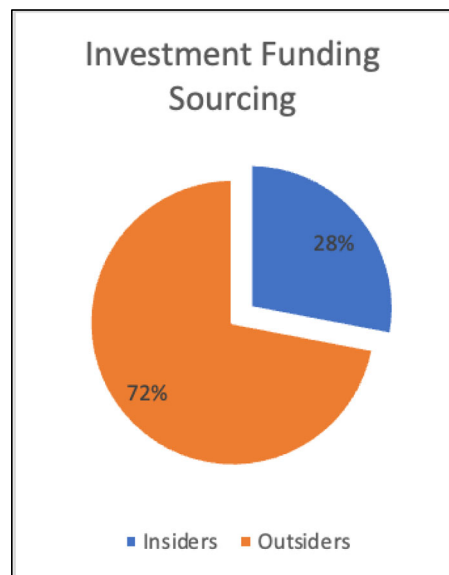


insurance costs to contribute a substantial portion of our expenses out of our initial investment. We estimated these costs at 20% of the year's projected total utilities, insurance, and rent, to account for year zero's handling fee of the expenses, resulting in a cost of about \$94,000 (*Appendix 46: SG&A Expense Tables*). Finally, as part of our commitment to quality we will be investing in employee training, costing 30% of their yearly wage in the process, for a total startup training cost of approximately \$160,000 (*Exhibit 21*). Only properly trained employees can deliver the quality service our customers are expecting and ensure they are familiarized with the skills required for their roles, making this a worthwhile investment.

Funding Request

The specific cost to start the business along with its expansion are covered

Based on the requirements listed previously, we will be incurring a total initial startup cost of approximately \$880,000 after tax deduction. In order to fulfill the projected trend of increasing demand, we decided to expand to the basement floor of our current location in year two. The expansion will allow us to meet the higher demand in years two and after, making this a critical necessity for the success of our business. For this we will require additional funding of about \$70,000 to finance the expansion (*Appendix 51: Cash Flow*), which has been accounted for with the net income we expect to generate in year two (*Appendix 49: Income Statement*). Because our expansion occurred at the original location's basement, we will not incur any additional expenses, like marketing awareness generation, on top of the projected cost in year two. The majority of our funding request is due to our planned renovations of the newly rented basement as well as the increase in insurance and cost of additional inventory.



Distribution of initial funding sourcing
Exhibit 22: Funding Source Ratio

Initial Funding Source

Breakdown of the sources of our initial funding

When it comes to the source of funding, we expect to receive \$250,000 from insiders, friends, and family, accounting for about 28% of the total initial investment needed to start the business (*Exhibit 22*). The remaining \$630,000 will be coming from outside investors that, given our positive NPV and IRR, we expect to fully pay back within the scope of this business plan.

Drivers for Revenue and Profit

Covers relevant drivers of revenue and profit along with an overview of EBIT breakeven

As a service, our revenue and profit models rely heavily on our flow of customers, brand recognition, and customers intentions to purchase and return. Services, compared to products, are more sensitive to customer counts rather than strictly profit margin. As such, some of the most important factors for our service are the awareness of our business within Boston, the purchase intent of the customers, and the overall segment size that we are aiming to target. Consequently, changes to those variables are likely to have the greatest impacts on our income streams, making them the key drivers of our revenue and profit growth.

EBIT Breakeven: When calculating the projected profitability of our business from years one to five, we were able to obtain key metrics that can be used to determine our success. As seen in our Finance Break-Even Analysis (*Appendix 52*:



Break-Even Analysis), our business will reach EBIT breakeven in year one as our projected customers served is greater than the number of customers needed for EBIT to equal zero, giving us a 7.63% margin of safety. This number continues to increase as it reaches year five, allowing us to have greater flexibility when encountering uncertainties, roadblocks, or economic downturns.

Investor Payback: By continuing to examine our financial models, we can also conclude when we will be able to achieve investor payback. We can assume that if our net cash flow for a year is positive, we will be paying out that amount as dividends and payback payments. Conversely, if the net cash flow is negative, we will require additional funding from outside sources. As can be seen in our cash flow statement, we expect to be able to fully pay back our total funded investment cost, needed in year zero's opening and year two's expansion, in year four. Here, we will have paid out approximately \$1.4 million to cover the total cost of \$0.98 million from year zero and year two, making us cash flow positive (*Appendix 51: Cash Flow*).

Required Rate of Return

Provides overview calculation of Fifth Dimension's minimum required rate of return

The process used to determine the Beta for our business involved taking the average Betas of our five most comparable firms: AMC, IMAX, Cinemark Holdings, Disney, and (importantly) Key to Amaze, our most direct competitor (*Exhibit 27*). By using this method, we were able to calculate the Beta of our business to be

Beta	1.37
10 Year T-bond Rate	3.75%
Expected Market Return	9.00%
Risk Premium	9.00%
Company Discount Rate	19.96%

Necessary components for discount rate

Exhibit 23 : CAPM Component

1.37 (*Exhibit 23*). Though this is not high, it does not guarantee our success. As we

are a non-traditional business, investors will rightly demand a higher risk premium of 9% (*Exhibit 23*) to justify the higher risk to start the business, consistent with the model of higher risks demanding higher returns. In order to calculate our discount rate as well as minimum required rate of return, we used the Capital Asset Pricing Model with the ten-year treasury bond rate as our risk-free rate and 9% for the expected market return. This computes our minimum required rate of return of 19.96% (*Exhibit 23*) which we will be using for comparison with competitors as well as our business discount rate.

Expected NPV and IRR

The relevant information surrounding Fifth Dimension's NPV and IRR is discussed

When laid out, our projected net cash flows from years zero to five combined with our discount rate can be used to calculate our net present value. Given our consistently positive cash flows, EBIT breakeven in year zero, and risk

NPV: Including TV	\$238,116
IRR: Including TV	27.46%

Forecasted NPV and for all five years

Exhibit 24 : Projected NPV and IRR

premium of 9%, we computed a matching positive NPV of \$238,000 in year zero (*Exhibit 24*), as well as a high internal rate of return of 27.5% (*Appendix 51: Cash Flow*). To better

understand and break down these numbers, we will need to compare them to our metrics for profitability. When looking at the NPV, it is an indication that this is a great business to financially invest in, despite the higher risk, as the NPV is far above \$0. Consequently, an IRR of 27.5%, which is greater than our business discount rate of 19.96%, further justifies our profitability and generates incentive to invest in the business.

Terminal Value

The details of Fifth Dimension's terminal value are mentioned

As our business is innovative and highly profitable we expect competition to come in quickly and heavily. As such, we project that our business will start to decline after year five, when we expect to be losing 40% of our revenue to competition (*Appendix 14: Bases Model*). With these assumptions in mind, we decided to use a declining perpetuity to determine our terminal value in year five. We will be using 20% for our declining rate after year five and are projected to reach the end of our service life cycle in year ten. Together, these values calculate a terminal value of approximately \$690,000 for our business, which was also used to account for the total net cash flow in year five and the NPV in year zero. If this expected decline takes place we plan to exit the market by liquidating our assets and returning leftover cash to investors as dividends.

Competitor and Industry Ratios

These ratios display information about Fifth Dimension's competitors and industry

Ratio Analysis: Our closest competitor is Key to Amaze, a Boston-based virtual reality entertainment service focusing on gaming offerings. However, as their business is private we were unable to find any key financial metrics. As such, we instead chose to use Cinemark and the average entertainment industry financial ratios as our benchmark comparison. When assessing our ability to meet financial obligations, we evaluated our quick ratio and debt to equity ratio against our benchmark. The quick ratio is the current asset minus inventory divided by current liabilities, and it shows our ability to meet short-term financial obligations. Our year five quick ratio is 10.76, which is astonishingly high compared to Cinemark and the average entertainment industry quick ratio of 1.18 and 0.68 respectively (*Exhibit 26*). This discrepancy is attributable to our comparably low

liabilities, as we are fully funded with no debt, whereas the average entertainment business is not. Our low levels of debt can be further examined through a debt to equity ratio analysis, which involves taking the total liabilities divided by total shareholder's equity. Since our business has no debt, our only liabilities are accounts payable, leading to a debt/equity ratio of 0.0256, whereas the industry average hovers around 0.17. This difference can be explained by the fact that our company is fully equity financed, whereas publicly traded companies use both equity and debt. Consequently, it makes sense that our debt to equity ratio is noticeably lower than the industry average.

Exhibit 25: Ratio Analysis

Key ratios of Fifth Dimension used for benchmarking

Current Assets Year 5	\$612,247	SG&A Year 5	\$2,173,129	Sales Year 5	\$3,324,448
Current Liabilities Year 5	\$19,470	Sales Year 5	\$3,324,448	Gross Profit	\$2,850,679
Inventory Year 5	\$402,762				
Quick Ratio	10.76	SG&A/Sales	65.37%	Gross Margin	85.75%
Net Income Year 5	\$285,324	Net Income Year 5	\$285,324	Sales Year 5	\$3,324,448
Total Equity Year 5	\$759,901	Taxes Year 5	\$105,531	Operating Profit Year 5	\$390,856
		Depreciation Year 5	\$190,693		
		EBITDA Year 5	\$581,548		
		Sales Year 5	\$3,324,448		
ROE	37.55%	EBITDA/Sales	17.49%	Operating Margin	11.76%

Exhibit 26: Benchmark Key Ratios*Key ratios of Fifth Dimension's benchmark comparisons*

Average Entertainment Industry Financial Ratios		Cinemark Financial Ratios
Quick Ratio	0.68	1.18
Debt/Equity	0.17	12.49
ROE	-28.64%	-62.76%
EBITDA/Sales	-4.36%	-0.29%
Gross Margin	77.16%	36.28%
Operating Margin	-8.97%	0.48%
SG&A /Sales	-	26.08%

EBIT Ratio Metrics: We compared our ROE, EBITDA/Sales, gross margin, and operating margin against our benchmarks to assess our profitability. Due to the impacts of the coronavirus, many of the average profitability ratios of our benchmarks are skewed by their poor performance in 2020. To assess how efficiently our equity generates returns, we performed a ROE analysis that divides the net income by the total shareholder's equity. In year five we are projected to have a ROE of 37.55%, considerably higher than the industry average of -28.64%. Since a negative ROE does not provide a meaningful comparison, we also compared it against the average ROE of the total market which is 18.12%³⁴. In this comparison our ROE is still significantly greater than the total market average, suggesting we have an efficient use of equity. The EBITDA to Sales ratio provides insight into how efficiently a business generates cash flow irrespective of how the company is funded, its depreciation, taxes, and amortization. In year five our EBITDA/Sales is projected to be 17.49% of sales. This is optimistic considering the EBITDA/Sales of the industry average is -4.36% and Cinemark -0.29%. The impact of the COVID-19 pandemic has been a major detriment to the entertainment

³⁴ Damodaran online: Home page for Aswath Damodaran, accessed December 6, 2022, <https://pages.stern.nyu.edu/~adamodar/>.



industry. Many of the public companies in this sector suffered major losses in consecutive quarters between 2020 and 2021. However, as the pandemic recedes and the U.S economy stabilizes, we expect the EBITDA/Sales average to revert back to a positive percentage. A more comparable metric may be the gross margin, calculated by dividing the gross income by the total revenue, which evaluates how efficient a business is at managing their cost of goods sold. Our gross margin as a percentage of sales computes to 85.75%, significantly greater than our benchmark competitor Cinemark of 36.28%. Similarly, the operating margin is calculated by dividing the operating income by the total revenue and evaluates how efficiently a business operates. Our operating margin as a percentage of sales is 11.76%, significantly greater than our benchmark competitor Cinemark of 0.48% and the industry average of -8.97%. According to our research most businesses should aim for an operating margin of at least 15%³⁵. Since our business model involves high gross margin and high SG&A costs, this naturally leads to a lower operating margin.

To assess our operating efficiency, we evaluated our SG&A as a percentage of sales by dividing the total SG&A expense by our total sales to understand what proportion of revenue is being used in operations. Our SG&A is 65.37% of sales, which is greater than our benchmark competitor Cinemark's SG&A to sales of 26.08%, as our business model involves low cost of goods sold and a high gross margin. This results in more allocation of cost to total overhead costs, resulting in a lower operating margin. Consequently, SG&A as a percentage of our sales will be much greater because our business naturally allocates a greater proportion of financing towards SG&A.

³⁵ "MacroTrends," MacroTrends, September 28, 2020, <https://macrotrends.net/definitions/terms/operating-margin/>.

Finance Conclusion

Provides summary of what was discussed in the finance section

Overall, we expect to require approximately \$630,000 from outside investors to fund Fifth Dimension during the start-up period. In year two, we will require an additional investment of \$70,000 from outside investors. By year five we expect a positive IRR of 27.46%, which is greater than our business discount rate of 19.96%, indicating we are a profitable business. Our margin of safety in year one is 7.63%, which continues to increase as it reaches year five. This allows us to have greater flexibility when encountering roadblocks, uncertainties, or market turmoil. The key highlights of our financial ratios analysis includes our quick ratio, ROE, and gross margin. Our fifth year quick ratio is 1.18, indicating that the company has no difficulties meeting short-term liabilities. ROE is expected to reach 37.55% (*Exhibit 25*), which is greater than average ROE of the total market. Finally, our gross margin also exceeds the entertainment industry average, meaning that we are more efficient at managing our cost of goods sold than comparable companies.

Exhibit 27: Competitor Table

Shows profit margin, ROE, and beta of comparable companies in the industry

Companies	AMC	IMAX	Key to Amaze	Cinemark Holdings	Disney
Profit Margin	-21.03%	-4.93%	11.30%*	-9.86%	6.55%
ROE	-41.47%	-3.70%	39.03%*	-75.00%	3.32%
Beta	1.9	1.28	0.32	2.14	1.23

*industry data

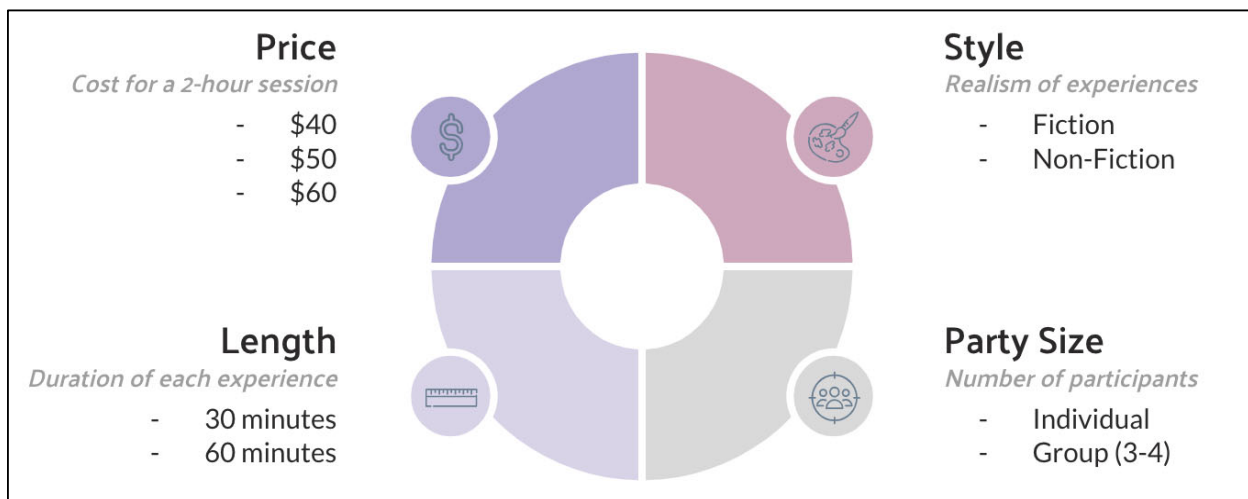
CROSS FUNCTIONAL RISK ASSESSMENT AND MANAGEMENT

Summary of the different factors that will impact Fifth Dimension's business potentials

QM Overview

Briefly goes over the different aspects that will be discussed in future sections

To aid us with our service design, our risk assessment and management analytics shaped our decisions across all other functions. Our conjoint analysis generated valuable insights, like the importance of group size and price as consumer preferences. Consequently, we made accommodations like adding a group discount for groups of five or more people. To determine risk, we brainstormed thirty variables and situations that could potentially create a risk for Fifth Dimension, and separated them into qualitative and quantitative risks. Next, we created a qualitative risk matrix with the most likely and highest risk situations located in the top right corner. Then, we determined the sensitivity of each of the quantitative variables by measuring their elasticities (% change in NPV due to 1% increase), showing segment size to have the highest elasticity. We were then able to create our quantitative risk matrix by using the elasticities as a proxy for the magnitude of impact, with the separation between high impact and low impact being 5% elasticity. Finally, we decided to run simulations on the variables segment



Shows four attributes and corresponding levels used in our conjoint analysis

Exhibit 28: Attributes and Levels



size (from marketing) and direct labor (from operations) to measure their impact on the entire business under unstable circumstances. This allowed us to conclude that segment size is our most impactful variable as it has the greatest impact on our NPV and highest simulated risk (*Exhibit 36*).

Conjoint Analysis

Shows what potential Fifth Dimension customers would value most from the given options

To understand how potential customers value different service attributes (price, style, length, and party size (*Exhibit 28*), we surveyed 81 people in person with 44 of the survey results falling within our target demographic and being deemed usable. After running a regression analysis on this data and calculating part-worth statistics (*Exhibit 29*), we learned that people prefer group experiences over individual experiences (*Exhibit 15*). As a result, we decided to prioritize an open floor plan over private rooms in our facility. The survey also indicated a preference for a low price of \$40 over medium of \$50 or high of \$60, giving us a better idea for target pricing and shaping our subsequent marketing survey. Finally, since people preferred that each experience last 60 minutes, rather than 30 minutes, we picked VR software that was more in depth, informative, and consequently closer to that longer duration (*Exhibit 26*).

Exhibit 29: Partworth Statistics

Customers prefer low price, longer experiences, group party size, and non-fictional style

	Mean	SD	Min	Max
Intercept*	4.29	1.30	1.00	7.00
Price High (\$60)	-0.20	0.63	-2.05	1.43
Price Low (\$40)	0.30	1.03	-3.25	2.25
Non-Fiction	0.13	0.87	-1.37	3.30
60-minutes	0.57	1.17	-1.80	4.60
Individual	-0.84	1.48	-4.13	2.93

*Price \$50, Fiction, 30-minutes, Group (3-4)

In the table (*Exhibit 30*), we can see each partworth's importance related to the intercept, calculated by using the range of the partworth and its percentage of importance relative to the total range. This identifies party size and experience length as having the highest importances of these partworths, 41%, and 28% respectively. In response, to capitalize upon big groups we plan to offer a discount for parties of five or more individuals. This table also shows that customers care greatly about the length of the VR experience. Therefore, we decided to give our customers a two hour period per \$40, during which they can freely explore any of our experiences at their own pace. Style was found to be the least important attribute, so we will offer both fictional and non-fictional experiences to customers based on software availability and seasonal trends.

Exhibit 30: Partworth Importance Table

Party Size is the most important attribute to customers

	Range	Importance
Price	0.50	25%
Style	0.13	6%
Length	0.57	28%
Party Size	0.84	41%
Sum of Ranges	2.04	100%

Next, we conducted a cluster analysis on the two most important partworths, party size and experience length, to see how different clusters of customers have differing priorities and preferences. By using the K-means algorithm, we identified three distinct clusters in the data (*Exhibit 32*). We concluded that these should be treated as three distinct groups of individuals within our target segment, each with unique views, described by their centroids in the table below (*Exhibit 31*).

Partworth Cluster Centroids Table		
	Length: 60 minutes	Party Size: Individual
Cluster 1	-0.64	0.76
Cluster 2	-0.19	-0.92
Cluster 3	1.36	0.44

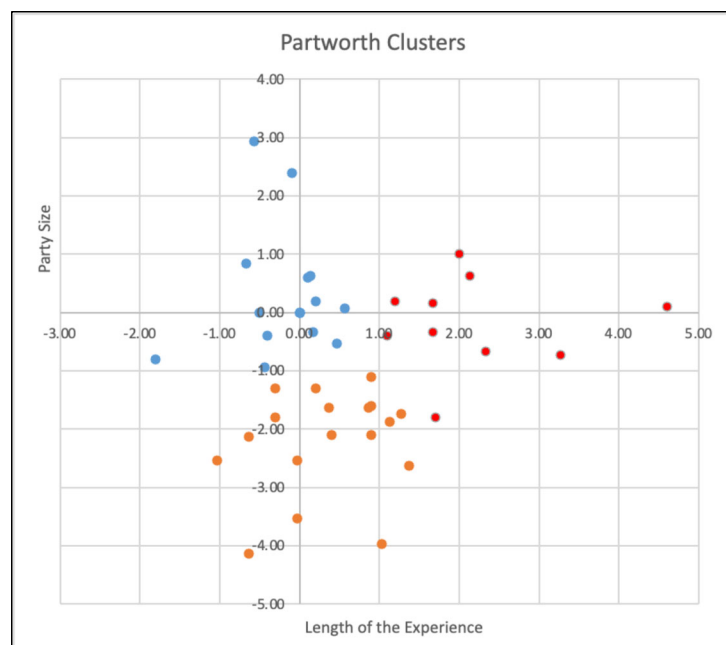
Cluster centroids of the partworths generated using k-means clustering

Exhibit 31: Partworth Cluster Centroids Table

As seen in the Partworth Cluster Centroids Table (*Exhibit 30*) the respondents belonging to cluster one tend to favor experiences shorter than 60 minutes, while strongly preferring to come as individuals rather than in groups. Cluster two, respondents tend to have a moderate preference for shorter experiences and instead strongly prefer to come as a group. Cluster three is particularly unique as the respondents within this group very strongly prefer the longer 60 minute experiences and have a moderate preference to come alone. As seen there are largely varying opinions on these two important variables, with strong interest for both shorter and longer experiences as well as a focus on individuals and groups. This led us to the decisions described previously to widen our offerings and empower the consumers by charging \$40 for two hours, rather than a specific experience. This option allows us to satisfy customers that want long or short experiences without limiting our selection.

Separate Quantitative Risks from Qualitative Risks

Provides insights into potential quantitative and qualitative risks



Visualized customer clusters generated using XLMiner

Exhibit 32: Partworth Clusters Graph

Unforeseeable events and other risks associated are often hard to estimate, but they can cause our revenue and profit to fluctuate. These risks can be separated into quantitative and qualitative risks. Quantitative risks are uncertainties that can be numerically quantified and analyzed, whereas qualitative risks can not be quantified by

numbers. Some of these risks are more probable and it is crucial for us to identify the chances of these risks happening, but more importantly to understand how they affect our bottom line and NPV.

Qualitative Risk Matrix

Risks that were considered and categorized for its magnitude and probability of occurring

High Impact	Virtual Reality Loses Popularity	Recession/Economic Down Turn
	Safety Accident	VR Headsets Become More Affordable
	Pandemic	Increased Augmented Reality Popularity Lowers Market Share
	Technology Sanctions Increase Semiconductor Costs	New Technology
		Increasing GPU Prices
Low Impact	Workers not Customer Friendly	Employee Absenteeism
	Technology Company Boycott	Customers Break the VR Headsets During the Experience
	Pipe Leakage	
	Inventory Supply Chain Shortages	
	Low Probability	High Probability

Illustrates potential qualitative risks categorized by high/low impact and high/low probability

Exhibit 33: Qualitative Risk Matrix

While qualitative risks cannot be measured numerically, we classified these risks based on high impact or low impact, then used our best judgment to determine the probability of occurring. According to *Exhibit 33*, the risks that have the highest chance of happening with a high magnitude of impact on our business are recession, VR headset pricing becoming more affordable, a new technology that lowers people's interests in VR, and the increase of GPU prices. Any high



impact and high probability risks are likely to have a direct impact on Fifth Dimension's profitability and operations.

One of the risks that was included in the high impact and low probability quadrant is the risk of safety accidents of the customers. One safety accident is enough to have a catastrophic impact on our business, there we utilized an open floor design so our onsite employees can oversee every customer and ensure that accidents are handled urgently.

Our IT assistants are properly trained to ensure every customer is having the best experience, but more importantly preventing accidents from happening in the first place.

Another important consideration is the risk of customers breaking our headsets during the session. We believe that even though this has a high probability of occurring, due to its low impact, we have decided to focus on other risks that have more significance. We assumed that whenever a customer damages a headset, our onsite IT assistants will fix and tweak small damages. In case the headsets are damaged beyond repair, we have accounted for backup headsets to ensure they can be replaced immediately.

Sensitivity of NPV by Variable

Method and process of performing the sensitivity analysis

After assessing qualitative risks, we used sensitivity analysis to classify our quantitative risks into a matrix based on probability and impact levels. First, we calculated the breakeven value of NPV for each of the variables to see how much the sensitivity of each variable can increase or decrease before our NPV drops below \$0. Next, we calculated the elasticity of the quantitative risks with the most direct (high) impact on our NPV. We then ran the sensitivity analysis simulating the effect of how the overall NPV will change based on a 1% increase in the associated

quantitative variable. After the simulations, we analyzed the results to calculate each variable's elasticity based on how much its change impacted the NPV. With the elasticities operating as a proxy for magnitude of impact we were finally able to place the quantitative variables on the quantitative risk matrix seen below (*Exhibit 34*).

Quantitative Risk Matrix

Quantitative risks ranked by impact on NPV and categorized into different probability

High Impact	Repeat Occasions	Segment Size
	Rent Per Square Foot	Repeat Rate
		Purchase Intent
		Awareness
Low Impact	Marketing Expenses	Direct Labor Year 1
	Probability of Groups of 5+ People	COGS
	Insurance Expenses	Beta
	Cost of Capital	Utility Cost Per Square Foot
		Risk Premium
	Low Probability	High Probability

Illustrates potential quantitative risks categorized by high/low impact and high/low probability

Exhibit 34: Quantitative Risk Matrix

We decided to use 5% elasticity as the threshold separating high impact from low impact variables as it was a natural break point in the data (*Appendix 53: Sensitivity Analysis*). Meaning, all the risks that had an elasticity of higher than 5%, were considered high impact and inversely those below were low impact. Based on these results it became clear that the variables our business should focus on are segment size, purchase intent, awareness, and repeat rate, as they had the highest elasticities ranging from 32% for the first to 23% for the last. This means that with a 1% increase in segment size, our NPV would increase by 32%, making it clearly an



impactful variable. Beyond these four we've also considered other variables that may have a high impact on our business, such as cost of capital. But as the majority of these have fixed percentages, like market return and risk premium, they have a low chance of fluctuating, making it unnecessary to focus time, resources, and funding toward mitigating their potential impacts. A more relevant quantitative variable that we also took into consideration is direct labor cost for year one. Even though the direct labor cost has high chance of fluctuating as people demand more pay, an elasticity of 3.63% will not be impactful on NPV.

Simulations

Provides an explanation behind how the simulations were conducted

Based on the results of the quantitative risk sensitivity analysis, we will use two of the most important variables, with each variable from marketing and operation to run a simulation of the uncertainty of those variables that will affect our NPV with given parameters occurring. The first variable we chose is segment size from the marketing functional group, then we chose to use direct labor for our second variable from the operation management group. In terms of choosing the variables to use for the simulation, we believe that using triangular distribution for both variables from different functional groups will simulate a more realistic effect that can occur across different functional groups.

The distribution parameters that we have selected for our two simulation variables will both be triangular-shaped because the distribution is bounded and can also be asymmetric. To calculate the upper and lower bound of segment size we first increased and decreased each of our segment variables by 1% (*Appendix 54: Segment Size Distribution Adjusted by 1%*). This method allows us to simulate the change in segment size in the event our calculation is 1% above or under. This results in a distribution with an upper bound of 104.9% and a lower bound of



96.3%. We continued to perform simulations using this distribution with a base number of one. While on the other hand when determining the parameters of the direct labor variable, we research the range of average salary wage for IT assistants and positioned ourselves into that range. In our business model, we are offer a salary our IT assistant \$55,000, while our research indicates that the upper bound of the salary is \$71,000 while the lower bound is \$52,000³⁶. We calculated our upper and lower bound parameters by dividing the upper and lower bound salary by the salary we offer, resulting in upper and lower bound parameters of 129.9% and 94.55%.

To conduct the full simulation process, we ran three simulations in total, first only changing the segment size variable by randomly adjusting the sensitivity for all five years of the segment variable within the set parameters, while keeping the direct labor variable constant to see the effects it has on NPV. Then we ran a second simulation by only changing the sensitivity of the direct labor variable while keeping the segmentation variable constant. Lastly, we ran the third final simulation, changing both the segment size and the direct labor variable by randomly adjusting both of their sensitivity for all five years. We conducted all three simulations with 1000 replications and captured the effect of those variables on the NPV.

³⁶ Salary.com, Site built by: "Information Technology Support Technician Salary in Massachusetts." Salary.com. Accessed December 6, 2022.
<https://www.salary.com/research/salary/recruiting/information-technology-support-technician-salary/ma#:~:text=How%20much%20does%20an%20Information, falls%20between%20%2452%2C068%20and%20%2471%2C252.>

Assumptions about Parameters

This section provides an explanation of how our parameters impacted things like NPV

Our simulation results were similarly aligned with our sensitivity analysis. During the first simulation where only the segment size variable was changed, we obtained a higher average NPV than our base case, a \$275,066 average NPV with a standard deviation of \$140,486 (*Exhibit 35*). The results suggest that on average we will be making a profit. However, the trade-off is an increase in standard deviation, which has a higher chance of causing the NPV to fluctuate by a substantial amount, which magnifies our gains and losses. When looking at the distribution in simulation one (*Appendix 55: Simulation One Results, Segment Size Simulation Input*

	Segment Size
Average	\$275,066
Median	\$265,614
Standard Deviation	\$140,486
%NPV < 0	1.10%

NPV generated by segment size variable

Exhibit 35: Segment Size vs NPV Results

Distribution), we see a triangular distribution for the inputs with a slight skew to the right. The statistical decision variable that calculates the probability of NPV below \$0 is 1.1% (*Exhibit 35*). In other words, when only

adjusting and accounting for the segment size variable uncertainty within the parameters, our NPV should be positive 98.9% of the time.

The second simulation evaluating the impacts of direct labor produced a result with an average NPV of \$170,393 and a standard deviation of \$66,442 (*Appendix 58: Means Distributions of Simulation Variables*). The standard deviation of direct labor is lower than that for segment size, indicating that direct labor has a much smaller effect on NPV. The result can be explained because the direct labor triangular distribution is extremely right skewed. This asymmetrical distribution of the variable creates a much greater chance for the cost of direct labor to exceed

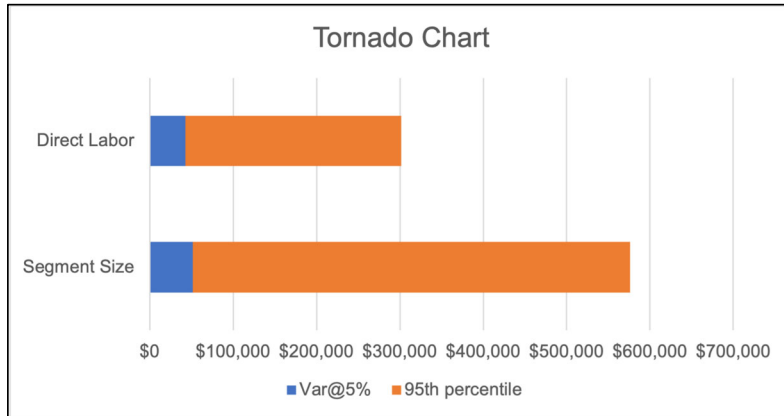


the base case. As such, even though the simulation between direct NPV resulted in a smaller standard deviation, the average NPV of the simulation is much lower.

When comparing the two histograms from the second simulation (*Appendix 56: Simulation Two Results*), we notice that the two graphs have opposing skewness because the distribution of direct labor is right skewed. This can be explained because the extreme positive direct labor value will cause the average NPV of the simulation to decrease, which causes more data of the NPV results to be left skewed. The probability of NPV being less than \$0 is 0.2% for the second simulation (*Appendix 58: Means and Distributions of Simulation Variables*), which reinforced the idea that direct labor has a much lower impact compared to segment size.

Moving on to the third simulation where we randomly adjusted both the sensitivity of the segment size and the direct labor variables within their set parameters. The result of the third simulation suggested an average NPV of \$202,271 with a standard deviation of \$148,841 (*Exhibit 35*). The increase in the average NPV from the second simulation is due to the fact that we included both variables into account when running the simulation, where it simulated an average NPV that is between the average of simulation one, \$275,066, and simulation two, \$170,393. Since both variables were taken into account, the NPV is interacting and facing more uncertainty with more risks, making the NPV fluctuate at a greater scale and become less constant. Thus, the standard deviation when all variables are included is the highest at \$148,841, with \$140,486 in simulation one and \$66,442 in simulation two. It is also due to the fact more uncertainties are being accounted for in the simulation, the probability of NPV being below \$0 also significantly jumped to 9% and made the business model much riskier. The output distribution for the third simulation also resembled a triangular distribution with a

tiny more simulation on the right of the graph, causing it to be not perfectly symmetrical (*Appendix 57: Simulation Three Results*).



Average NPV in the 5th and 95th percentile of the simulation variables

Exhibit 36: Simulation Tornado Chart

Tornado Chart Analysis

Summarizes the findings of the tornado chart

A way to visualize the effects of our simulation variables on our NPV is by creating and analyzing a tornado chart with the 5th and 95th percentile of the NPVs

generated by our simulations. The purpose of the tornado chart is for us to have a better understanding of the spread of the NPVs in extreme cases, at the 5th and the 95th percentile. By looking at the chart, we can also see the effects of the standard deviations of the variables at the percentiles. From the tornado chart we generated (*Exhibit 36*), we can see that the 5th percentile NPV generated with the direct labor variable is lower than the NPV generated with segment size. This is the same case as the average NPV of the two variables, due to the uniquely more skewed distribution of the direct labor variable, the NPVs generated with direct labor are being exposed to a higher cost, thus causing the average NPV to be low and have a lower starting position. While we are examining the 95th percentile of both variables, we can see that the NPV for segment size is much greater than direct labor at the 95th percentile. This can be explained by the significantly greater standard deviation that the segment size variable brings to the NPV, causing the NPV to fluctuate at a greater magnitude and have a higher chance of going to the extreme in either way. Thus giving us the conclusion that solidifies the idea that



the segment size variable is a greater influencer when computing and determining the NPV.

QM Conclusion

Summarizes the most relevant parts of the previous sections

After figuring out our calculations for Marketing, Operations, and Finance, we assessed the risks involved with variables that may contain some uncertainty. During the process, we looked at the elasticities of each variable that had a high impact on the NPV, and we considered and grouped them into high and low probability. The variables that were grouped into the high probability and high impact are segment size, repeat rate, purchase intent, as well as awareness. The main drivers and factors of our business are segment size because it has the highest elasticity affecting NPV and the amount of people coming to our service, thus the profitability model of our whole service is oriented based on segment size. While the simulation results also suggest that we spend more attention on mitigating and maintaining a good position for segment size as the segment size has a higher standard deviation and has more power affecting our NPV.



CONCLUSION

Fifth Dimension is a virtual reality entertainment service with educational and informative offerings specializing in immersive virtual reality travel to historic locations around the world, through space and time. For \$40 customers can rent a VR headset for a two-hour period and freely use any of our pre-installed experiences during their time. This started with our target market: From a robust marketing survey we established our purchase intent and retail price of \$40, culminating in a BASES model that projects a revenue peak of \$3.7 million in year three before sustaining over \$3 million in the following years. Due to our inherently digital business model, our operating costs are comparably low and variable unit costs virtually nonexistent. As such, despite high SG&A and direct labor costs, we expect an immediate EBIT breakeven at year one, NPV of \$240,000, and IRR of 27% in year five, well above even our aggressive risk premium of 9% and required rate of return of 19.96%. Similarly, our current ratio of 10.76 surpasses the industry average by over tenfold, and our gross margin of 86% in year five also outperforms the industry average of 77%. To ensure stability we have investigated possible qualitative and quantitative risks associated with our business and developed mitigation plans for each, including diversification tangential to reduce systematic risk exposure, monthly new experiences to attract and retain target customers, and unique guerilla marketing campaigns to increase brand awareness. Above all we intend to remain as we began: customer-centric, and in doing so maintain high profitability and entertainment disruption throughout these five years and beyond.

APPENDICES

Appendix 1: VR Educational Workshop

Total increase of 1.4% in CSR Lift from educational workshop

VR Educational Workshop CSR				
	DB + PB	% DB + PB	CSR Increase	Total CSR Lift
Significantly More	27	39.1%	2.5%	1.0%
Slightly More	19	27.5%	1.5%	0.4%
Will Not Change	23	33.3%	0.0%	0.0%
				1.4%

Appendix 2: Make A Wish VR Sessions

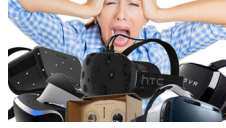
Total increase of 1.5% in CSR Lift from Make A Wish initiative

Make A Wish CSR				
	DB + PB	% DB + PB	CSR Increase	Total CSR Lift
Significantly More	25	35.7%	2.5%	0.9%
Slightly More	26	37.1%	1.5%	0.6%
Will Not Change	19	27.1%	0.0%	0.0%
				1.5%



Appendix 3: Segmentation Grid

Our target segment being adventurous VR enthusiasts who do not own a VR headset

	Target Segment	Non-Target Segment	Non-Target Segment	Non-Target Segment
Segment name	Adventurous VR enthusiasts without VR headsets	People under the age of 20 and over the age of 50	VR Headset Owners	Non-VR Enjoyers
Image of Identified Target Consumer				
Segment Size	0.81M	0.28M	0.27M	0.38M
Demographics	Boston residents with income >\$35K aged 20 to 50	Boston residents with income >\$35K who are under 20 years old or over the age of 50	Boston residents with income >\$35K who already own a VR headset	Boston residents with income >\$35K who do not enjoy VR
Psychographics	Interested in VR, adventurous	Not interested in VR	Interested in VR	Not interested in VR
Maximum Retail Price Segment Will Pay	\$55	\$30	\$10	\$0
Key Influencers (if any)	Friends, social media influencers, celebrity	Friends, cable ads	Friends, social media influencers, online ads	None
Other Service Segment Buys	Technology services (ex. AR entertainments, video games)	Museums, movies	Technology services (ex. AR entertainments, video games)	Non-technology products and services
Key Service Attributes & Benefits Sought By Segment	Unique experiences, affordable, immersive technology, educational, interactive, group experiences	Stability, simplicity, entertainment, educational	Immersive technology, group experiences, interactive, educational	Group experiences, interactive, educational
Segment's Information Sources	Friends, social media, TV commercials, billboards, website	Cable ads, parents, child-programs, billboards, website	Friends, social media, billboards	Friends, social media, billboards
Best Distribution Outlets	Physical location in Boston	Physical location in Boston	Physical location in Boston	Physical location in Boston

Appendix 4: Unused Variables in Segmentation Tree

Variables that were passed as they did not provide as accurate representation of our target

Unused Variables	Yes	No
Access to a Car	2.34M (66.2%)	1.20M (33.8%)
Access to Internet	3.08M (87.1%)	0.46M (12.9%)
Owns a Game Console	2.23M (63%)	1.31M (37%)
Visited a Museum the Previous Year	0.85M (24%)	2.69M (76%)
Play Video Games	2.30M (65%)	1.24M (35%)
Owns a Smartwatch	0.38M (21%)	1.43M (79%)
# of Household that go on Vacation	1.42M (40%)	2.12M (60%)
Education Level (bachelor's degree)	1.82M (51.3%)	1.72M (48.7%)

Appendix 5: Summary of Interviews Table

Main points generated from marketing interviews

One-on-One Interviews
• Most interviewees have heard of VR from friends, social media, or events. • Majority have used VR with mixed/circumstantial interest in further usage. • Emphasis on immersion, accessibility, comfort, uniqueness, and education. • Most would search on Google/Yelp or ask informed friends. • High price strongly dissuades potential customers (\$60). • Some would go alone but most in groups (3-5 people). • Peak hours during weekend afternoons/evenings. • Must be within 35 min travel distance. • Interested in visiting exotic/impossible places (ex. underwater, ancient, poles, cities, venice, iceland, paris, madagascar).

Appendix 6: Assumptions

Shows key project inputs for sales, capacity, inventory, labor and finance

Year	Startup	1	2	3	4	5
Sales						
Segment Size		810,000	806,715	790,808	759,141	719,809
Segment Growth Rate			2.00%	2.00%	2.00%	2.00%
Purchase Intent		22.46%	22.79%	22.79%	23.74%	24.69%
Awareness		10.71%	17.43%	26.35%	30.25%	32.45%
Peak Awareness Year	5					
Retail Price		\$40.00	\$40.00	\$40.00	\$38.00	\$36.00
Group (5+) Discount %		10.00%	15.00%	15.00%	15.00%	15.00%
Group (5+) Discounted Price		\$36.00	\$34.00	\$34.00	\$32.30	\$30.60
% Groups of 5+		32.86%	32.86%	32.86%	32.86%	32.86%
Repeat Rate		25.00%	25.00%	25.00%	25.00%	25.00%
Repeat Occasions		5	5	5	4	4
Marketing Expenses	\$29,625	\$98,750	\$148,994	\$193,971	\$257,424	\$298,594
% of Year 1 Marketing Expenses in Year 0	30.00%					
Capacity						
Size of Location (ft2, Ground Floor)	2,645	2,645	2,645	2,645	2,645	2,645
Size of Location (ft2, Basement)			1,800	1,800	1,800	1,800
Total ft2	2,645	2,645	4,445	4,445	4,445	4,445
Price/ft2 (yearly)	\$150	\$150	\$150	\$150	\$150	\$150
% Space for VR Activities	75.00%					
ft2/VR Customer	49					
Customer Setup Time (minutes)	5					
Customer Transaction Time (minutes)	2					
Peak Demand Increase	120.00%					
Non-Peak Demand Decrease	80.00%					
Cost of Renovation/ft2	\$150	\$150	\$150	\$150	\$150	\$150
Utility Cost/ft2 (yearly)	\$15	\$15	\$15	\$15	\$15	\$15
Insurance Cost (yearly)	\$20,000	\$20,000	\$20,000	\$20,000	\$20,000	\$20,000
Average Cleaning Supply Cost/ft2	\$4.70	\$4.70	\$4.70	\$4.70	\$4.70	\$4.70
Depreciation of Original Headsets	3	Years, straight-line				
Depreciation of Future New Headsets	2	Years, straight-line				
Depreciation of Original Renovation	5	Years, straight-line				
Depreciation of Future New Renovation	2	Years, straight-line				
Inventory						
Inventory Week Count	50					
Inventory Days/Week	5					
Inventory Day Count	250					
Inventory Maximum %	120.00%					
Inventory Minimum %	80.00%					
Target Service Level	97.50%					
Target Service Level Z-Score	1.96					
Lead Time (days)	5					
COGS/unit	-	\$3.49	\$4.18	\$3.55	\$3.34	\$3.67
Raw Materials Inflation Rate	2.50%					
Labor						
Working Weeks/Year	50					
Working Days/Week	5					
Working Hours/Day	10					
Time/Headset Wipe (seconds)	10					
Setup Time/Customer (minutes)	5					
Direct Labor Rate/Hour		\$3.26	\$3.95	\$3.32	\$3.11	\$3.44
Financials						
Beta	1.37					
10 Year T-bond Rate	3.75%					
Expected Market Return	9.00%					
Risk Premium	9.00%					
Company Discount Rate	19.96%					
Tax Rate (combined Federal & State)	27.00%					

Appendix 7: Pivot Tables for Segmentation Variable Test on *Income*

Our cumulative purchase intent including people who have yearly income less than \$35,000

Income Segmentation Variable						
Before Excluding Low Income						
Price	DNB	PNB	NS	PB	DB	Total
25 - 34.99	5	13	18	12	9	57
35 - 44.99	2	4	15	16	5	42
45 - 54.99	2	10	19	17	4	52
55 - 64.99	5	4	3	7	8	27
65 - 74.99	1	2	3	2	6	14
Total	15	33	58	54	32	192

Price \$40	Raw	Adj.	PI	Cumu. PI
PB	42	12.60	6.56%	16.15%
DB	23	18.40	9.58%	

Our cumulative purchase intent excluding people who have yearly income less than \$35,000

After Excluding Low Income						
Price	DNB	PNB	NS	PB	DB	Total
25 - 34.99	3	11	13	10	8	45
35 - 44.99	2	4	13	13	5	37
45 - 54.99	2	9	15	15	4	45
55 - 64.99	5	4	3	7	8	27
65 - 74.99	1	2	3	2	6	14
Total	13	30	47	47	31	168

Price \$40	Raw	Adj.	PI	Cumu. PI
PB	37	11.10	6.61%	17.56%
DB	23	18.40	10.95%	

Appendix 8: BASES Models for Segmentation Variable Test on *Income*

Table shows revenue generated before and after income level adjusting

Before Income Adj	
	Year 1
Segment Size	1020000
Cumulative PI	16.15%
Awareness	8.00%
Trial HH's	13175
Selling Price	\$40
Trial Revenue	\$527,000
Repeat	
Repeat %	25.00%
# of Repeats	5
Repeat HH's	16469
Repeat Selling Price	\$40
Repeat Revenue	\$658,750
Total Revenue	\$1,185,750

After Income Adj	
	Year 1
Segment Size	820000
Cumulative PI	17.56%
Awareness	12.00%
Trial HH's	17279
Selling Price	\$40
Trial Revenue	\$691,143
Repeat	
Repeat %	25.00%
# of Repeats	5
Repeat HH's	21598
Repeat Selling Price	\$40
Repeat Revenue	\$863,929
Total Revenue	\$1,555,071

Appendix 9: Pivot Tables for Segmentation Variable Test on Age

Table shows our cumulative purchase intent including people who are younger than 20 years old

Before Excluding Age						
Price	DNB	PNB	NS	PB	DB	Total
25 - 34.99	5	13	18	12	9	57
35 - 44.99	2	4	15	16	5	42
45 - 54.99	2	10	19	17	4	52
55 - 64.99	5	4	3	7	8	27
65 - 74.99	1	2	3	2	6	14
Total	15	33	58	54	32	192

Price \$40	Raw	Adj.	PI	Cumu. PI
PB	42	12.60	6.6%	16.15%
DB	23	18.40	9.6%	

Table shows our cumulative purchase intent excluding people who are younger than 20 years old

Excluding <20						
Price	DNB	PNB	NS	PB	DB	Total
25 - 34.99	4	10	12	8	8	42
35 - 44.99	1	2	12	12	3	30
45 - 54.99	2	9	16	15	4	46
55 - 64.99	5	3	3	7	6	24
65 - 74.99	1	2	2	1	6	12
Total	13	26	45	43	27	154

Price \$40	Raw	Adj.	PI	Cumu. PI
PB	35	10.50	6.82%	16.69%
DB	19	15.20	9.87%	

Table shows our cumulative purchase intent excluding people who are older than 50 years old

Excluding 50+						
Price	DNB	PNB	NS	PB	DB	Total
25 - 34.99	4	12	14	11	9	50
35 - 44.99	2	3	14	16	4	39
45 - 54.99	2	9	18	17	4	50
55 - 64.99	5	4	3	7	8	27
65 - 74.99	1	2	3	2	6	14
Total	14	30	52	53	31	180

Price \$40	Raw	Adj.	PI	Cumu. PI
PB	42	12.6	7.0%	16.8%
DB	22	17.6	9.8%	

Our cumulative purchase intent excluding people who were between 20 - 50 years old

Excluding <20 & 50+						
Price	DNB	PNB	NS	PB	DB	Total
25 - 34.99	4	10	12	9	9	44
35 - 44.99	2	2	12	13	3	32
45 - 54.99	2	9	16	15	4	46
55 - 64.99	5	4	3	7	7	26
65 - 74.99	1	2	2	2	6	13
Total	14	27	45	46	29	161

Price \$40	Raw	Adj.	PI	Cumu. PI
PB	37	11.1	6.9%	16.8%
DB	20	16	9.9%	

Appendix 10: BASES Model for Segmentation Variable Test on Age

Table shows revenue generated before and after adjusting for age

Before Excluding Age	
	Year 1
Segment Size	2,040,000
Cumulative PI	16.15%
Awareness	7.00%
Trial HH's	23056
Selling Price	\$40
Trial Revenue	\$922,250
Repeat	
Repeat %	25.00%
# of Repeats	5
Repeat HH's	28820
Repeat Selling Price	\$40
Repeat Revenue	\$1,152,813
Total Revenue	\$2,075,063

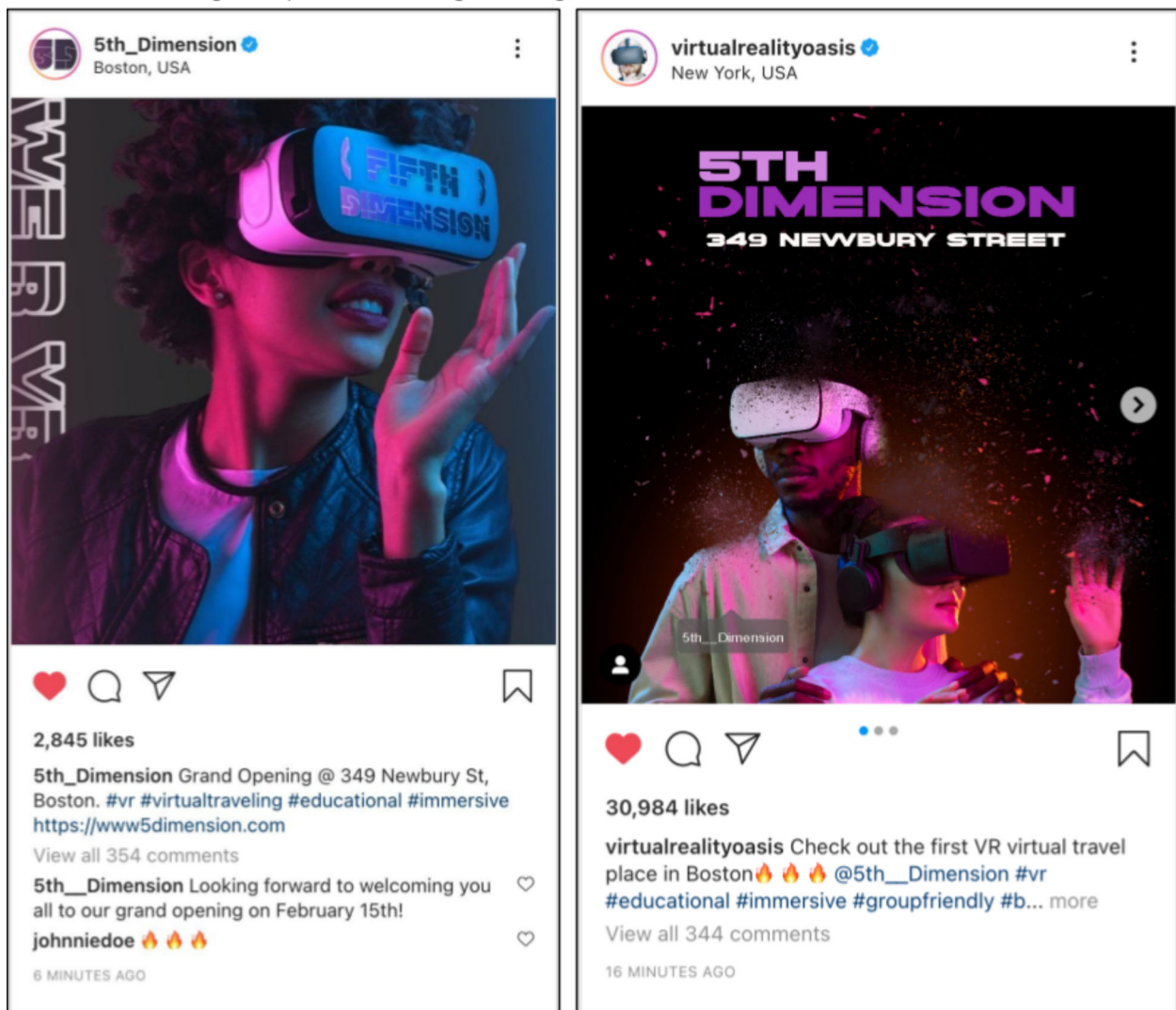
Excluding <20	
	Year 1
Segment Size	1,590,000
Cumulative PI	16.69%
Awareness	9.00%
Trial HH's	23881
Selling Price	\$40
Trial Revenue	\$955,239
Repeat	
Repeat %	25.00%
# of Repeats	5
Repeat HH's	29851
Repeat Selling Price	\$40
Repeat Revenue	\$1,194,049
Total Revenue	\$2,149,288

Excluding 50+	
	Year 1
Segment Size	1,280,000
Cumulative PI	16.78%
Awareness	9.00%
Trial HH's	19328
Selling Price	\$40
Trial Revenue	\$773,120
Repeat	
Repeat %	25.00%
# of Repeats	5
Repeat HH's	24160
Repeat Selling Price	\$40
Repeat Revenue	\$966,400
Total Revenue	\$1,739,520

Excluding <20 & 50+	
	Year 1
Segment Size	820,000
Cumulative PI	16.83%
Awareness	12.00%
Trial HH's	16563
Selling Price	\$40
Trial Revenue	\$662,519
Repeat	
Repeat %	25%
# of Repeats	5
Repeat HH's	20704
Repeat Selling Price	\$40
Repeat Revenue	\$828,149
Total Revenue	\$1,490,668

Appendix 11: Social Media Post

In-house Instagram post including our logo and color scheme



Appendix 12: Influencer Post

Hired Instagram post including our logo and color scheme



Digital Draft of signage that will be placed in front of our location
Appendix 13: Signage

Appendix 14: Bases Model

Table shows us our revenue generated and total yearly visits through years 1-5

Year	1	2	3	4	5
Target Market Size	810,000	806,996	793,026	761,832	722,683
Adjusted purchase intent	22.46%	22.79%	22.79%	23.74%	24.69%
Awareness	10.56%	16.37%	26.04%	30.08%	32.47%
Trial					
Trial individuals	19,204	30,110	47,054	54,386	57,936
Retail selling price	\$39	\$38	\$38	\$36	\$34
Trial Revenue	\$742,929	\$1,145,028	\$1,789,392	\$1,964,809	\$1,982,897
Retention of customers across years					
Retention rate within trial year	70%	70%	70%	70%	70%
Retention for year 1 individuals	0	3,361	2,353	1,647	1,153
Retention for year 2 individuals	0	0	5,857	4,100	2,870
Retention for year 3 individuals	0	0	0	9,671	6,770
Retention for year 4 individuals	0	0	0	0	12,216
Retention for year 5 individuals	0	0	0	0	0
Total retained customers	0	3,361	8,210	15,418	23,008
Retail selling price	\$39	\$38	\$38	\$36	\$34
Revenue from retained customers	\$0.00	\$127,804	\$312,208	\$557,009	\$787,476
Repeats					
Trial and Retained Group	19,204	33,471	55,264	69,804	80,945
Repeat %	25%	25%	25%	25%	25%
Repeats individuals	4,801	8,368	13,816	17,451	20,236
# occasions	5	5	5	4	4
Retail selling price	\$39	\$38	\$38	\$36	\$34
Revenue from repeat customers	\$928,661	\$1,591,040	\$2,627,000	\$2,521,819	\$2,770,373
Total revenue before competition	\$1,671,590	\$2,863,872	\$4,728,600	\$5,043,637	\$5,540,747
Competition adjustment (% of sales lost to competition)	-	10%	20%	30%	40%
Revenue lost to competition	-	\$286,387	\$945,720	\$1,513,091	\$2,216,299
Total revenue after competition adjustment	\$1,671,590	\$2,577,485	\$3,728,880	\$3,530,546	\$3,324,448
Total visits	43,210	75,125	110,528	118,667	129,511

Appendix 15: Price vs Purchase Intent Tables*Customer's purchase intent vs the price we offer*

Price	DNB	PNB	NS	PB	DB	Total
\$25 - \$34.99	6%	26%	26%	20%	23%	100%
\$35 - \$44.99	6%	10%	35%	39%	10%	100%
\$45 - \$54.99	5%	20%	30%	35%	10%	100%
\$55 - \$64.99	19%	15%	12%	27%	27%	100%
\$65 - \$74.99	8%	15%	15%	15%	46%	100%
Total	8%	18%	26%	29%	19%	100%

Purchase intent of respondents grouped by price ranges

Price	DNB	PNB	NS	PB	DB	Total
\$25 - \$34.99	17%	35%	24%	17%	29%	24%
\$35 - \$44.99	17%	12%	30%	29%	11%	21%
\$45 - \$54.99	17%	31%	32%	33%	14%	28%
\$55 - \$64.99	42%	15%	8%	17%	25%	18%
\$65 - \$74.99	8%	8%	5%	5%	21%	9%
Total	100%	100%	100%	100%	100%	100%

Distribution of customers within each category of purchase intent based on the price ranges

Price	Probably Buy	Definitely Buy
\$25 - \$34.99	7	8
\$35 - \$44.99	12	3
\$45 - \$54.99	14	4
\$55 - \$64.99	7	7
\$65 - \$74.99	2	6
Grand Total	42	28

Appendix 16: Revenues Corresponding to Retail Price

Table shows revenue generated at different price levels

Price	Adjusted PB	Adjusted DB	PB * .3	DB * .8	Sum Across	Cumulative PI	Revenue
\$30	4.83%	5.52%	1.45%	4.41%	5.86%	24.14%	\$5,865,517
\$40	8.28%	2.07%	2.48%	1.66%	4.14%	18.28%	\$5,921,379
\$50	9.66%	2.76%	2.90%	2.21%	5.10%	14.14%	\$5,725,862
\$60	4.83%	4.83%	1.45%	3.86%	5.31%	9.03%	\$4,390,759
\$70	1.38%	4.14%	0.41%	3.31%	3.72%	3.72%	\$2,111,586

Appendix 17: Adjusted PI at Optimal Price Table

Table shows an optimal pricing and adjusted purchase intent including CSR initiatives

\$40	
# Definitely Buy	20
% Definitely Buy (80%)	11%
# Probably Buy	35
% Probably Buy (30%)	7.2%
Total CSR% Increase (DB & PB)	2.9%
Total Adjusted PI (including CSR)	21.1%



Appendix 18: IMC Schedule Year 1

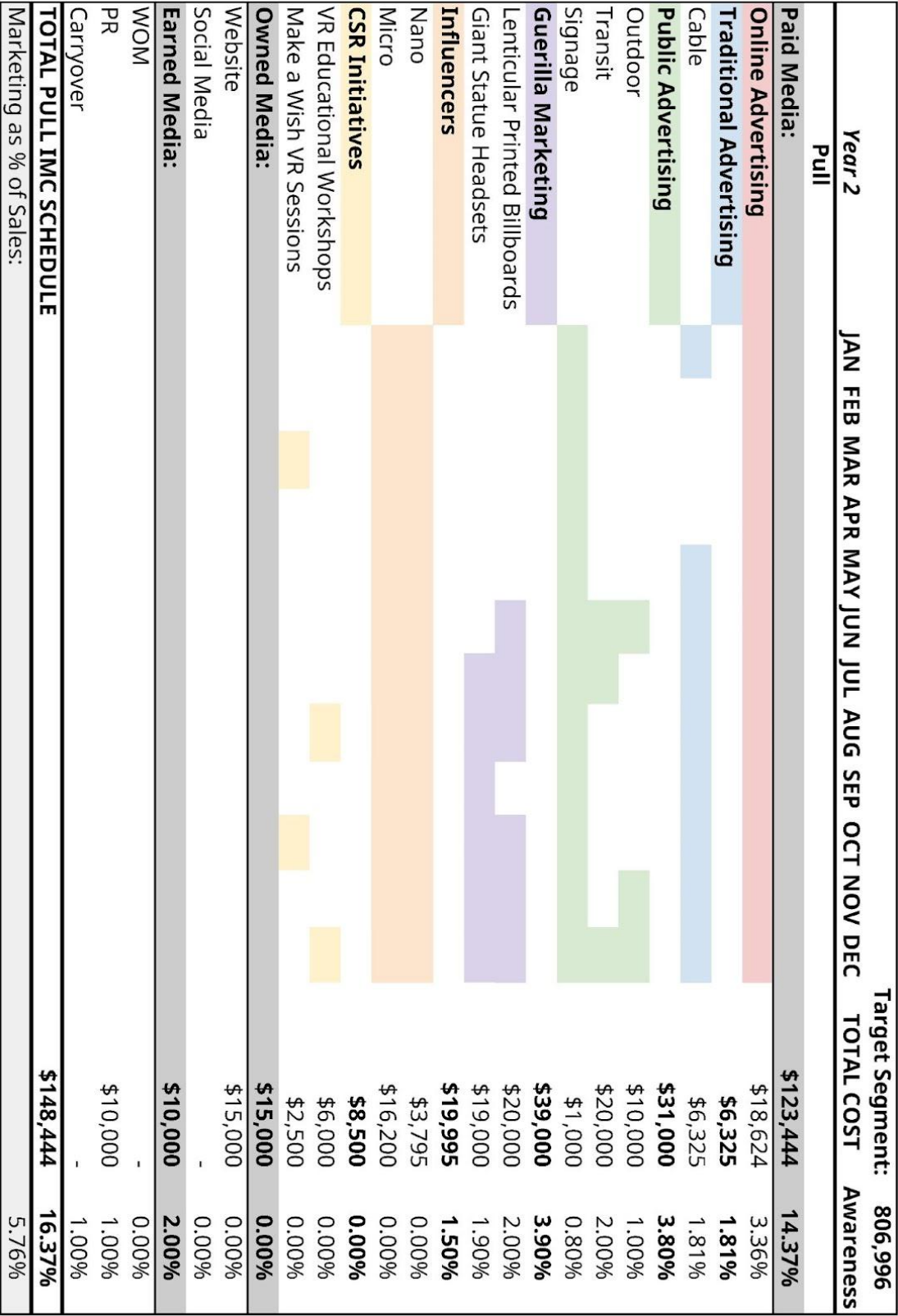
Shows how much awareness was generated and its cost in year 1

Target Segment: 810,000											
Year 1											
Pull											
Paid Media:											
Online Advertising											\$9,875 1.98%
Traditional Advertising											\$4,025 1.77%
Cable											\$4,025 1.77%
Public Advertising											\$22,000 3.00%
Outdoor											\$10,000 1.00%
Transit											\$10,000 1.20%
Signage											\$2,000 0.80%
Guerilla Marketing											\$23,000 2.30%
Lenticular Printed Billboards											\$12,000 1.20%
Giant Statue Headsets											\$11,000 1.10%
Influencers											\$10,000 0.50%
Nano											\$2,200 0.00%
Micro											\$7,800 0.00%
CSR Initiatives											\$4,500 0.00%
VR Educational Workshops											\$3,000 0.00%
Make a Wish VR Sessions											\$1,500 0.00%
Owned Media:											\$15,000 0.00%
Website											\$15,000 0.00%
Social Media											- 0.00%
Earned Media:											\$10,000 1.00%
WOM											- 0.00%
PR											\$10,000 1.00%
Carryover											- 0.00%
TOTAL PULL IMC SCHEDULE											\$98,400 10.56%
Marketing as % of Sales:											5.89%



Appendix 19: IMC Schedule Year 2

Shows how much awareness was generated and its cost in year 2



Appendix 20: IMC Schedule Year 3

Shows how much awareness was generated and its cost in year 3

Year 3												Target Segment: 793,026	
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC												TOTAL COST	Awareness
Pull													
Paid Media:												\$168,421	20.04%
Online Advertising												\$29,096	4.68%
Traditional Advertising												\$6,325	3.56%
Cable												\$6,325	3.56%
Public Advertising												\$31,000	3.80%
Outdoor												\$10,000	1.00%
Transit												\$20,000	2.00%
Signage												\$1,000	0.80%
Guerilla Marketing												\$55,000	5.50%
Lenticular Printed Billboards												\$28,000	2.80%
Giant Statue Headsets												\$27,000	2.70%
Influencers												\$30,000	2.50%
Nano												\$1,100	0.00%
Micro												\$2,400	0.00%
Mid-Tier												\$7,000	0.00%
Macro												\$19,500	0.00%
CSR Initiatives												\$17,000	0.00%
VR Educational Workshops												\$12,000	0.00%
Make a Wish VR Sessions												\$5,000	0.00%
Owned Media:												\$15,000	0.00%
Website												\$15,000	0.00%
Social Media												-	0.00%
Earned Media:												\$10,000	6.00%
WOM												-	3.00%
PR												\$10,000	1.00%
Carryover												-	2.00%
TOTAL PULL IMC SCHEDULE												\$193,421	26.04%
Marketing as % of Sales:													5.11%



Appendix 21: IMC Schedule Year 4

Shows how much awareness was generated and its cost in year 4

Year 4												Target Segment:
Pull												761,832
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC												TOTAL COST
Awareness												
Paid Media:												
												\$231,874
												24.08%
Online Advertising												\$45,049
												5.20%
Traditional Advertising												\$6,325
												3.77%
Cable												\$6,325
												3.77%
Public Advertising												\$46,000
												5.30%
Outdoor												\$15,000
												1.50%
Transit												\$30,000
												3.00%
Signage												\$1,000
												0.80%
Guerrilla Marketing												\$63,000
												6.30%
Lenticular Printed Billboards												\$32,000
												3.20%
Giant Statue Headsets												\$31,000
												3.10%
Influencers												\$40,000
												3.50%
Nano												\$1,100
												0.00%
Micro												\$3,900
												0.00%
Mid-Tier												\$9,000
												0.00%
Macro												\$26,000
												0.00%
CSR Initiatives												\$31,500
												0.00%
VR Educational Workshops												\$24,000
												0.00%
Make a Wish VR Sessions												\$7,500
												0.00%
Owned Media:												\$15,000
												0.00%
Website												\$15,000
												0.00%
Social Media												-
												0.00%
Earned Media:												\$10,000
												6.00%
WOM												-
												3.00%
PR												\$10,000
												1.00%
Carryover												-
												2.00%
TOTAL PULL IMC SCHEDULE												\$256,874
												30.08%
Marketing as % of Sales:												7.28%



Appendix 22: IMC Schedule Year 5

Shows how much awareness was generated and its cost in year 5

Year 5												Target Segment: 722,683			
Pull												TOTAL COST		Awareness	
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC															
Paid Media:												\$273,044	26.47%		
Online Advertising												\$59,719	5.70%		
Traditional Advertising												\$6,325	3.98%		
Cable												\$6,325	3.98%		
Public Advertising												\$46,000	5.30%		
Outdoor												\$15,000	1.50%		
Transit												\$30,000	3.00%		
Signage												\$1,000	0.80%		
Guerrilla Marketing												\$71,000	6.99%		
Lenticular Printed Billboards												\$36,000	3.49%		
Giant Statue Headsets												\$35,000	3.50%		
Influencers												\$50,000	4.50%		
Nano												\$1,100	0.00%		
Micro												\$23,400	0.00%		
Mid-Tier												\$6,000	0.00%		
Macro												\$19,500	0.00%		
CSR Initiatives												\$40,000	0.00%		
VR Educational Workshops												\$30,000	0.00%		
Make a Wish VR Sessions												\$10,000	0.00%		
Owned Media:												\$15,000	0.00%		
Website												\$15,000	0.00%		
Social Media												-	0.00%		
Earned Media:												\$10,000	6.00%		
WOM												-	3.00%		
PR												\$10,000	1.00%		
Carryover												-	2.00%		
TOTAL PULL IMC SCHEDULE												\$298,044	32.47%		
Marketing as % of Sales:													8.97%		

Appendix 23: Influencers Table

Lists influencers from IMC by their social media username, follower count, and cost per post.

	Nano	Micro	Mid-Tier	Macro
Username	Instagram: <i>Within</i> 	Instagram: <i>virtualrealityoasis</i> 	YouTube: <i>On Chain Gaming</i> 	YouTube: <i>ShearsyVR</i> 
Followers	7,400	22,000	130,000	601,000
Cost	\$55/post	\$300/post	\$1000/post	\$3250/post

Appendix 24: Mode of Transportation

Shows most customers prefer to get to 349 Newbury Street by taking T/Bus or Walk

Willingness to Travel						
	T / Bus	Walk	Car	Uber/Lyft	Other*	Total
Count	49	41	31	21	4	145
% of total	33.79%	28.28%	21.38%	14.48%	2.76%	100%



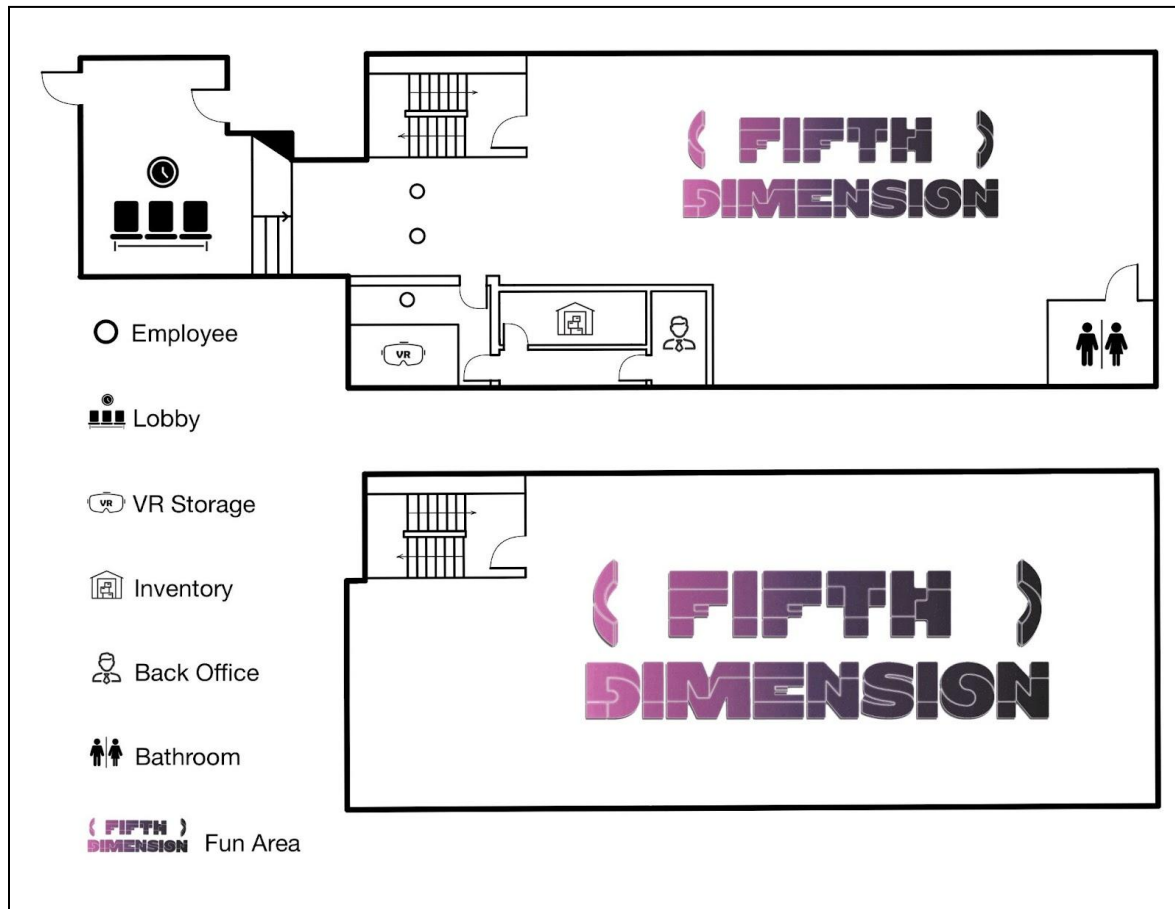
Appendix 25: Cannibalization

Table shows a percentage and number of people who are going to switch to our second location

Overlapping Percentage
28.91%
Overlapping Customers Switch to the Second Location
50%
Yr1 Targeted Segment Size
810,000
Cannibalization
117,086

Appendix 26: Facility Layout

Shows a layout of the Fifth Dimension building



Appendix 27: House of Quality

Shows customer attributes which are rated on importance for customers

		Engineering Characteristics										Competitive Evaluation				
		headset performance	headset durability	# of experiences	UI usability	roomscale space	# of employees	hours of operation	# of refreshments			Z = Us A = Competitor A B = Competitor B (5 is best)				
												1	2	3	4	5
Customer Attributes																
group friendly	5			+		+	+	+	+			B			Z	A
immersive technology	5	++		+	+	++										
affordable	4	<	<	<		<<<	<		++					AZ		B
educational	4	+		+										Z	A	B
interactive	4											A		Z		B
unique experiences	3	+		+++									A	Z		B
good service	3			<	++		+++		++			B		Z		A
comfortable	3		++		+	+	+	+	++						ZA	B
safe environment	2		+			++	+								ZA	B
accessible	2				+			++				A	Z			B
Unit of measure		mhZ	#	#	#	ft ²	#	#	#							
Competitor A's value (KTA)		7012	8	34	7	32	5	7	0							
Competitor B's value (MFA)		0	0	8	0	51k	990	7	156							
Target value		587	8	10	8	32	5	8	20							

Appendix 28: Demand Uncertainty Table

Table illustrates our demand distribution in different scenarios

Demand Uncertainty					
Year	Year 1	Year 2	Year 3	Year 4	Year 5
Best Case	51,852	85,350	132,633	142,401	155,414
Most Likely	43,210	71,125	110,528	118,667	129,511
Worst Case	34,568	56,900	88,422	94,934	103,609
Mean					
Yearly	43,210	71,125	110,528	118,667	129,511
Weekly	864	1,422	2,211	2,373	2,590
Daily	173	284	442	475	518
Standard Deviation					
Yearly	2,881	4,742	7,369	7,911	8,634
Weekly	407	671	1,042	1,119	1,221
Daily	182	300	466	500	546

Appendix 29: Capacity Analysis

Shows the capacity and capacity utilization through years 1-5

Year	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
Capacity (yearly)	-	101,212	170,089	170,089	170,089	170,089
Average Demand (yearly)	-	43,210	71,125	110,528	118,667	129,511
Average Demand Fulfilled (yearly)	-	43,210	71,125	110,528	118,667	129,511
Average Demand Fulfilled (hourly)	-	17.28	28.45	44.21	47.47	51.80
Average Capacity Utilization (yearly)	-	42.69%	41.82%	64.98%	69.77%	76.14%
Peak Demand (yearly)	-	51,852	85,350	132,633	142,401	155,414
Peak Demand Fulfilled (yearly)	-	51,852	85,350	132,633	142,401	155,414
Peak Demand Fulfilled (hourly)	-	20.74	34.14	53.05	56.96	62.17
Peak Capacity Utilization (yearly)	-	51.23%	50.18%	77.98%	83.72%	91.37%
Non-Peak Demand (hourly)	-	34,568	56,900	88,422	94,934	103,609
Non-Peak Demand Fulfilled (yearly)	-	34,568	56,900	88,422	94,934	103,609
Non-Peak Demand Fulfilled (hourly)	-	13.83	22.76	35.37	37.97	41.44
Non-Peak Capacity Utilization (yearly)	-	34.15%	33.45%	51.99%	55.81%	60.91%
Average Direct Labor (yearly)						
IT Assistants	0	1.49	2.45	3.81	4.09	4.46
IT Assistants (rounded up)	0	2	3	4	5	5
Peak Direct Labor (yearly)						
IT Assistants	0	1.79	2.94	4.57	4.90	5.35
IT Assistants (rounded up)	0	2	3	5	5	6
Non-Peak Direct Labor (yearly)						
IT Assistants	0	1.19	1.96	3.05	3.27	3.57
IT Assistants (rounded up)	0	2	2	4	4	4
Equipment						
Headsets	0	20.74	34.14	53.05	56.96	62.17
Headsets (rounded up)	0	21	35	54	57	63

Appendix 30: Peak Weekends Scheduling Year One

Shows how many employees should be hired in order to meet our demand in year one

Peak Weekends (Sat - Sun)		Hour 1	Hour 2	Hour 3	Hour 4	Hour 5	Hour 6	Hour 7	Hour 8	Hour 9	Hour 10	Total
Units demand		21	21	21	21	21	21	21	21	21	21	210
Regular DL service rate	20	40	40	40	40	40	40	40	40	40	40	405
Part-time DL service rate	0	0	0	0	0	0	0	0	0	0	0	0
Overtime DL service rate	0	0	0	0	0	0	0	0	0	0	0	0
Service Capacity Available: cust/hour		40	40	40	40	40	40	40	40	40	40	
Idle Capacity (if +)/Unserved Customers (if -)		-19	-19	-19	-19	-19	-19	-19	-19	-19	-19	
Capacity Utilization		51.98%	51.98%	51.98%	51.98%	51.98%	51.98%	51.98%	51.98%	51.98%	51.98%	
\$ cost per DL regular fully-loaded hour	\$29											
\$ cost per DL part-time fully-loaded hour	\$0											
\$ cost per DL overtime fully-loaded hour	\$0											
Sum of service units provided:												
Number Regular DL employees scheduled:		2	2	2	2	2	2	2	2	2	2	
Number Part-time DL employees scheduled:		0	0	0	0	0	0	0	0	0	0	
Number Overtime DL employees scheduled:		0	0	0	0	0	0	0	0	0	0	
Total employees scheduled:		2	2	2	2	2	2	2	2	2	2	
Regular DL fully-loaded cost		\$57	\$57	\$57	\$57	\$57	\$57	\$57	\$57	\$57	\$57	\$572
Part-time DL fully-loaded cost		\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
OT DL fully-loaded cost		\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Total labor cost per hour		\$57	\$57	\$57	\$57	\$57	\$57	\$57	\$57	\$57	\$57	\$572

Appendix 31: Peak Weekends Scheduling Year Five

Shows how many employees should be hired in order to meet our demand in year five

Peak Weekends (Sat - Sun)		Hour 1	Hour 2	Hour 3	Hour 4	Hour 5	Hour 6	Hour 7	Hour 8	Hour 9	Hour 10	Total
Units demand		62	62	62	62	62	62	62	62	62	62	622
Regular DL service rate	11	68	68	68	68	68	68	68	68	68	68	680
Part-time DL service rate	0	0	0	0	0	0	0	0	0	0	0	0
Overtime DL service rate	0	0	0	0	0	0	0	0	0	0	0	0
Service Capacity Available: cust/hour		68	68	68	68	68	68	68	68	68	68	
Idle Capacity (if +)/Unserved Customers (if -)		-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	
Capacity Utilization		91.49%	91.49%	91.49%	91.49%	91.49%	91.49%	91.49%	91.49%	91.49%	91.49%	
\$ cost per DL regular fully-loaded hour	\$30											
\$ cost per DL part-time fully-loaded hour	\$0											
\$ cost per DL overtime fully-loaded hour	\$0											
Sum of service units provided:												
Number Regular DL employees scheduled:		6	6	6	6	6	6	6	6	6	6	
Number Part-time DL employees scheduled:		0	0	0	0	0	0	0	0	0	0	
Number Overtime DL employees scheduled:		0	0	0	0	0	0	0	0	0	0	
Total employees scheduled:		6	6	6	6	6	6	6	6	6	6	
Regular DL fully-loaded cost		\$178	\$172	\$172	\$172	\$172	\$172	\$172	\$172	\$172	\$172	\$1,723
Part-time DL fully-loaded cost		\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
OT DL fully-loaded cost		\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Total labor cost per hour		\$178	\$172	\$172	\$172	\$172	\$172	\$172	\$172	\$172	\$172	\$1,723

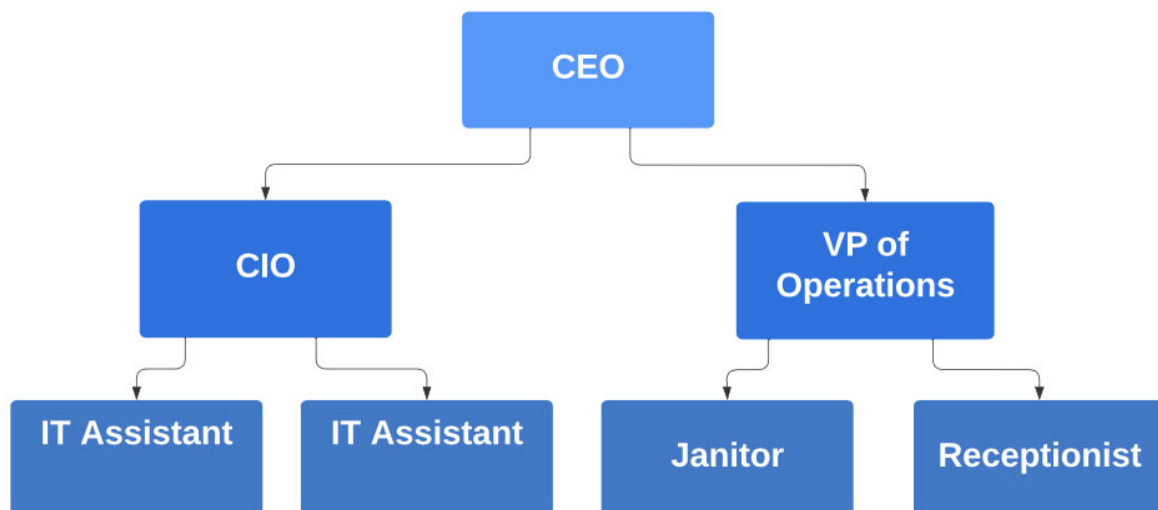
Appendix 32: Service Blueprint

Illustrates processes which gives insights on how create a better user experience

Customer actions line of information	Arrive	Pay	Preparation for Experience	Experience	Leave	
Contact persons line of visibility	Greeted by receptionist who checks availability.	Receptionist calculates expenses, receives payment.	IT Assistant brings customers to the main floor for the experiences. IT Assistants help the consumer pick experience and put on the headset.	IT Assistant checks on customers. IT Assistant helps customers switch experiences.	Receptionist thanks customers.	
Backstage contacts line of internal interaction						Headsets cleaned by IT Assistant and janitor.
Support	Business phone line, reservation system.	POS system.	Software rental.			Sanitization supplies.

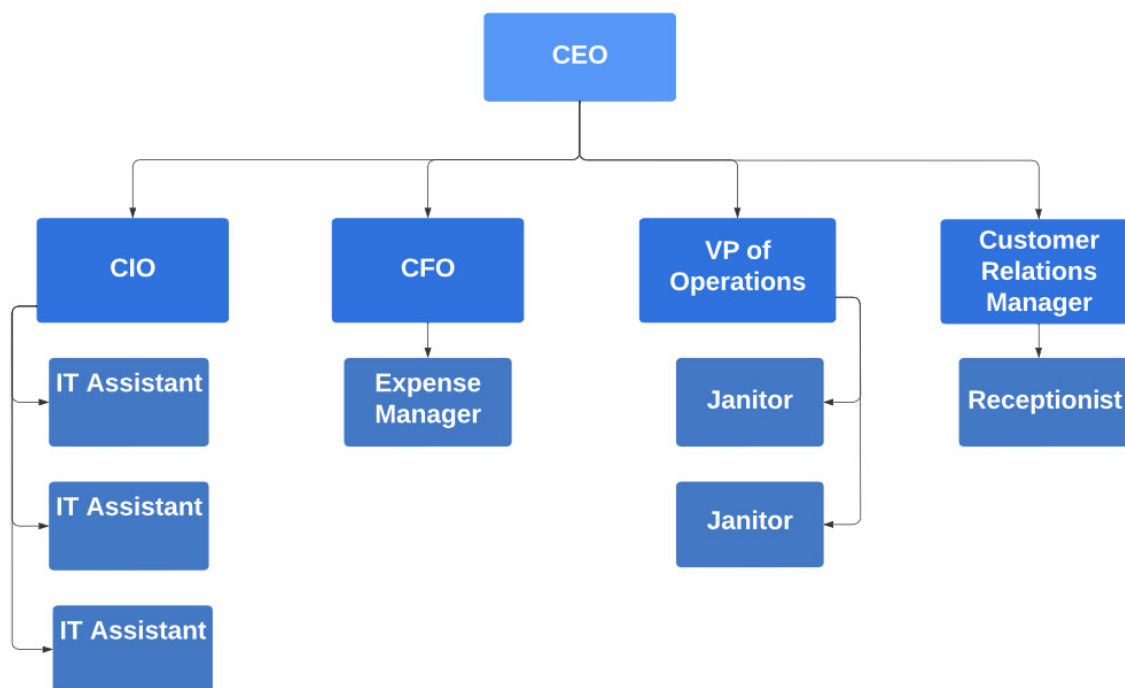
Appendix 33: Organization Chart Year One

Illustrates how Fifth Dimension is structured in year one



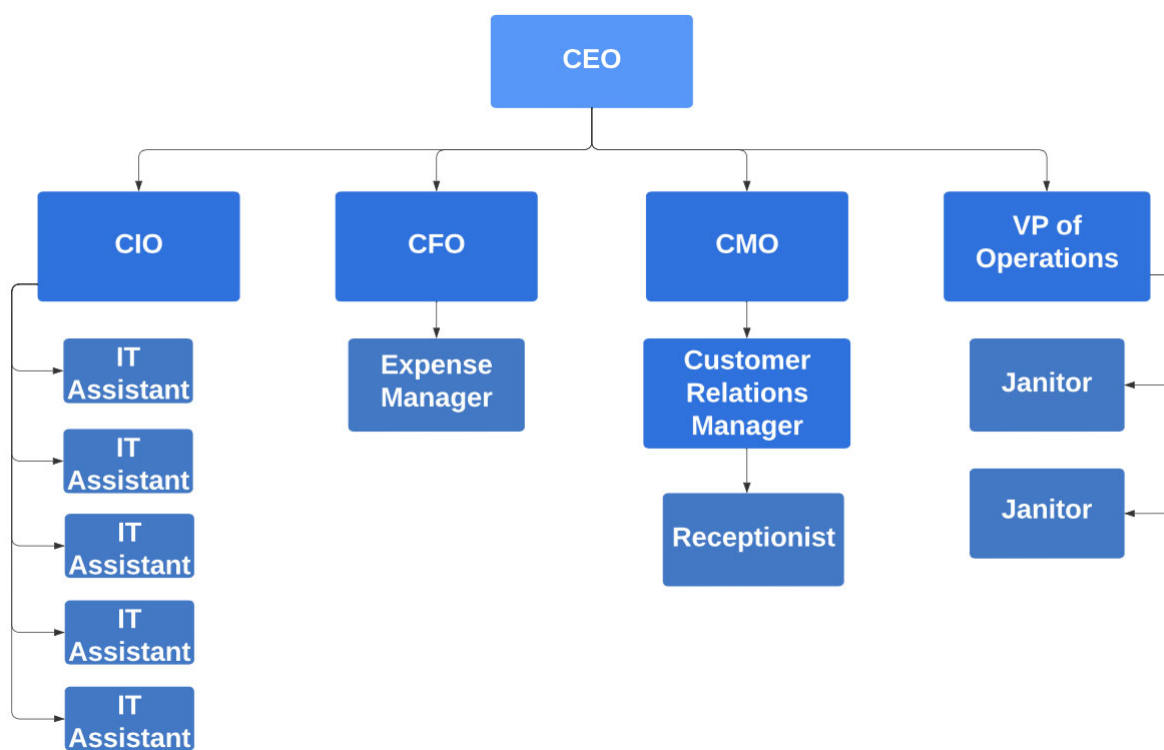
Appendix 34: Organization Chart Year Two

Illustrates how Fifth Dimension is structured in year two



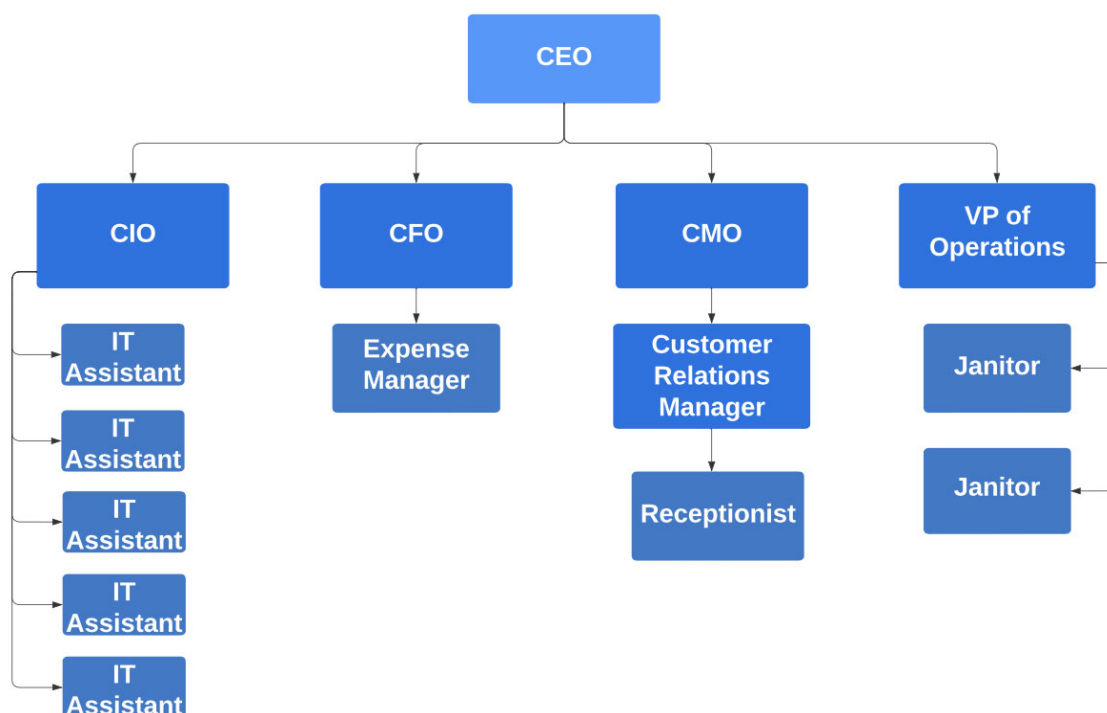
Appendix 35: Organization Chart Year Three

Illustrates how Fifth Dimension is structured in year three



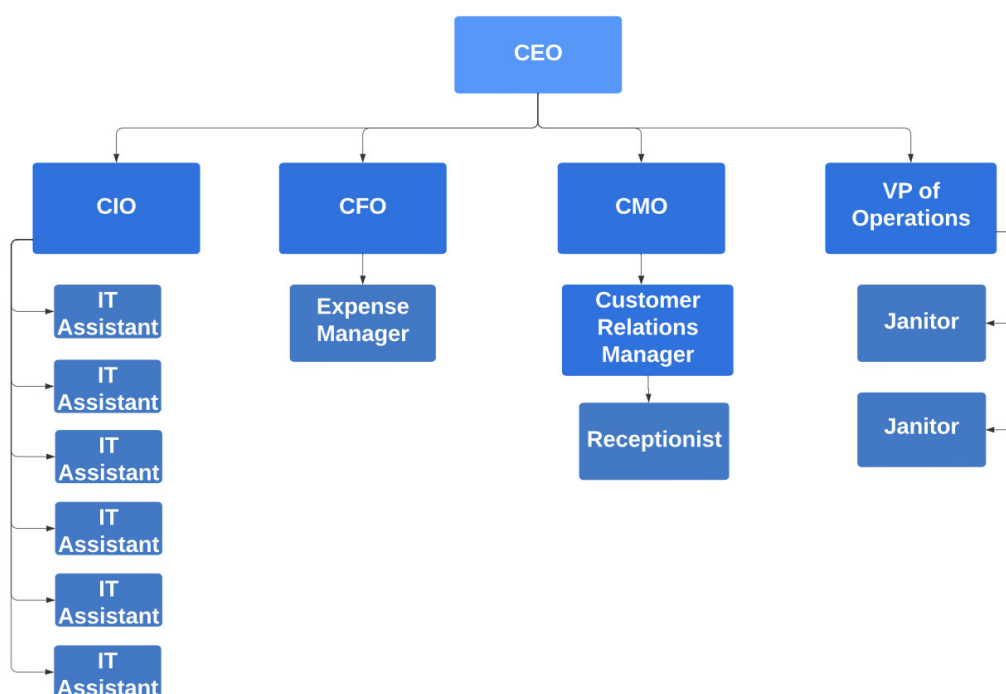
Appendix 36: Organization Chart Year Four

Illustrates how Fifth Dimension is structured in year four



Appendix 37: Organization Chart Year Five

Illustrates how Fifth Dimension is structured in year five





Appendix 38: OM Labor Year One

Table shows the number of employees used and company's expense on employees in year one

Year				1
Employee Role	Count	Training	Wage	Expense
Indirect Labor				
CEO	1	-	\$143,000	\$143,000
CMO	0	-	-	-
CIO	1	-	\$123,500	\$123,500
CFO	0	\$28,500	-	\$28,500
VP of Operations	1	-	\$93,150	\$93,150
Customer Relations Manager	0	\$18,000	-	\$18,000
Receptionist	1	-	\$65,000	\$65,000
Expense Manager	0	\$18,000	-	\$18,000
Janitor	1	\$7,500	\$32,500	\$40,000
Total Indirect	5	\$72,000	\$457,150	\$545,650
Direct Labor				
IT Assistant	2	\$16,500	\$143,000	\$159,500
Total Direct	2	\$16,500	\$143,000	\$143,000
Total Direct per Unit				\$3
Total Labor Expense	7			\$688,650

Appendix 39: OM Labor Year Five

Table shows the number of employees used and company's expense on employees in year five

Year				5
Employee Role	Count	Training	Wage	Expense
Indirect Labor				
CEO	1	-	\$167,290	\$167,290
CMO	1	-	\$144,478	\$144,478
CIO	1	-	\$144,478	\$144,478
CFO	1	-	\$144,478	\$144,478
VP of Operations	1	-	\$106,892	\$106,892
Customer Relations Manager	1	-	\$89,507	\$89,507
Receptionist	1	-	\$73,158	\$73,158
Expense Manager	1	-	\$89,507	\$89,507
Janitor	2	-	\$70,358	\$70,358
Total Indirect	10	-	\$1,030,144	\$1,030,144
Direct Labor				
IT Assistant	6	-	\$444,015	\$444,015
Total Direct	6	-	\$444,015	\$444,015
Total Direct per Unit				\$3
Total Labor Expense	16			\$1,474,159

Appendix 40: Inventory Year One

Table shows the quantity and expense of inventory purchased in year one

Year 1 (weekly)					
	Perishable	Non-Perishable			Total Inventory
Material	Bottled Water	Lysol Disinfectant Wipes	Softsoap Hand Soap	Miscellaneous	
Cost of product	\$3.69	\$16.49	\$5.97	\$100	\$126.15
Transportation Cost	\$0.37	\$1.65	\$0.60	\$10	\$12.62
Units in Order	24	480	50	50	604
Unit Cost	\$0.17	\$0.04	\$0.13	\$2.20	\$0.23
Demand per customer	1	1	0.16	1	3.16
Mean demand (weekly)	864	864	138	864	2731
SD demand (weekly)	407	407	65	407	1287
Mean LT	0.71	0.71	0.71	0.71	2.86
SD LT	0	0	0	0	0
Cycle Stock (Quantity)	432	432	69	432	1365
Average Cycle Stock (\$)	\$73.08	\$16.33	\$9.08	\$951	\$1,049
			\$25.41		
Safety Stock (Quantity)	675	675	108	675	2132
Average Safety Stock (\$)	\$114.13	\$25.50	\$14.18	\$1,485	\$1,638
			\$39.68		
Total Weekly Inventory (Quantity)	1107	1107	177	1107	3498
Total Weekly Inventory (\$)	\$187.21	\$41.83	\$23.26	\$2,435	\$2,688
			\$65.09		
Total Yearly Inventory (Quantity)	55346	55346	8855	55346	174893
Total Yearly Inventory (\$)	\$9,360	\$2,091	\$1,163	\$121,761	\$134,376
			\$3,255		

Appendix 41: Inventory Year Two

Table shows the quantity and expense of inventory purchased in year 2

Year 2 (weekly)					
	Perishable	Non-Perishable			Total Inventory
Material	Bottled Water	Lysol Disinfectant Wipes	Softsoap Hand Soap	Miscellaneous	
Order Cost	\$3.69	\$16.49	\$5.97	\$100	\$126.15
Transportation Cost	\$0.37	\$1.65	\$0.60	\$10	\$12.62
Units in Order	24	480	50	50	604
Unit Cost	\$0.17	\$0.04	\$0.13	\$2.20	\$0.23
Demand per customer	1	1	0.16	1	3.16
Mean demand (weekly)	1422	1422	228	1422	2731
SD demand (weekly)	671	671	107	671	1287
Mean LT	0.71	0.71	0.71	0.71	2.86
SD LT	0	0	0	0	0
Cycle Stock (Quantity)	711	711	114	711	2248
Average Cycle Stock (\$)	\$120.29	\$26.88	\$14.95	\$1,565	\$1,727
			\$41.82		
Safety Stock (Quantity)	1111	1111	178	1111	3510
Average Safety Stock (\$)	\$187.86	\$41.98	\$23.34	\$2,444	\$2,697
			\$65.32		
Total Weekly Inventory (Quantity)	1822	1822	292	1822	5758
Total Weekly Inventory (\$)	\$308.15	\$68.85	\$38.29	\$4,008	\$4,424
			\$107.14		
Total Yearly Inventory (Quantity)	91102	91102	14576	91102	1555
Total Yearly Inventory (\$)	\$15,408	\$3,443	\$1,914	\$200,424	\$221,188
			\$5,357		

Appendix 42: Inventory Year 3

Table shows the quantity and expense of inventory purchased in year three

Year 3 (weekly)					
	Perishable	Non-Perishable			Total Inventory
Material	Bottled Water	Lysol Disinfectant Wipes	Softsoap Hand Soap	Miscellaneous	
Order Cost	\$3.69	\$16.49	\$5.97	\$100	\$126.15
Transportation Cost	\$0.37	\$1.65	\$0.60	\$10	\$12.62
Units in Order	24	480	50	50	604
Unit Cost	\$0.17	\$0.04	\$0.13	\$2.20	\$0.23
Demand per customer	1	1	0.16	1	3.16
Mean demand (weekly)	2211	2211	354	2211	2731
SD demand (weekly)	1042	1042	167	1042	1287
Mean LT	0.71	0.71	0.71	0.71	2.86
SD LT	0	0	0	0	0
Cycle Stock (Quantity)	1105	1105	177	1105	3493
Average Cycle Stock (\$)	\$186.93	\$41.77	\$23.23	\$2,432	\$2,684
			\$64.99		
Safety Stock (Quantity)	1726	1726	276	1726	5455
Average Safety Stock (\$)	\$291.94	\$65.23	\$36.27	\$3,798	\$4,191
			\$101.50		
Total Weekly Inventory (Quantity)	2831	2831	453	2831	8947
Total Weekly Inventory (\$)	\$478.87	\$107.00	\$59.50	\$6,229	\$6,875
			\$166.50		
Total Yearly Inventory (Quantity)	141572	141572	22651	141572	0
Total Yearly Inventory (\$)	\$23,943	\$5,350	\$2,975	\$311,458	\$343,726
			\$8,325		

Appendix 43: Inventory Year 4

Table shows the quantity and expense of inventory purchased in year four

Year 4 (weekly)					
	Perishable	Non-Perishable			Total Inventory
Material	Bottled Water	Lysol Disinfectant Wipes	Softsoap Hand Soap	Miscellaneous	
Order Cost	\$3.69	\$16.49	\$5.97	\$100	\$126.15
Transportation Cost	\$0.37	\$1.65	\$0.60	\$10	\$12.62
Units in Order	24	480	50	50	604
Unit Cost	\$0.17	\$0.04	\$0.13	\$2.20	\$0.23
Demand per customer	1	1	0.16	1	3.16
Mean demand (weekly)	2373	2373	380	2373	2731
SD demand (weekly)	1119	1119	179	1119	1287
Mean LT	0.71	0.71	0.71	0.71	2.86
SD LT	0	0	0	0	0
Cycle Stock (Quantity)	1187	1187	190	1187	3750
Average Cycle Stock (\$)	\$200.70	\$44.84	\$24.94	\$2,611	\$2,881
			\$69.78		
Safety Stock (Quantity)	1853	1853	297	1853	5856
Average Safety Stock (\$)	\$313.43	\$70.03	\$38.95	\$4,077	\$4,500
			\$108.98		
Total Weekly Inventory (Quantity)	3040	3040	486	3040	9606
Total Weekly Inventory (\$)	\$514.13	\$114.88	\$63.88	\$6,688	\$7,381
			\$178.76		
Total Yearly Inventory (Quantity)	151997	151997	24320	151997	0
Total Yearly Inventory (\$)	\$25,707	\$5,744	\$3,194	\$334,394	\$369,038
			\$8,938		

Appendix 44: Inventory Year Five

Table shows the quantity and expense of inventory purchased in year five

Year 5 (weekly)					
	Perishable	Non-Perishable			Total Inventory
Material	Bottled Water	Lysol Disinfectant Wipes	Softsoap Hand Soap	Miscellaneous	
Order Cost	\$3.69	\$16.49	\$5.97	\$100	\$126.15
Transportation Cost	\$0.37	\$1.65	\$0.60	\$10	\$12.62
Units in Order	24	480	50	50	604
Unit Cost	\$0.17	\$0.04	\$0.13	\$2.20	\$0.23
Demand per customer	1	1	0.16	1	3.16
Mean demand (weekly)	2590	2590	414	2590	2731
SD demand (weekly)	1221	1221	195	1221	1287
Mean LT	0.71	0.71	0.71	0.71	2.86
SD LT	0	0	0	0	0
Cycle Stock (Quantity)	1295	1295	207	1295	4093
Average Cycle Stock (\$)	\$219.04	\$48.94	\$27.22	\$2,849	\$3,144
		\$76.16			
Safety Stock (Quantity)	2023	2023	324	2023	6391
Average Safety Stock (\$)	\$342.08	\$76.43	\$42.50	\$4,450	\$4,911
		\$118.94			
Total Weekly Inventory (Quantity)	3318	3318	531	3318	10484
Total Weekly Inventory (\$)	\$561.11	\$125.38	\$69.72	\$7,299	\$8,055
		\$195.10			
Total Yearly Inventory (Quantity)	165887	165887	26542	165887	0
Total Yearly Inventory (\$)	\$28,056	\$6,269	\$3,486	\$364,951	\$402,762
		\$9,755			

Appendix 45: Capital Expenditure Tables

Table shows total planned expenditures and gross fixed assets through years 1-5

Schedule of Planned Expenditures						
Item	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
Renovation Cost	\$396,750		\$270,000		\$111,125	
VR Program Cost						
VR Headsets	\$55,781			\$111,562		\$111,562
Yearly Total	\$452,531	-	\$270,000	\$111,562	\$111,125	\$111,562
Gross Fixed Asset	\$452,531	\$452,531	\$722,531	\$834,093	\$889,437	\$730,999

Table shows depreciation and accumulated depreciation through years 1-5

Schedule of Depreciation						
Item	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
Original VR Headsets	-	\$18,594	\$18,594	\$18,594	-	-
New VR Headsets	-	-	-	-	\$55,781	\$55,781
Renovation	-	\$79,350	\$79,350	\$79,350	\$79,350	\$79,350
Add Renovation	-	-	-	\$135,000	\$135,000	\$55,563
Yearly Total	-	\$97,944	\$97,944	\$232,944	\$270,131	\$190,693
Accumulated	-	\$97,944	\$195,887	\$428,831	\$643,181	\$563,874

Appendix 46: SG&A Expense Tables

Table shows total general & administrative expenses through years 1-5

General & Administrative Expense Schedule						
General & Administrative Expense Item	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
Rent	-	\$396,750	\$712,556	\$712,556	\$712,556	\$712,556
Utilities	-	\$39,675	\$66,675	\$66,675	\$66,675	\$66,675
Cleaning Supplies	-	\$12,432	\$20,892	\$20,892	\$20,892	\$20,892
Insurance	-	\$20,000	\$40,000	\$40,000	\$40,000	\$40,000
Total General & Administrative Expenses	-	\$468,857	\$840,122	\$840,122	\$840,122	\$840,122

Table shows sales & marketing expense through years 1-5

Sales & Marketing Expense Schedule						
Sales & Marketing Expense Items	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
Medium Size Billboard	-	\$10,000	\$10,000	\$10,000	\$15,000	\$15,000
Bus Exterior Ads	-	\$10,000	\$20,000	\$20,000	\$30,000	\$30,000
Signicades	-	\$2,000	\$1,000	\$1,000	\$1,000	\$1,000

Appendix 47: IS Expenses Table

Table shows total information system expenses through years 1-5

IS Item	Monthly Cost	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
POS System	\$100		\$1,200	\$1,200	\$1,200	\$1,200	\$1,200
ERP System	\$250		\$3,000	\$3,000	\$3,000	\$3,000	\$3,000
AT&T Internet	\$50		\$600	\$600	\$600	\$600	\$600
Business Phone	\$20						
Apple iMac	-	\$1,200	\$1,200				
Website Design	-	\$10,000					
Cordless Phone	-	\$90	\$20	\$20	\$20	\$20	\$20
Total Information Systems Expenses		\$11,290	\$6,020	\$4,820	\$4,820	\$4,820	\$4,820

Operating Cost	Monthly Cost	Start-up	Year 1	Year 2	Year 3	Year 4	Year 5
Software Rental	\$8,000	\$96,000	\$96,000	\$96,000	\$96,000	\$96,000	\$96,000

Appendix 48: FE Assumptions

Shows key financial inputs used in Balance Sheet, Income Statement and Cash Flow

(In US Dollars)	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
Initial Startup Costs (Startup Column Only)						
INITIAL INVESTMENT IN FIXED ASSETS						
Long Term Balance Sheet Items	\$452,531					
INITIAL INVESTMENT IN NET WORKING CAPITAL						
Cash	\$24,600					
Start-Up Inventory	\$26,875					
INITIAL OPERATING EXPENSES						
Facilities Operating Expenses	\$96,000					
Pre-Marketing Expenses	\$29,520					
Utilities, Insurance, Rent	\$93,771					
Subtotal Start Up Operating Expenses	\$219,291					
Operating Information (Year 1 - 5)						
Customers Served						
At Retail Location 1		43,210	71,125	110,528	118,667	129,511
On-line		-	-	-	-	-
Total Customers Served		43,210	71,125	110,528	118,667	129,511
BUSINESS SELLING PRICE						
Weighted Average Sale Price Location 1		\$38.69	\$38.03	\$38.03	\$36.13	\$34.23
Average Sale Price Online		-	-	-	-	-
COST OF SOLD						
Variable Cost per unit		\$3.54	\$3.35	\$3.58	\$3.35	\$3.66
Total Variable Costs		\$3.54	\$3.35	\$3.58	\$3.35	\$3.66
Fixed Costs to provide the service		-	-	-	-	-
Total Cost of Sales		\$3.54	\$3.35	\$3.58	\$3.35	\$3.66
SELLING & GENERAL ADMINISTRATIVE EXPENSES						
Production costs not included in Cost of Sales	-	\$468,857	\$840,122	\$840,122	\$840,122	\$840,122
Fixed Administrative Costs	\$157,500	\$545,650	\$858,220	\$958,882	\$1,010,360	\$1,030,144
Marketing Expenses (Push and Pull)		\$98,400	\$148,444	\$193,421	\$256,874	\$298,044
IS EXPENSES	\$11,290	\$6,020	\$4,820	\$4,820	\$4,820	\$4,820
Total SG&A Expenses		\$1,118,927	\$1,851,607	\$1,997,245	\$2,112,177	\$2,173,130
Balance Sheet Information (Year 1-5)						
Accounts Receivable (Days AR Outstanding)	5	Days				
Accounts Payable (Days Payable Outstanding)	15	Days				
Inventory		\$134,376	\$221,188	\$343,726	\$369,038	\$402,762
Additional Investment in Fixed Assets		-	-	-	-	-
OTHER ASSUMPTIONS						
Discount Rate	19.96%	19.96%	19.96%	19.96%	19.96%	19.96%
Tax Rate	27%					
Cash as a Percentage of Sales	5%					
Credit Payment as a Percentage of Sales	95%					
Terminal Value Method	Declining					
Declining Rate	20%					

Appendix 49: Income Statement

Table shows key financial metrics and net income through years 1-5

Income Statement						
(In US Dollars)	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
TOTAL REVENUES	-	\$1,671,590	\$2,577,485	\$3,782,880	\$3,530,546	\$3,324,448
Variable Costs	-	\$152,927	\$238,348	\$395,406	\$397,276	\$473,769
TOTAL COST OF SALES	-	\$152,927	\$238,348	\$395,406	\$397,276	\$473,769
GROSS PROFIT	-	\$1,518,663	\$2,339,137	\$3,387,474	\$3,133,271	\$2,850,679
Startup Expenses	\$219,291					
Administrative Overhead	\$157,500	\$545,650	\$858,220	\$958,882	\$1,010,360	\$1,030,144
General Operating	-	\$468,857	\$840,122	\$840,122	\$840,122	\$840,122
Marketing Expenses	\$29,520	\$98,400	\$148,444	\$193,421	\$256,874	\$298,044
IS Expenses	\$11,290	\$6,020	\$4,820	\$4,820	\$4,820	\$4,820
Software Rental Expense	\$96,000	\$96,000	\$96,000	\$96,000	\$96,000	\$96,000
Total Yearly Depreciation		\$97,944	\$97,944	\$232,944	\$270,131	\$190,693
Earnings Before Tax	(513,601)	\$205,793	\$293,586	\$1,061,286	\$654,963	\$390,856
Taxes	(138,672)	\$55,564	\$79,268	\$286,547	\$176,840	\$105,531
NET INCOME	(374,929)	\$150,229	\$214,318	\$774,739	\$478,123	\$285,325
Statement of Retained Earnings						
Net Income	(374,929)	\$150,229	\$214,318	\$774,739	\$478,123	\$285,325
Dividends	-	\$66,223	-	\$704,081	\$627,794	\$346,863
Change in Retained Earnings	(374,929)	\$84,005	\$214,318	\$70,658	(149,671)	(61,539)
Income Statement Financial Ratios						
COGS % Sales	0.00%	9.15%	9.25%	10.45%	11.25%	14.25%
Gross Margin % Sales	0.00%	90.85%	90.75%	89.55%	88.75%	85.75%
Operating Margin % Sales	0.00%	12.31%	11.39%	28.05%	18.55%	11.76%
Net Income % Sales	0.00%	8.99%	8.32%	20.48%	13.54%	8.58%
SG&A % Sales (including depreciation)	0.00%	66.94%	71.84%	52.80%	59.83%	65.37%
Marketing Expenses % Sales	0.00%	5.89%	5.76%	5.11%	7.28%	8.97%

Appendix 50: Balance Sheet

Table shows our total assets, liability, equity and key financial ratios through years 1-5

Balance Sheet						
(In US Dollars)	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
Current Assets						
Cash reserves	\$24,600	\$83,579	\$128,874	\$189,144	\$176,527	\$166,222
Accounts Receivable	-	\$21,754	\$33,543	\$49,229	\$45,945	\$43,263
Inventory	\$26,875	\$134,376	\$221,188	\$343,726	\$369,038	\$402,762
Current Assets	\$51,475	\$239,709	\$383,605	\$582,099	\$591,511	\$612,248
Fixed Assets						
Gross Fixed Assets	\$452,531	\$452,531	\$722,531	\$834,093	\$889,437	\$730,999
Accumulated Depreciation	-	\$97,944	\$195,887	\$428,831	\$643,181	\$563,874
Net Fixed Assets	\$452,531	\$354,587	\$526,644	\$405,262	\$246,256	\$167,124
TOTAL ASSETS	\$504,006	\$594,296	\$910,249	\$987,361	\$837,767	\$779,372
Current Liabilities						
Accounts Payable	-	\$6,285	\$9,795	\$16,250	\$16,326	\$19,470
Current Liabilities	-	\$6,285	\$9,795	\$16,250	\$16,326	\$19,470
Equity						
Paid in Capital	\$878,935	\$878,935	\$977,059	\$977,059	\$977,059	\$977,059
Retained Earnings	(374,929)	(290,924)	(76,606)	(5,948)	(155,619)	(217,158)
Total Equity	\$504,006	\$588,011	\$900,454	\$971,112	\$821,441	\$759,902
TOTAL LIABILITIES AND EQUITY	\$504,006	\$594,296	\$910,249	\$987,361	\$837,767	\$779,372
Net Working Capital Balance	\$51,475	\$233,424	\$373,810	\$565,850	\$575,185	\$592,778
Balance Sheet Financial Ratios						
Accounts Receivable Days		4.75	4.75	4.75	4.75	4.75
Accounts Payable Days		15	15	15	15	15
Inventory Days COGS		365	365	365	365	365

Appendix 51: Cash Flow

Table shows net present value, internal rate of return and total cash generated through years 1-5

Cash Flow						
(In US Dollars)	Startup	Year 1	Year 2	Year 3	Year 4	Year 5
Initial Investment	-	-	-	-	-	-
Net Income	(\$374,929)	\$150,229	\$214,318	\$774,739	\$478,123	\$285,325
+ Depreciation	-	\$97,944	\$97,944	\$232,944	\$270,131	\$190,693
- Change in Net Working Capital	\$51,475	\$181,949	\$140,386	\$192,040	\$9,335	\$17,593
- Capital Spending	\$452,531	-	\$270,000	\$111,562	\$111,125	\$111,562
Net Free Cash Flow	(\$878,935)	\$66,223	(\$98,124)	\$704,081	\$627,794	\$346,863
Terminal Value of Business						\$694,360
Total Cash Flow	(\$878,935)	\$66,223	(\$98,124)	\$704,081	\$627,794	\$1,041,224
Net Present Value: Including TV	\$238,116		IRR: Including TV	27.46%		

Appendix 52: Breakeven Analysis

Table shows EBIT and cash breakeven units and EBIT and cash margin of safety through years 1-5

EBIT Breakeven					
Year	1	2	3	4	5
FC	\$1,304,777	\$2,018,957	\$2,197,688	\$2,323,369	\$2,395,830
Depreciation	\$97,944	\$97,944	\$232,944	\$270,131	\$190,693
P/unit	\$39	\$38	\$38	\$36	\$34
VC/unit	\$3.54	\$3.35	\$3.58	\$3.35	\$3.66
Break-even units	39,911	61,046	70,553	79,120	84,617
Units sold	43,210	71,125	110,528	118,667	129,511
Margin of Safety	7.63%	14.17%	36.17%	33.33%	34.66%
Cash Breakeven					
Year	1	2	3	4	5
FC	\$1,304,777	\$2,018,957	\$2,197,688	\$2,323,369	\$2,395,830
P/unit	\$39	\$38	\$38	\$36	\$34
VC/unit	\$3.54	\$3.35	\$3.58	\$3.35	\$3.66
Break-even units	30,901	48,791	52,822	58,857	63,242
Units sold	43,210	71,125	110,528	118,667	129,511
Margin of Safety	28.49%	31.40%	52.21%	50.40%	51.17%

Appendix 53: Sensitivity Analysis

Quantitative risk variables ranking from the highest elasticities with NPV to the lowest

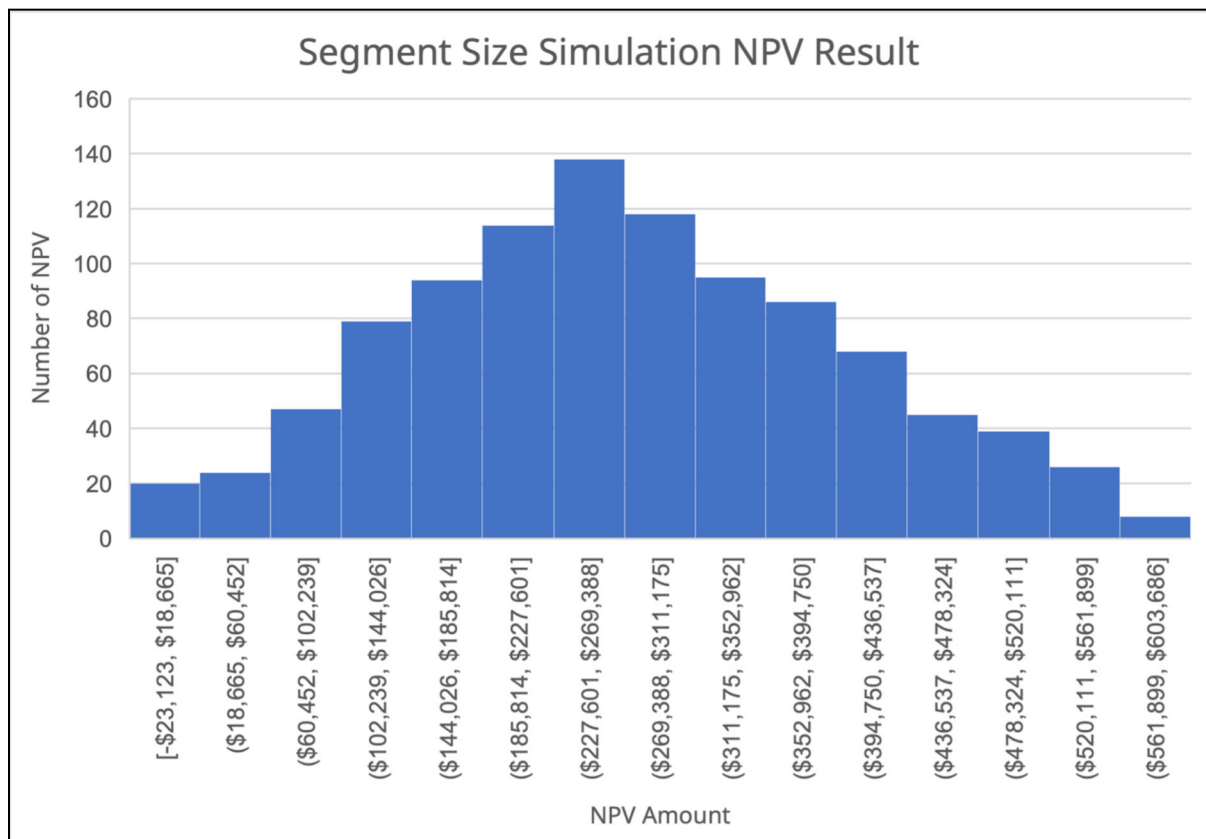
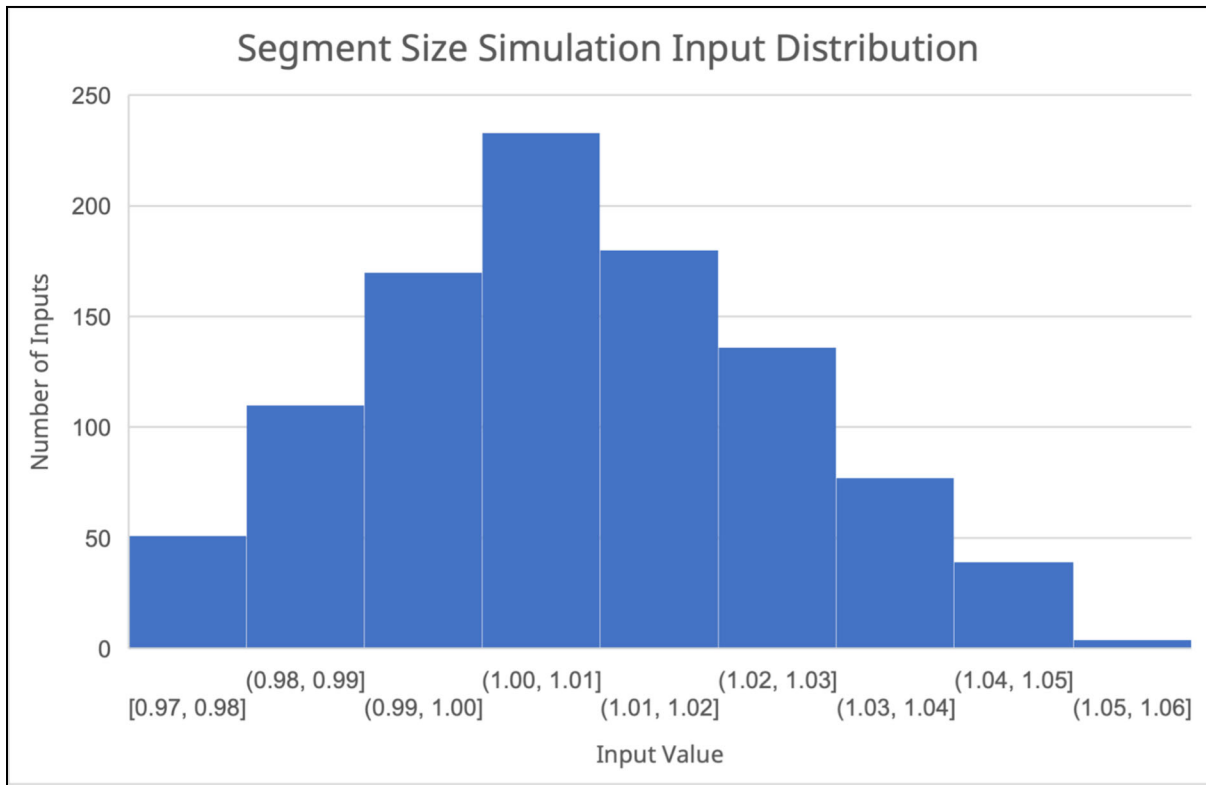
	Year 1 Value (Base case)	Breakeven Value	Breakeven % Change	Elasticities (% change due to 1% increase)
Segment Size	810,000	784,595	-3.1%	31.88%
Purchase Intent	22.46%	21.69%	-3.4%	29.17%
Awareness	10.56%	10.20%	-3.4%	29.17%
Repeat Rate	25.00%	23.92%	-4.3%	23.42%
Repeat Occasions	5	4.71	-5.9%	17.01%
Rent per sq ft	\$150	\$169.39	12.9%	-7.73%
COGS	\$3.54	\$4.45	25.8%	-3.88%
Cost of Capital	19.96%	26.38%	32.2%	-3.68%
Direct Labor Year 1	\$3.31	\$4.22	27.6%	-3.63%
Marketing Expenses	\$98,400	\$136,340	38.6%	-2.59%
Risk Premium	9.00%	15.42%	-71.3%	-1.66%
Beta	1.37	2.60	89.0%	-1.33%
Utility Cost sq ft	\$15	\$35.50	136.6%	-0.73%
Probability of Groups of 5+ People	32.86%	85%	158.7%	-0.53%
Insurance Expenses	\$20,000	\$66,840	234.2%	-0.43%

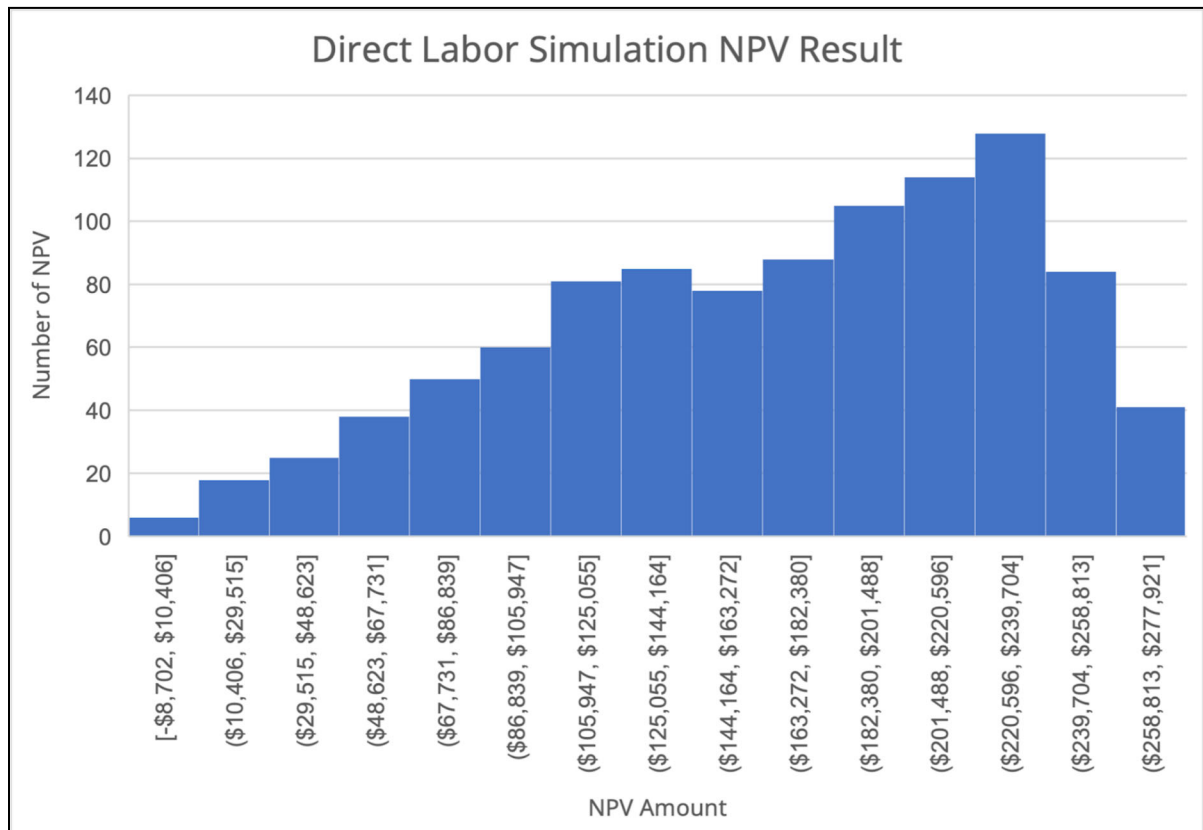
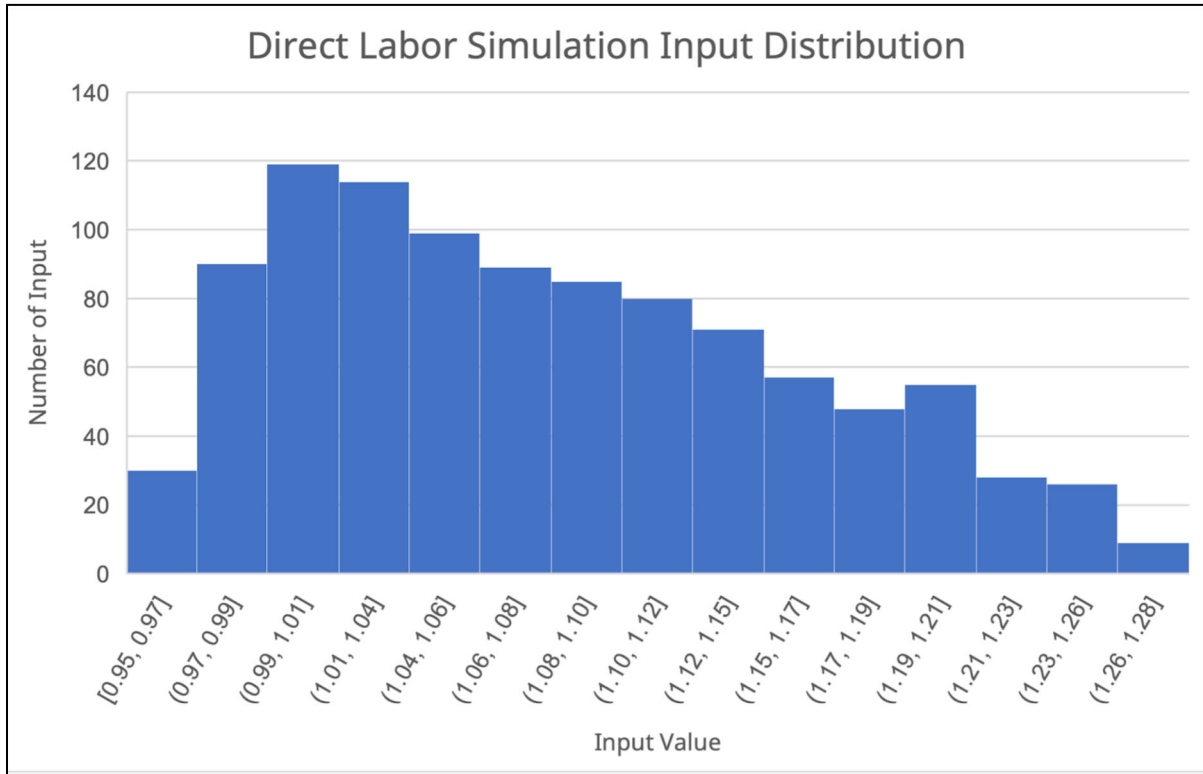


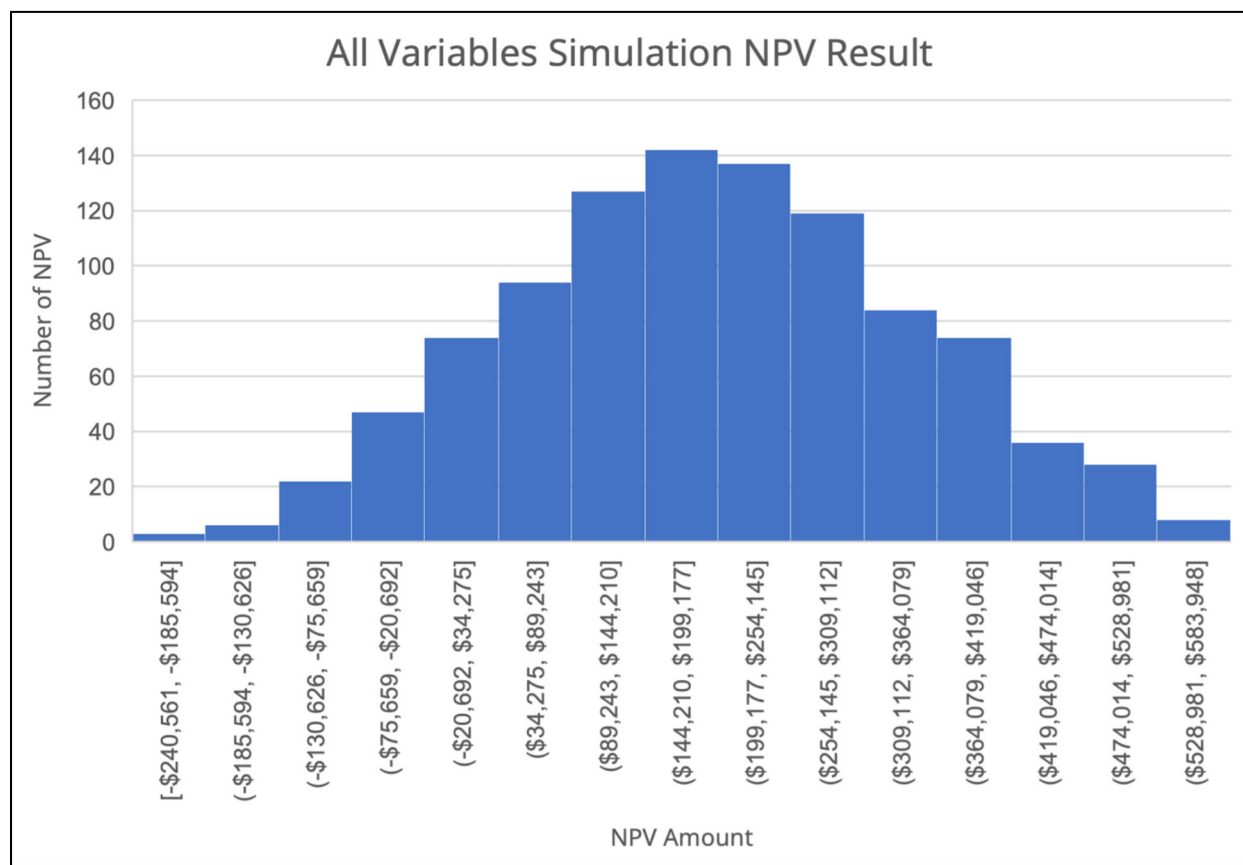
Appendix 54: Segment Size Distribution Adjusted by 1%

The lower and upper bound parameters are calculated by adjusting each variable in our segmentation tree by 1%

Segment Size Distribution Adjusted By 1%			
Base Starting Segment Size	1,600,000		
Percentage Adjustment	1% Less	Base	1% More
Income>35K, Aged 20-30 in Greater Boston	99%	100%	101%
Does Not Own Virtual Reality Headsets	87%	88%	89%
Interested in Virtual Reality	75%	76%	77%
Considers Themselves Adventurous	75%	76%	77%
Target Segment Size	780,000	810,000	850,000
Distribution Parameters	0.963	1.00	1.049

Appendix 55: Simulation One Results*Histograms of the input and output distributions for Simulation One*

Appendix 56: Simulation Two Results*Histograms of the input and output distributions for Simulation Two*

Appendix 57: Simulation Three Results*Histogram of the output distributions for Simulation Three***Appendix 58: Means and Distributions of Simulation Variables***Key metrics results of all three Simulations*

	Segment Size	Direct Labor	All Variables
Average	\$275,066	\$170,393	\$202,271
Median	\$265,614	\$181,516	\$196,167
Standard Deviation	\$140,486	\$66,442	\$148,841
%NPV < 0	1.10%	0.20%	9.00%



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Marketing Consumer Survey

New Service Survey

We are a team of Boston University students working on a new service project for a class. Please answer the questions as carefully and as honestly as possible. We will keep your responses confidential.

Q1 Do you own a Virtual Reality (VR) headset? (ex. Oculus Quest 2)

- ☐ No
- ☐ Yes



Description: We are a Virtual Reality service offering educational and entertaining experiences for individuals and groups. This service offers immersive VR adventures, such as traveling to popular and historic locations around the globe, through space, and time. A separate dining area is offered during breaks from the experiences or while waiting for a headset to become available. Customers pay per 2-hour session and can freely switch between experiences during their time.



For the following questions, please check the box that best describes your opinion.

Q2 Now that you have reviewed our service concept, how interested would you be in this service if it became available?

- ☐ Not at all Interested
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ Very Interested

Q3 At what price would you expect this service to be priced per person for 2 hours?

- ☐ \$25-\$34.99
- ☐ \$35-44.99
- ☐ \$45-\$54.99
- ☐ \$55-\$64.99
- ☐ \$65-\$74.99



Q4 At the price you just indicated, what is the likelihood that you will buy/use the service?

- ☐ Definitely Not Buy
- ☐ Probably Not Buy
- ☐ Not Sure
- ☐ Probably Buy
- ☐ Definitely Buy

Q5 How long would you expect a VR experience to last in minutes?

- ☐ <5 min
- ☐ 5-14 min
- ☐ 15-29 min
- ☐ 30-44 min
- ☐ 45-60 min
- ☐ 60+ min

Q6 What kind of food would you expect to see at a VR entertainment place? (select 1 or multiple)

- ☐ No Food
- ☐ Candy/Snacks
- ☐ Fast Food
- ☐ Full Meals
- ☐ Other _____

Q7 For the option(s) you just indicated, what is the maximum you would be willing to spend per person?

- ☐ \$0-3
- ☐ \$4-9
- ☐ \$10-15
- ☐ \$16-21
- ☐ \$22-27

Q8 How many new experiences per year would you like us to introduce to make you want to revisit our service? (0-10)



Q9 In the next year, how many times would you come to/use this VR entertainment service?

Q10 How do you expect to get to the service if it is located on Newbury Street?

- ☐ Car
- ☐ T/Bus
- ☐ Walk
- ☐ Uber/Lyft
- ☐ Other _____

Q11 How long of a wait time would be acceptable for you in minutes?

- ☐ 0 min
- ☐ 1-9 min
- ☐ 10-19 min
- ☐ 20-29 min

Q12 What party size would you expect to come to the service with?

- ☐ 1 person
- ☐ 2 people
- ☐ 3 people
- ☐ 4 people
- ☐ 5+ people

Q13 What day of the week and time frame would you be most likely to visit our service?

	Monday- Wednesday	Thursday	Friday	Saturday	Sunday
Before 11:00 am	☐	☐	☐	☐	☐
11:01am - 3:00pm	☐	☐	☐	☐	☐
3:01 pm - 7:00 pm	☐	☐	☐	☐	☐
7:01 pm - 11:00 pm	☐	☐	☐	☐	☐
After 11:00pm	☐	☐	☐	☐	☐

Q14 If the company selling this service were to host educational and interactive workshops with VR headsets for underprivileged communities, would this change your intent to purchase?

- ☐ I would be significantly more likely to buy this service
- ☐ I would be slightly more likely to buy this service
- ☐ My purchase intention would not change

Q15 If the company selling this service contributed to Make a Wish to bring VR experiences for free to children's hospitals, would this change your intent to purchase?

- ☐ I would be significantly more likely to buy this service
- ☐ I would be slightly more likely to buy this service
- ☐ My purchase intention would not change

Q16 Which of the following names do you think would best fit a service like this?

- ☐ Fifth Dimension
- ☐ Museum of Virtual History
- ☐ We R



- ☐ Virtual Travel
- ☐ Other _____

Q17 How many times do you go to an entertainment service (movie theater, museum, etc) per year?

- ☐ less than once
- ☐ 1-3 times
- ☐ 4-7 times
- ☐ 7-10 times
- ☐ Other (# of times) _____

Q18 'When I go to an entertainment service, I prefer to do so with a group.'

- ☐ Strongly disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Strongly agree

Q19 For classification purposes, please answer how you identify:

- ☐ Female
- ☐ Male
- ☐ Non-Binary
- ☐ Prefer not to Answer
- ☐ Other

Q20 What is your current age range?

- ☐ <20 years
- ☐ 20-29
- ☐ 30-39
- ☐ 40-49
- ☐ 50+

Q21 Neighborhood/place of residence? (ex. Allston, Somerville, etc.)

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Q22 Annual household income:

- ☐ less than \$35,000
- ☐ \$35,000 - \$79,999
- ☐ \$80,000 - \$119,999
- ☐ \$120,000 - \$159,999
- ☐ \$160,000 or above